

Representation of Familiarity in the Face Space

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Abstract

The visual system can successfully differentiate thousands of faces despite the fact that all faces share the same overall configuration. The face space theory suggests the visual system overcomes this homogeneity problem by encoding faces in a multidimensional space based on their perceived similarity. The face space theory is able to explain how recognition performance for faces differs based on their orientation, distinctiveness, their membership in an other-group, and caricaturing. Recent advancements in the field of face recognition research revealed that familiarity yields important recognition and perceptual differences in face processing. Specifically, face familiarity allows the ability to amalgamate the variability of a person's face from one encounter to the next so that we can recognize them when they change their appearance. Face recognition, then, relies on the ability to both differentiate similar-looking people and amalgamate dissimilar views of the same person. To investigate how the visual system achieves this remarkable ability, I investigated how familiarity, and by extension, within-person variability, is represented in the face space. I found that familiarity organizes faces in memory by decreasing the psychological distance between different views of the same person and by encoding them in reference to an identity-specific prototype. Familiarity, however, may not organize faces in memory by increasing distances between identities. Together, I show that the assumptions of the face space theory can account for recent findings in the literature and can provide a unifying account for the visual system's ability to represent between-people homogeneity and within-person variability.

Keywords: face recognition, face perception, face space, perceptual organization, visual memory

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Dedication

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Transparency Statement

I only used ChatGPT to debug code, and generate random person names as materials for a couple of the studies, which are cited below appropriately. No generative AI tools were used anywhere else in the creation of this thesis.

1 Introduction

You're so exactly like other people... the two eyes, so (marking their places in the air with his thumb) nose in the middle, mouth under. It's always the same. Now if you had the two eyes on the same side of the nose, for instance—or the mouth at the top—that would be some help. (Carroll, 1946)

As quoted in Rhodes (1996), Humpty-Dumpty's observation above highlights a remarkable capability of the visual system. All faces share the same configuration; they have the same basic components (i.e., two eyes, a nose, and a mouth) arranged in the same way. Yet, we are able to differentiate thousands of faces from one another. In fact, a modern human knows about 5000 faces (Jenkins et al., 2018). That is 5000 faces that the visual system can individuate. Just as fascinating is our ability to recognize a familiar face across changes in viewpoint and context, such as lighting, expression, age, and makeup. How does the visual system overcome the homogeneity problem while being able to tolerate the variability across different views of a familiar face? We must assume that any account of face recognition should explain how we organize faces in memory in a way that can serve to differentiate faces from one another, while encoding identity-specific information that can serve to recognize familiar individuals across several contexts. In this dissertation, I aim to reconcile a well-founded theory of differentiating faces (i.e., the face space) with relatively newer findings in the literature concerning the importance of face familiarity, which is concerned with amalgamating different views of the same face.

The Introduction is organized as follows. First, the face space theory will be reviewed with findings in the literature that the face space theory serves to unify. Namely, I will discuss the inversion effect, the distinctiveness effect, the caricature effect, the other-race effect, and

face-specific after-effects. These phenomena all provide insights into the architecture of face representations in memory that provide an account of how faces are differentiated. Second, I will argue that the face space theory has some theoretical and methodological limitations that have been revealed by more recent literature. These limitations emphasize the importance of face familiarity, with the operational definition of face familiarity as tolerance to within-person variability; that is, our ability to recognize a familiar face across several views despite the fact that faces highly vary in unsystematic ways. A summary of the most relevant studies discussed in this section can be found in Appendix A.

The aim of the current research program is to address these limitations of the face space by investigating how familiarity, and by extension, within-person variability, is represented in the face space. Specifically, I show that familiarity organizes within-person variability in the face space by increasing the perceived similarity between images of the same person, but not necessarily by increasing the perceived dissimilarity among different individuals. I also found that familiarity organizes different images of the same person around an identity-specific prototype as opposed to a general gender-matched prototype. With these findings, I hope to provide a unified account of face recognition that provides insights into how faces are organized in memory.

1.1 Telling Faces Apart

Models of representation, classification, and recognition of stimuli represented in a multidimensional space have been advanced for non-face stimuli (e.g., Nosofsky, 1986; see Valentine, 2001). In the early 90s, the face space was proposed as a theory for representing faces in a multidimensional representational space where faces are encoded according to their perceived characteristics (Valentine, 1991). According to the face space theory, each face

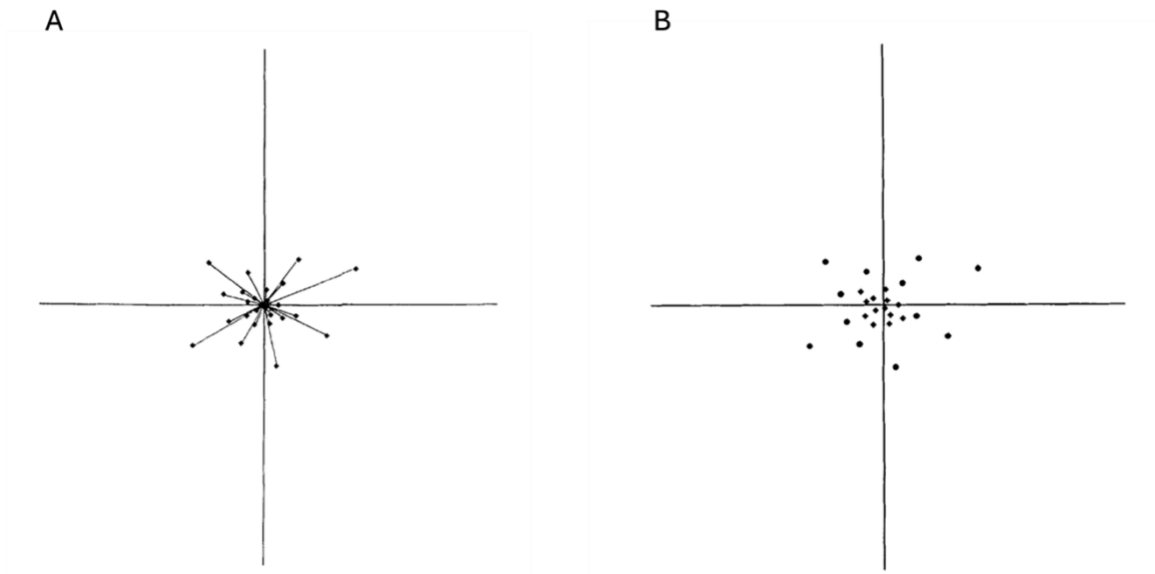
corresponds to a location in the face space. It is unclear how many dimensions there are in the face space, and what constitutes those dimensions, but they serve to differentiate faces from one another. The face space has been widely accepted as a unifying concept that could account for several findings in the literature, namely the inversion effect, the distinctiveness effect, the caricature effect, and the other-race effect (Valentine, 1991; Valentine et al., 2016).

An important property of the face space is that the origin of the face space is the central tendency of these dimensions, and previously encountered faces are normally distributed around the origin. Valentine proposed two models in which faces are encoded in the face space: the norm-based model and the exemplar-based model. The main distinction between these two models is the role of a prototype. The norm-based model suggests that an abstracted norm, or a “prototype” sits at the origin of the face space. This prototype is an idealized representation of all previously-encountered faces and does not correspond to any of the individual instances. Importantly, this model supposes that faces are encoded based on the relative distance from the prototype, so each face is an n -dimensional vector in an n -dimensional face space (Figure 1.1 **Error! Reference source not found.**A). In other words, each face is encoded based on how it differs from the prototypical face, which in turn helps overcome the homogeneity problem by representing faces as the way in which they deviate from a prototype face. According to the norm-based model, there are two steps that are assumed to be involved in the recognition of a face: (a) a stimulus is encoded as an n -dimensional vector, and (b) some decision process is used to determine whether or not this vector matches with the vector of a familiar face. This decision process will depend on the error associated with the vector (poor viewing conditions will allow for more error), and a measure of similarity between the new vector and the nearest familiar face’s vector. Although

it is not known how this similarity is calculated, it is implicitly assumed to be a Euclidean distance.

Figure 1.1

Illustration of Face Space Models



Note. (A) A two-dimensional norm-based model of the face space taken from Valentine (1991). Each encoded face is represented as a vector. At the origin is the prototype face, which is assumed to be a central tendency of all faces. (B) A two-dimensional illustration of the exemplar-based model taken from Valentine and Endo (1992). In this model, each face is encoded as an absolute point as opposed to a vector. There is no prototype at the origin, but since faces are assumed to be normally distributed, they are clustered more densely around the central tendency. Reused with permission from Sage Publications (License Number: 6211051180813).

The exemplar-based model, on the other hand, does not necessitate the abstraction of a prototype and suggests that faces are located as individual points in the face space (Figure 1.1B). The origin of the face space, then, is only where the highest density of faces is found

and does not include a prototype that plays a role in encoding faces. The similarity between two points (i.e., faces) in the face space is simply a monotonic function of the distance between them. The decision process, then, will depend on the estimate of error associated with encoding (similar to the norm-based model), and the distance between the location of the encountered stimulus and the nearest familiar face.

Valentine (2001) distinguishes the norm-based model from the prototype models of semantic categorization in that the norm-based model still assumes that exemplars are used in recognition, it is only the way faces are encoded that differs from the exemplar-based model. Therefore, the exemplar-based models and the norm-based models often make similar predictions regarding how faces are differentiated, but they offer different underlying mechanisms for several phenomena in the literature. In the following sections, phenomena that are unified within the face space theory will be described, and their implications for the face space theory along with their relationship with the norm-based and the exemplar-based models will be discussed.

1.1.1 The Inversion Effect: Looking at Upside-Down Faces

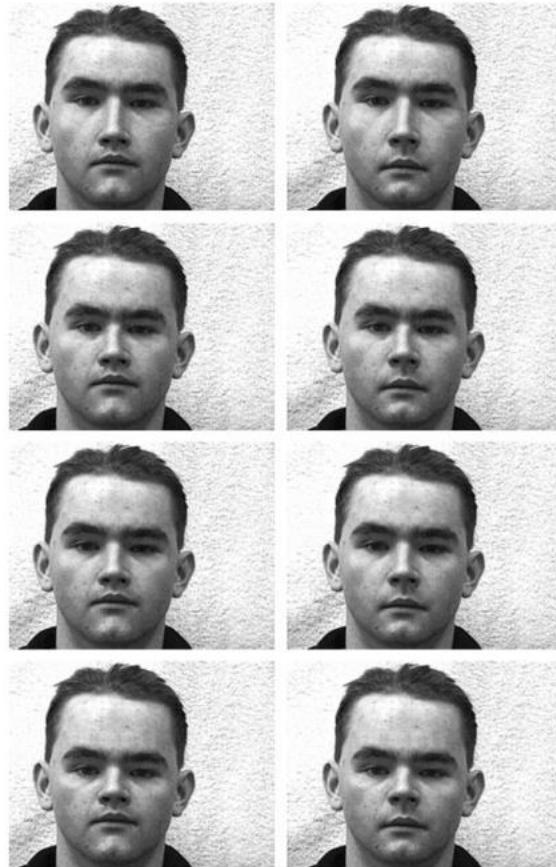
Recognition of faces, compared to other classes of mono-oriented (i.e., viewed in one orientation) visual stimuli, suffer disproportionately from inversion (e.g., Diamond & Carey, 1986; Fraire et al., 2000; Valentine, 1988; Yin, 1969). Yin (1969) conducted a study where observers studied images of faces, houses, airplanes, and stick figure men in motion. Following the study phase, observers were given a two alternative force choice (2AFC) recognition test where they were asked to indicate which image appeared in the study list. For some stimuli, the orientation of the study and the test images matched (i.e., upright-upright, inverted-inverted), and for others the orientation of the study and the test images mismatched

(i.e., upright-inverted, inverted-upright). In all conditions that involved inversion, recognition performance was worse, but importantly, inversion was disproportionately detrimental to face recognition performance, despite all the image categories being mono-oriented. These results revealed that although all mono-oriented objects suffer from inversion, faces suffer from inversion even more.

Yin (1969) argued that the disproportionate inversion effect of faces indicates that faces are a special class of visual objects. This speculation raised the question regarding what property or properties of faces make them special and also makes them vulnerable to the inversion effect. The most prevalent hypothesis is that faces suffer from inversion because they are processed holistically (e.g., Young et al., 1987) – all faces share the same configuration, so the ability to differentiate faces must rely on detecting subtle differences in the overall configuration of faces. This cannot be accomplished if faces are decomposed into parts, since faces with different configurations appear as different identities even if the individual parts are held constant (Figure 1.2). Inversion is assumed to disrupt holistic processing (Diamond & Carey, 1986), which makes faces more difficult to recognize when inverted.

Figure 1.2

Faces That Share Different Configuration with Constant Features



Note. Taken from Freire et al. (2000). Reused with permission from Sage Publications (License Number: 6211051358763).

Evidence for holistic processing of faces came from Young et al.'s (1987) composite face illusion. When the top half and the bottom half of two different identities are used to create a composite face, the perception of a third identity emerges (Figure 1.3). Moreover, when observers are shown a composite face and are asked to identify one of the constituent parts, recognition slows down because the top and the bottom halves interfere with each other. This impairment in recognition is eliminated when the top and the bottom halves are misaligned or when the faces are inverted, suggesting that faces are processed in a holistic manner, and inversion interferes with holistic processing.

Figure 1.3

A Composite Face Illusion Using Margaret Thatcher and Shirley Williams



Note. Taken from Young et al. (1987). Reused with permission from Sage Publications (License Number: 6211080148259).

Farah et al. (1995) investigated whether holistic processing in general is impaired by inversion. In their experiment, they created dot patterns that only differed in subtle configuration. Observers studied these dot patterns, some observers saw mono-coloured dot patterns (i.e., holistic processing) and some saw dot patterns with multiple colours to encourage part decomposition by using the Gestalt similarity principle. During the recognition test, these dot patterns were shown either upright or inverted. As predicted, recognition of mono-coloured dot patterns was disrupted by inversion to a greater extent than multi-coloured dot patterns. In another experiment, Farah et al. (1995) applied this to faces by either presenting whole faces or faces decomposed into parts. When recognition was tested using whole faces (upright or inverted) they found that the recognition performance of observers who studied whole faces suffered from inversion more than the ones who studied parts of faces. These results suggest that inversion disrupts holistic processing for non-face stimuli, and face recognition is particularly vulnerable to inversion due to holistic processing.

It appears that inversion disrupts holistic processing not only for face recognition but also for face perception. Freire et al. (2000) presented observers with a pair of images and asked them whether the two images depicted the same person (i.e., matching task). They found that matching performance was impaired by inversion when the pair of faces shared the same parts (e.g., eyes, nose, mouth) with varying configuration (as in Figure 1.2), but not when the pair of faces shared the same configuration with varying parts.

Diamond and Carey (1986) examined whether there are other classes of visual stimuli that use holistic processing. They argued that expertise would mediate the ability to detect subtle differences in configuration (called second-order configuration), so any visual object would suffer from inversion by experts of that visual object. To test this prediction, they gathered dog experts and novices and gave them a recognition task using dog images. They found that dog experts' recognition of dogs was impaired by inversion to a larger extent than novices' recognition of inverted dogs. The authors suggested that expertise in any visual object class that shares the same overall configuration (e.g., two eyes above a nose above a mouth) suffers from inversion, as long as they differ based on their second-order configurational features. Therefore, they argued that faces are not inherently special, but rather our expertise for faces makes them a special class of stimuli.

1.1.1.1 Implications for the Face Space. The face space is agnostic towards the inversion effect (Valentine, 2016). Goldstein and Chance (1980; Valentine, 1988) argue that a face prototype develops with extensive experience with upright faces. This face prototype improves the ability to recognize upright faces, but it comes at a cost of inflexibility in recognizing faces with transformations such as inversion. This prototype rigidity is similar to the face prototype proposed by the norm-based model. According to the norm-based model,

the first step of recognition involves encoding the face as an n-dimensional vector, and the second step involves a decision process to determine whether the face is familiar. A factor that contributes to this decision process is the error associated with encoding the stimulus, which will be greater with more difficult viewing conditions. The exemplar-based account makes a similar assumption, where a face is first encoded as a point in the n-dimensional space, and the decision process is influenced by the error associated with encoding the stimulus. Inversion, then, is simply a transformation in the stimulus that increases the encoding error. Therefore, both models make a similar prediction that inverting a face will make it more difficult to recognize. Importantly, both models also predict that the effect of inversion will be greater for typical faces than for distinctive faces, because the increased error in encoding an inverted typical face will make it more likely to activate a nearby representation. This prediction was confirmed in Valentine (1991) who found that inversion caused a greater impairment in recognizing typical faces compared to distinctive faces.

1.1.2 The Distinctiveness Effect: The Role of Inter-Item Similarity

Superior recognition performance is observed for distinctive faces compared to typical faces (e.g., Bartlett et al., 1984; Cohen & Carr, 1975; Johnston et al., 1997; Shepherd et al., 1991; Valentine & Bruce, 1986a; 1986b). For example, Going and Read (1974) asked observers to rate unfamiliar faces on “uniqueness” (compared to the general population; i.e., absolute distinctiveness), and later found recognition accuracy to be higher for the faces that were rated high in uniqueness compared to low in uniqueness. Cohen and Carr (1975) replicated this effect by asking observers to rank faces based on distinctiveness (within the stimulus set; i.e., relative distinctiveness) and found lower misses and false alarms for recognizing distinctive faces. Light et al. (1979) found that “unusual” faces were more

accurately and confidently recognized under both incidental and intentional learning conditions, and with presentation times varying between three seconds to 15 seconds per item. Using personally familiar and famous faces, Valentine and Bruce (1986a; 1986b) found that distinctive faces were accepted as familiar faster than typical faces. Shepherd et al. (1991) also found that distinctive faces were recognized with higher hit rates, higher sensitivity (d'), and shorter reaction time.

It is challenging to define distinctiveness for faces compared to other classes of visual stimuli for which distinctiveness can be described on a single dimension (e.g., colour; Valentine, 2001). Nevertheless, the effect seems to be robust regardless of how distinctiveness is operationalized in the literature. Some researchers have adopted terms like “atypical” (e.g., Light et al., 1979; Vokey & Read, 1992), “unusual” (e.g., Bartlett et al., 1984), “unique” (e.g., Going & Read, 1992), or “distinctive” (e.g., Valentine & Bruce, 1986a; 1986b). Commonly, distinctiveness is measured through asking observers to rate the faces based on how easy it is to spot them in a crowd (e.g., at a busy railroad station; Valentine & Bruce, 1986a; 1986b; Valentine & Endo, 1992), to rate the faces that look more similar to a typical high school male student (Light et al., 1979), or simply to rate or rank faces for distinctiveness (e.g., Cohen & Carr, 1975). Some present observers with pairs of faces and ask them to rate the similarity of the pair (e.g., Johnston et al., 1997; Light et al., 1979), so items that are rated as most dissimilar are described as most distinctive. Although these ratings are subjective, they have high inter-rater agreement (Cohen & Carr, 1975; Valentine, 2001). In fact, Valentine and Endo (1992) found cross-cultural agreement on distinctiveness. British and Japanese observers rated the distinctiveness of British and Japanese faces, and there was high agreement about which faces were more distinctive irrespective of the observers’ and the

faces' race. Subjective ratings can also be predicted by more objective measures of distinctiveness. For example, Bruce et al. (1994) gathered eccentricity measures by computing distances between facial features (e.g., nose length, the beakiness of the nose) and found that subjective ratings of distinctiveness correlated with the eccentricity measures. Therefore, although it is difficult to pin down what exactly constitutes distinctiveness for faces, different ways of measuring and describing distinctiveness seem to refer to the same psychological construct of typicality.

To investigate what accounts for the distinctiveness effect, Light et al. (1979) investigated the possibility that the distinctiveness effect was mediated by greater attention being paid to distinctive faces, in line with the depth of processing account proposed by Craik and Tulving (1975). Accordingly, they presented observers with distinctive and typical faces and asked them to judge their likeability (elaborate processing) or gaze direction (shallow processing) under both incidental and intentional learning conditions. They hypothesized that if the distinctiveness effect is mediated by depth of processing, there would be an interaction between typicality and the type of judgment (likeability or gaze direction) in that the recognition advantage for distinctive faces would be reduced when measured in likeability ratings. They did find such an interaction for false alarms, but it fell short of statistical significance and therefore was inconclusive.

Shepherd et al. (1991) explored whether the distinctiveness effect is a result of encoding distinctive features. They hypothesized that shortening the presentation time during study would prevent observers from encoding distinctive features and in turn would diminish the effect. Although Light et al. (1979) also manipulated presentation durations, they allowed for about three seconds per item, which may have been long enough for distinctive features to be

encoded. As such, Shepherd et al. (1991) shortened the presentation time to 1 second per item. They replicated the distinctiveness advantage for hits, false alarms, and reaction times. However, they did not find an interaction between distinctiveness and presentation time, thus, shortening the presentation duration did not reduce the distinctiveness effect. Further, Valentine and Bruce (1986b) presented observers with distinctive and typical faces either intact or jumbled and asked them to indicate whether the stimulus was a face or not (i.e., face classification task). They found that intact typical faces were classified as faces faster than intact distinctive faces. If the distinctiveness effect was based on greater attention being paid to distinctive faces or distinctive features, they would be classified as faces faster than typical faces. These studies show that the distinctiveness effect is not a result of attention.

Bartlett et al. (1984) suggested that the distinctiveness effect was solely based on “context-free familiarity” information. They claimed that previously-unseen typical faces elicit a sense of familiarity due to their resemblance to most previously-encountered faces, and this in turn results in more false alarms for typical faces. Therefore, a pre-familiarization process with all faces should eliminate or even reverse the recognition advantage for distinctive faces. One set of observers rated unfamiliar typical and distinctive faces based on how familiar they felt (e.g., may have seen before on a city street), and found that typical faces indeed received higher familiarity ratings. Another set of observers were presented with unfamiliar typical and distinctive faces and were asked to rate them on friendliness followed by a recognition test on those faces. Some of these observers were additionally exposed to all the faces prior to the recognition test (i.e., “familiarized”). They found that familiarizing observers with the faces prior to the recognition test reduced the recognition advantage for distinctive faces. In other words, exposure to a distinctive face resulted in a higher increment

of familiarity than exposure to a typical face. However, this finding was observed only for false positives.

Valentine and Bruce (1986b) further tackled the role of context-free familiarity on the distinctiveness effect by using a different experimental paradigm that looked into response latencies. They argued that if context-free familiarity forms the basis of the distinctiveness effect in that distinctive faces gain more familiarity from each additional exposure, already familiar distinctive faces should feel more familiar than familiar typical faces. This would be reflected as a positive correlation between distinctiveness and familiarity ratings. To test this prediction, they first collected distinctiveness ratings from unfamiliar observers. They then presented those faces to familiar observers and gave them a familiarity decision task (i.e., Is this face familiar?) while measuring their reaction time. At the end of their experiment, the same familiar observers were asked to make familiarity judgments on the faces (i.e., which of these faces are you most familiar with?). They found that distinctive familiar faces, and the faces that were rated as most familiar, were classified as familiar faster than typical or less familiar faces. Critically, however, distinctiveness ratings and familiarity ratings were not correlated. In other words, they found that the faces that were most distinctive were not necessarily the most familiar, disconfirming the context-free familiarity hypothesis. This was also demonstrated in Vokey and Read (1988), where general familiarity judgments and the distinctiveness of faces contributed to recognition performance, but familiarity and distinctiveness factors were not related to each other. Therefore, context-free familiarity does not seem to be the only factor underlying the distinctiveness effect.

In Light et al.'s (1979) study, inter-item similarity ratings (of pairs of faces) were correlated with distinctiveness ratings, so distinctiveness must have its structural basis in

inter-item similarity. They explained this by proposing a two-component (i.e., schematic and specific) memory model. Typical faces activate the prototype face (i.e., schematic face) for faces due to their resemblance with most previously encountered faces. This schematic memory is in turn responsible for increased false alarm rates for typical faces. Distinctive faces, on the other hand, deviate from this schematic representation and thus access a “specific” memory component, resulting in higher hit rates and reduced false alarm rates for distinctive faces. In line with this theory, Vokey and Read (1992) proposed that distinctiveness was composed of two orthogonal components: general familiarity (i.e., context-free familiarity) that results in reduced differentiation for typical faces (increased false alarms), and memorability that results in increased differentiation for distinctive faces (decreased false alarms and increased hits). They asked observers to provide ratings of attractiveness, typicality, likability, familiarity, and memorability on a set of faces. Applying Principal Components Analysis (PCA) on the ratings, they found that the familiarity and memorability factors completely overlapped with the variance explained by the typicality factor. In another experiment, they gave observers a recognition test and found that the familiarity component was reflected exclusively in false alarms and not in hits, whereas the memorability component was mostly reflected in hits. Importantly, when familiarity and memorability were factored into the regression, adding typicality did not contribute to the model, suggesting that typicality could be described as a summation of familiarity and memorability.

In summary, distinctive faces are more readily recognized than typical faces. Distinctiveness of a face has its structural basis in the inter-item similarity in the population of face representations (Light et al., 1979), and is reflected by two orthogonal components:

familiarity and memorability (Vokey & Read, 1992). The distinctiveness effect reveals a noteworthy property of face representations: similarity of faces plays an important role in encoding and recognizing faces. Typical faces are similar to one another, and they access a prototypical representation of faces. Distinctive faces are dissimilar to this schematic representation; therefore, they access a more specific memory component which in turn facilitates their recognition.

1.1.2.1 Implications for the Face Space. Distinctiveness, then, is a function of inter-item similarity within the population of face representations (Light et al., 1979) and is described by general familiarity (reflected by typical faces) and memorability (reflected by distinctive faces; Vokey & Read, 1992). In line with the face space theory, where faces are also encoded in dimensions that serve to differentiate them, typical faces are clustered near the origin where the highest exemplar density is found (i.e., high general familiarity), and distinctive faces are encoded in more isolated regions where the exemplar density is low (i.e., high memorability). Typical faces, then, have several neighbours, which make them more likely to be confused with one another, resulting in higher false alarm rates. Distinctive faces, on the other hand, are more isolated, which makes them more likely to be accurately recognized and less likely to be mistakenly recognized as having been seen before (lower false alarm rates).

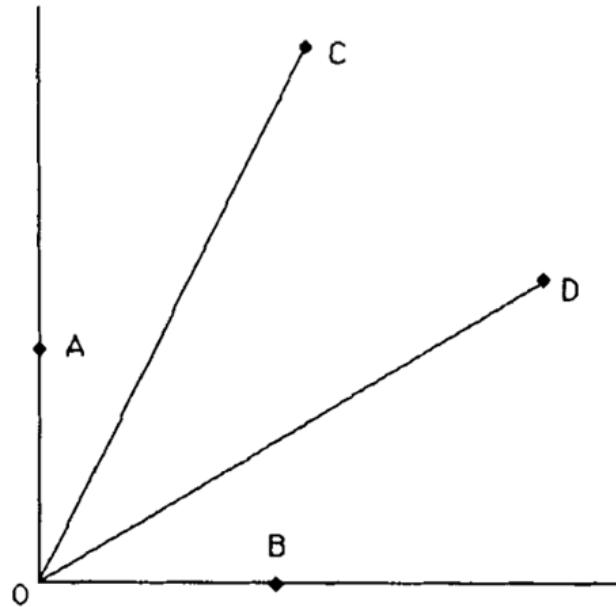
Since the distinctiveness effect is assumed to be a result of exemplar density, both the norm-based and the exemplar-based models make similar predictions about the distinctiveness effect. When a distinctive familiar face is encountered, it falls closer to the representation of the target face than another neighbour, making it faster and more accurate to recognize the face. When a typical familiar face is encountered, it falls close to the target representation but also to a near neighbour, making it slower and less accurate to recognize.

When an unfamiliar distinctive face is encountered, it will be encoded in a location where there are not many near neighbours, decreasing the possibility of false alarms. When a typical unfamiliar face is encountered, it will fall into a dense location, making a near face more likely to be falsely recognized. Exemplar density also provides an account for why typical faces are classified as faces faster (Valentine & Bruce, 1986b). Assuming that the classification task is done by calculating the similarity between the target and the central tendency of the category, when a typical face is encountered, it falls closer to the central tendency, making it faster to classify it as a face.

However, the mechanisms underlying the distinctiveness effect are described differently by the two models. According to the norm-based model, the similarity of two faces that are separated by the same distance depends on the distance from the norm. If two faces are closer to the norm, the angle that separates them is larger, whereas if two faces are further away from the norm, the angle that separates them is smaller (Figure 1.4). This leads to the prediction that two distinctive faces (that are further away from the norm) that are separated by the same distance from two typical faces would be more difficult to differentiate. Since this prediction goes against what would be predicted based on the distinctiveness effect, we can assume that the effect of exemplar density on recognition is larger than the opposing effect of the distance from the norm. Therefore, the exemplar-based model may provide a more parsimonious account for the distinctiveness effect, where the effect is based solely on the exemplar density in the face space.

Figure 1.4

Two Sets of Faces that Are Equidistant but Are Not Equally Dissimilar



Note. The distances between AB and CD are the same, but the OA and OB are separated by a larger angle than OC and OD. Reused with permission from Sage Publications (License Number 6211080362169).

1.1.3 The Caricature Effect: Not All Distortions Are the Same

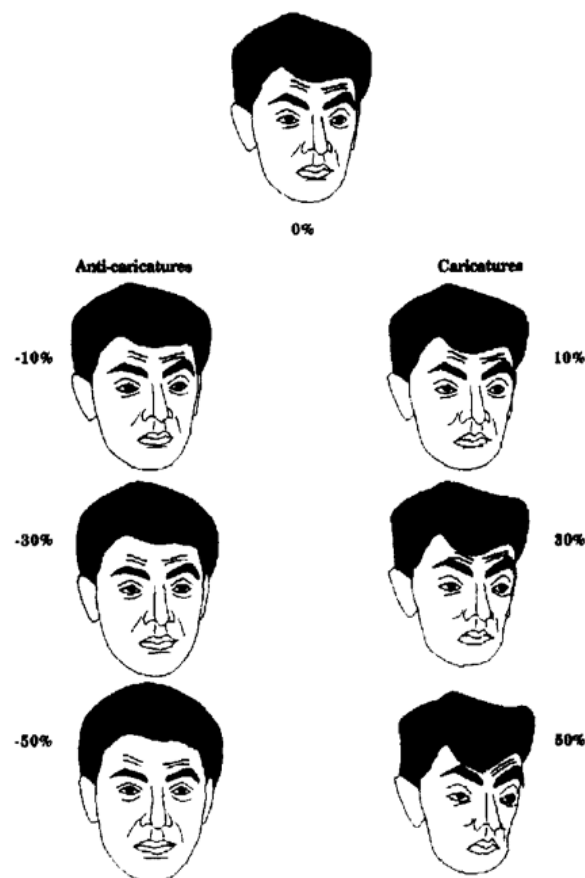
Can someone's distorted image be more recognizable than an undistorted one?

Caricatures - images where distinctive characteristics of a face are exaggerated - are recognized at least as well as undistorted images, and are better recognized than anti-caricatures, images where distinctive features of a face are de-emphasized. (e.g., Benson & Perrett, 1991; Gibling & Bennett, 1994; Mauro & Kubovy, 1992; Rhodes et al., 1987). Caricatures are either artistic, which are created through caricature artist drawings, or computer-made, where the differences between the target's face and an average (i.e., norm) face are enhanced. For example, if a face has smaller than average eyes, they are made even

smaller since the caricaturing process increases the difference between the prototype and the target face (Figure 1.5). For computer caricatures, the degree of enhancement is commonly varied in experiments, with greater percentages reflecting greater deviation from the norm. Anti-caricatures, then, are created by *decreasing* the difference between the image and the prototype, making the face look more typical in appearance. Therefore, caricatures (and anti-caricatures) represent a deviation from (or towards) a prototype, similar to distinctiveness of faces, with the important distinction that distinctiveness in caricatures is varied *within* a face instead of across faces (Rhodes et al., 1987).

Figure 1.5

A Veridical Drawing of Rowan Atkinson (0%) and Its Varying Levels of Caricatures and Anti-caricatures.



Note. Understanding face recognition: Caricature effects, inversion, and the homogeneity problem, Rhodes and Tremewan, *Visual Cognition*, © copyright #1994, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>

In this section, I will review the mixed evidence regarding the caricature effect. I will discuss image-based factors (e.g., photographs and line drawings), task-based factors (e.g., recognition, goodness of likeness ratings), and face-specific factors (e.g., distinctiveness) contributing to the caricature effect. I will then discuss the memory mechanisms responsible for the caricature effect. Specifically, I will discuss whether the caricature effect stems from a short-term or long-term memory mechanism, and whether long-term memory representations of familiar faces are caricatured. Lastly, I will discuss the representation of different types of caricatures (e.g., oblique caricatures) in the face space, and how different models of the face space make predictions about the recognition of different types of caricatures.

It is unclear whether caricatures are better recognized than veridical images (i.e., caricature advantage). For example, Hagen and Perkins (1983) did not find a caricature advantage for unfamiliar faces. Observers were shown images of unfamiliar faces to study, then were asked to recognize those individuals from a larger pool of images. These images were either artistic caricatures, three-quarter-view photographs, or profile photographs. In the same-medium condition the observers saw the same type of images during study and test (e.g., caricature-caricature, profile-profile). In the different-medium condition they saw different types of images during study and test (e.g., caricature-profile, three quarter-profile). As expected, they found that observers in the same-medium condition outperformed those in the different-medium condition. Within the different-medium condition, they found more

false alarms for conditions involving caricatures compared to veridical photographs, suggesting a lack of caricature advantage. Hagen and Perkins (1983) argued that the caricature advantage may result from better access to a face's representation in long-term memory. Since unfamiliar faces do not have a stored representation, the caricature advantage could be exclusive to familiar faces.

Unlike Hagen and Perkins' (1983) prediction, however, Tversky and Baratz (1985) did not find a caricature advantage using familiar faces. They collected photographs of famous politicians and obtained artistic caricatures of them. Observers saw either only caricatures, only photographs, or a mix of caricatures and photographs. In one experiment, they were asked to rate the images on goodness of likeness followed by a recall task. In another experiment, they were given a semantic verification task in which an image was paired with a name, and they indicated whether the image matched the name while their reaction time was measured. They found no caricature advantage for goodness of likeness ratings, recall performance, and semantic verification performance – photographs that were rated as better likenesses were recalled more accurately, and were identified more quickly.

One reason why Hagen and Perkins (1983) and Tversky and Baratz (1985) did not find a caricature advantage could be due to the fact that photographs simply contain more information than line drawings, indicating that photographs and line drawings may not yield a fair comparison. To address this, Rhodes et al. (1987) used a computer program to generate line drawings of veridical images, caricatures, and anti-caricatures (i.e., where the distinctive features are diminished instead of enhanced, so the image is made closer to the prototype) of familiar faces. In Experiment 1, they asked observers to identify the famous people depicted in veridical line drawings, caricatures, and anti-caricatures and collected likeness ratings.

They found no difference in accuracy, but caricatures were identified the fastest, followed by veridical line drawings, followed by anti-caricatures. Caricatures were also rated as better likenesses than veridical drawings (by interpolation, the best likeness caricature was at 16%). In Experiment 2, they obtained likeness judgments from unfamiliar observers by showing them the original photo upon which the drawings were based. Observers were asked to rate the likeness of the photo to the drawings either by direct comparison to the photo or by asking them to rely on their immediate memory. They found that veridical line drawings were rated as the best likenesses, followed by caricatures and anti-caricatures which did not differ in likeness ratings. Overall, they found a caricature advantage when identifying line drawings, but not when matching the likeness of a photograph to line drawings.

Benson and Perrett (1991) used photographic caricatures instead of line drawings to investigate whether the caricature advantage is limited to line drawings, or whether the advantage could be enhanced by using photographic caricatures. They created caricatures, veridical images (created as a 0% caricature to eliminate bias due to pixel anomalies), and anti-caricatures of familiar faces. In Experiment 1, observers were shown all images and were asked to provide familiarity ratings, choose the best likeness of the person among the images, and rate the goodness of likeness for the selected photo. They found that the goodness of likeness ratings peaked at the positive caricature (by interpolation, 4.4%), and caricatures were rated as better likenesses than anti-caricatures. When ratings were analyzed for each target face ($n = 7$), they found that three targets received better likeness ratings for caricatures than veridical images, whereas the ratings for veridical images and caricatures did not differ for the remaining four targets. Importantly, they found a correlation between familiarity and the degree of caricaturing on the image chosen – the more familiar a face, the more likely that

a caricature was the best likeness. In Experiment 2, they used a name-face matching task while measuring reaction time. The fastest RT was found for caricatures (except for the 48% caricature that did not differ from -48% anti-caricature). However, this RT advantage was limited to mismatch trials, meaning that observers were quicker to reject a name that did not match the caricature. Overall, the authors found that there was a caricature advantage for likeness ratings given to photographs and rejecting mismatched names, and the caricature advantage increased as a function of familiarity.

Benson and Perrett (1994) showed observers line drawings of famous faces and asked them to adjust the image using a slider to achieve the truest likeness of the person. Since the slider was continuous, they were able to overcome the limitations of categorically increasing the caricature level. They found that the best likeness was about 42% over the veridical image. Interestingly, when analyzed by each face, they found that distinctive faces required less caricaturing to achieve the best likeness. In Experiment 2, they replicated Experiment 1 and found that the best caricature represented a 54% exaggeration. When the best likeness caricatures were given to a separate set of observers in a naming task, they also found that caricatures were named faster and more accurately than veridical images. Curiously, this advantage was completely eliminated when the same task was given to observers with the external features (e.g., ears, hair) of the faces removed. It is unclear why that is the case, however the authors speculated that external and internal (e.g., eyes, nose, mouth) features may play different roles in identifying faces, as was found in Ellis et al., (1979), and recently confirmed by Kramer et al., (2018).

The interaction between the distinctiveness of a face and the degree of caricaturing needed to achieve best likeness found in Benson and Perrett (1994) is intriguing as it may

provide an account for the mixed findings in the literature. Rhodes et al. (1996) aimed to further explore whether the caricature advantage is larger for typical faces than for distinctive faces. Observers were trained to learn unfamiliar faces with varying distinctiveness followed by a recognition test of those faces with varying levels of caricaturing. They found that distinctive faces were recognized better than typical faces, and that caricatures were recognized better than anti-caricatures. However, they did not find an interaction between the degree of caricaturing and distinctiveness. Considering the face space metaphor, the authors speculated that the amount of facilitation produced by a caricature can be described as a function of the relative difference between the *advantage* from caricaturing that results from moving the face to a less dense location in the face space (nominator) and the *disadvantage* from caricaturing that results from distorting the face (denominator). To cancel out the benefit of moving a typical face to a less dense location (the *advantage*), the degree of distortion needed would be relatively large (the *disadvantage*). For a highly distinctive face, the face is already in a less dense location (the *advantage*), and distorting the face (the *disadvantage*) to a large extent could result in a non-face-like appearance. Therefore, caricaturing would facilitate recognition for typical and highly distinctive faces to a lesser extent than faces that are moderately distinctive, indicating a curvilinear relationship between typicality and the most optimal degree of caricaturing to recognize a face.

Taken together, the literature suggests that caricatures of faces seem to be recognized at least as well as undistorted images, and indeed better than anti-caricatures. Although Hagen and Perkins (1983) and Tversky and Baratz (1985) did not find a caricature effect, they used photographs where textural information may override the benefits gained from exaggerating facial features. It also seems like not all faces benefit from caricaturing the same way. For

example, Benson and Perrett (1991) found that whether the caricature was rated as best likeness varied from one face to another. Similarly, Rhodes and Tremewan (1994) found the caricature advantage for familiar faces in a naming task, but the effect was not replicated with a different set of faces. Benson and Perrett (1994) found that distinctiveness plays a role in the degree of caricaturing required for faces to achieve best likeness with typical faces needing higher levels of caricaturing. It is also evident that the effect is more reliably demonstrated with familiar faces, with mixed findings for unfamiliar faces.

Since the caricature advantage was primarily observed with familiar faces, Rhodes et al. (1987) argued that it must stem from a long-term memory mechanism. It is possible that familiar faces are stored as caricatures. It is also possible that familiar faces are stored veridically, but caricatured inputs activate the stored representation better than veridical inputs because caricatures do not activate as many similar competing representations as veridical inputs. In the former case, the caricature advantage would occur because the caricature is a better match to the representation of the face in memory. In the latter case, caricatures provide better access to the mental representation of a face, because distinctive features of the face are exaggerated. In turn, although the caricature is not as similar to the veridical representation as the veridical input itself, it is less similar to competing representations in memory. Indeed, Benson and Perrett (1991) found that observers were quicker to reject a name that mismatched the caricature compared to the veridical image of the same person. These findings provide support for the interpretation that caricatures provide better access to the veridical representations due to their increased dissimilarity to other faces. Benson and Perrett (1991) speculated that there could be multiple layers of representation, one level concerned with details representing faces in a metrically accurate way (i.e.,

veridical) that is useful for matching different inputs of a single identity, and another higher-level layer of representation concerned with differentiating faces that emphasizes distinctive features in comparison to a prototype (i.e., telling people apart).

If long-term memory representations are caricatured, it raises the question of at which point of the cognitive process caricaturing occurs. There is evidence to suggest that faces are caricaturized in short-term memory. For example, Mauro and Kubovy (1992) argued that distinctive features are emphasised as deviations from the prototype during encoding. In their experiment, observers were given a sequential matching task and were asked to indicate whether the two images were the same or not. They found that when a face preceded its caricature, observers were slower to respond “different” compared to when a caricature preceded its face. Accordingly, the authors claimed that when a face is encountered it is encoded in a caricatured form, making it slower to distinguish it from the caricature because it resembles the face’s representation in short-term memory. The authors also argued that because faces get caricaturized during encoding, there is no point in time where a face representation is veridical. However, Rhodes et al. (1987) found that even when observers were asked to rely on their immediate memory, they chose the veridical images of unfamiliar as best likeness over caricatures. Had they been encoding the face as a caricature in short-term memory, it would be expected that they would choose the caricature as the best likeness because it would be a better match to the representation of the face in short-term memory.

There is also evidence to suggest that face representations may not ever be caricatured, even in long-term memory. Since evidence regarding the caricature advantage is mixed, it is difficult to discern whether face representations are caricaturized or veridical. For example, Rhodes and Tremewan (1994) found that caricatures of familiar and unfamiliar faces were

recognized as accurately as veridical images, and both were recognized better than anti-caricatures. However, they also found that at some caricaturing levels for some faces, caricatures were recognized better than veridical images. Lee et al. (2000) obtained similar results in a naming task where some target faces showed a caricature advantage and some did not, but they found an overall caricature advantage when data were collapsed across targets. Benson and Perrett (1991) found a recognition advantage for caricatures, but their likeness results were mixed; some targets' caricatures received higher likeness ratings, whereas some targets' veridical images received higher likeness ratings. Therefore, even though Mauro and Kubovy's (1992) findings provide compelling evidence that face representations are caricaturized in memory, they do not completely rule out the possibility that caricatures simply provide better access to undistorted representations.

More recently, caricature effects have been tested using photographic caricatures rather than line drawings (e.g., Allen et al., 2009; Hancock & Little, 2011; Kaufmann & Schweinberger, 2008). Although caricature effects are more pronounced and consistent for line drawings (e.g., Rhodes et al., 1987), they have been weaker and less consistent for photographic caricatures (e.g., Benson & Perrett, 1994), possibly due to the textural information provided by photographs. In fact, more recent findings seem to demonstrate a preference towards the anti-caricature. For example, Allen et al. (2009) showed observers images of the observers themselves and of their close friends. Observers were allowed to scroll through the images to manipulate the degree of caricaturing, then they reported the image they thought was the best likeness. The results revealed a preference for the anti-caricature (about -12%). Since observers needed to differentiate different levels of caricaturing, the preference for the anti-caricature could be an artefact of differentiation. Due

to distinctiveness, faces that deviate away from the norm (i.e., caricatures) will be easier to differentiate than faces that deviate towards the norm (i.e., anti-caricatures). Therefore, preference for the anti-caricature could be due to its closer resemblance to the veridical image. Attractiveness could also play a role in choosing the anti-caricature as the best likeness. Faces closer to the average are usually perceived as more attractive, so a person's perceptions of their own and their friends' attractiveness may have influenced the preference for the anti-caricature.

Kaufmann and Schweinberger (2008) investigated the neural correlates of the caricature effect using event-related potentials (ERP). Observers viewed familiar (personally and celebrities) and unfamiliar faces while doing a familiarity classification task, as well as best likeness ratings for familiar faces. They found no effect of caricature for the familiarity classification task, reflecting a large range of tolerance for different levels of distortion. For the likeness ratings, however, caricatures received the worst likeness rating with no difference between veridical images and anti-caricatures. Their ERP data also showed no effect of caricaturing for familiar faces. For unfamiliar faces, interestingly, caricatures elicited larger N250 amplitudes (related to personal familiarity). This is possibly due to the fact that observers were becoming familiar with caricaturized unfamiliar faces to a greater extent in the course of the experiment since they are more distinctive.

The caricature effect could also be an artefact of adaptation, since most procedures involving caricatures require several seconds of viewing a face. Adaptation is a recalibration of the visual system after continually viewing a stimulus that biases the appearance of a subsequent stimulus in the opposite direction of the adapted stimulus (e.g., adapting to a male face makes an androgynous face appear more feminine; Webster et al., 2004). Hancock and

Little (2011) investigated the role of adaptation in the caricature effect. For example, if one adapts to an anti-caricature, it may influence the preference for the caricature as a veridical representation. In their experiment, observers were given celebrity faces that they adjusted to match the best likeness. This was done before and after adapting to an average face. The results revealed a preference for the anti-caricature before adaptation, and this effect was strengthened with adaptation to the average face. Therefore, some findings in the literature that found a preference for the caricature can be due to adaptation effects.

In summary, findings on the caricature effect are quite inconsistent, which could be due to several factors. First, caricature effects are stronger for line drawings. For example, Rhodes et al. (1987) and Benson and Perrett (1994) found that caricatures (around 50% and 42%, respectively) were chosen as the best likeness and were recognized faster and more accurately. However, when photographic caricatures are used, Benson and Perrett (1991) found the best likeness caricature to be at around 4.4%. Allen et al. (2009) found a preference for the anti-caricature using photographic stimuli. Importantly, introducing colour and texture information with photographic stimuli can exaggerate identity-irrelevant factors, like lighting or facial rotation, which in turn can reduce the caricature effect as observers may choose a smaller degree of caricature or an anti-caricature to reduce those identity-irrelevant factors (Hancock & Little, 2011). However, in the studies by Hancock and Little (2011) and by Allen et al. (2009) such factors were controlled in making the caricatures, and they still found a preference for the anti-caricature. Second, the type of task used can influence the result. Some studies used a recognition task, but many used goodness of likeness ratings, which may tap into different memory mechanisms. Recent research shows that likeness ratings are idiosyncratic and highly dependent on specific experiences with individuals (Ritchie et al.,

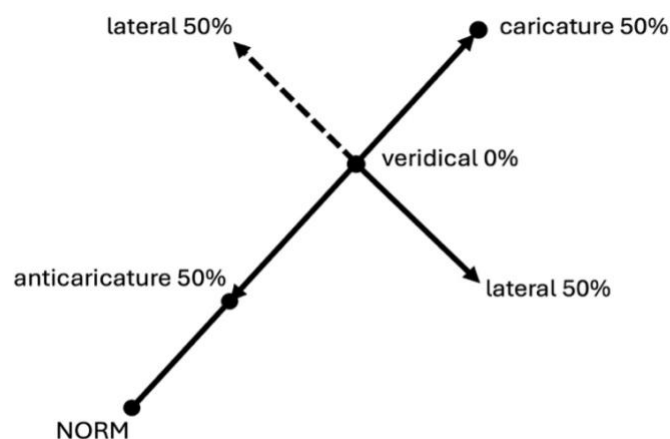
2018). Indeed, Allen et al.'s (2009) finding regarding the preference for the anti-caricature could be influenced by perceptions of attractiveness. Therefore, likeness ratings may have too much variation to sufficiently capture caricature effects. Third, not every face may benefit from caricaturing to the same extent. For example, Benson and Perrett (1994) found that distinctive faces achieve their best likeness with smaller caricaturing than typical faces. Rhodes and Tremewan (1994) found the caricatures to be effective for some faces but not others. Nevertheless, the caricature effect poses an interesting question regarding how faces are represented in memory.

1.1.3.1 Implications for the Face Space. The caricature effect is often used as evidence for the norm-based model of the face space over the exemplar-based model because there is a clear role of the prototype face in producing the effect. Caricatures are created by increasing the differences between a face and a prototype face, which emphasizes features of a face that deviate from the norm. And caricatures, although not always better recognized than veridical images, are evidently more readily recognized than anti-caricatures where the face is altered to be closer to the prototype (e.g., Rhodes et al., 1987; Rhodes & Tremewan, 1994). This suggests that faces are represented according to their deviation from the norm. However, the exemplar model could also explain the caricature effect solely in terms of a distinctiveness advantage; exaggerating facial features makes a face more distinctive and carries it to a more isolated region in the face space. The challenge for the exemplar model then would be to explain how the identity is preserved when the absolute point of the representation is moved. Therefore, it is assumed that the advantage from increasing distinctiveness would offset the disadvantage from distorting the face, making the face more recognizable.

Lateral caricatures, images in which a face is shifted “sideways” in the face space (Rhodes, 1996; Figure 1.6; Figure 1.7), have been used to understand the construction of the face space as they provide a unique case in which the amount of distortion can be the same as caricatures and anti-caricatures, but the direction of distortion is now orthogonal to the identity vector. The exemplar-based model and the norm-based model can make opposing predictions for the recognition of lateral caricatures. The exemplar-based model predicts that lateral caricatures will be better recognized than anti-caricatures because lateral caricatures are located in a region where lower exemplar density is found compared to anti-caricatures. The norm-based model predicts that lateral caricatures will be recognized more poorly than anti-caricatures, because not only the degree of distortion but the *direction* of distortion is important for recognition. However, the norm-based model can still account for better recognition of lateral caricatures than anti-caricatures if the distinctiveness of laterals offsets the impact of orthogonal distortion from a face’s vector (Rhodes et al., 1998).

Figure 1.6

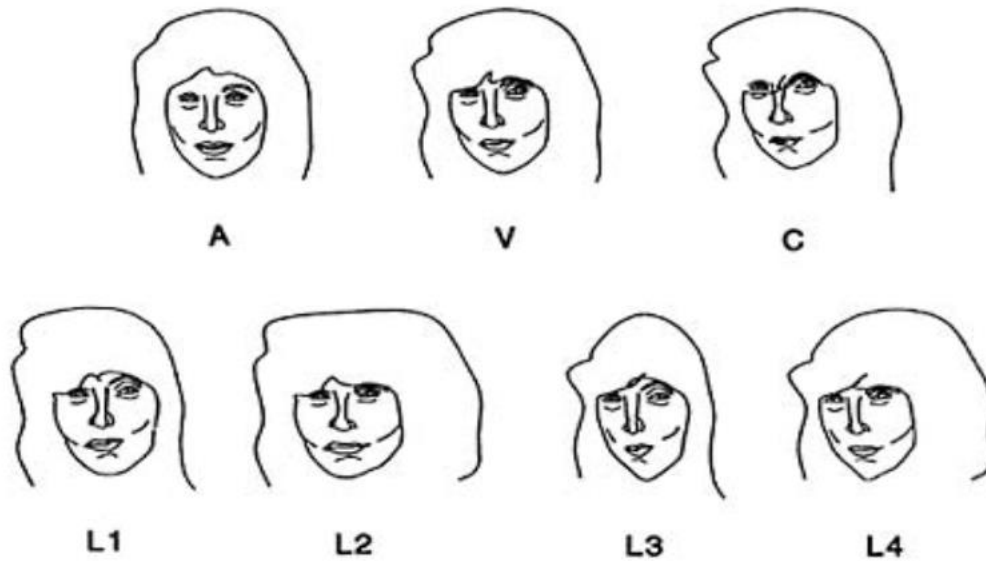
An Illustration of the Direction of Vectors for Caricatures, Anti-caricatures, and Lateral Caricatures in a Norm-based Model of the Face Space



Note. Adapted from Rhodes (1996).

Figure 1.7

50% Distortions and a Veridical Line Drawing of Cher



Note. Taken from Rhodes (1996). A = Anti-caricature, V = Veridical, C = Caricature, L1-4 = Lateral Caricatures. Reproduced with permission of Informa UK Limited through PLSclear (License Number: 9780863773983).

Unpublished data from Carey et al. (1992; cited in Carey, 1992; Rhodes et al., 1998) showed that lateral caricatures were recognized more poorly than anti-caricatures, providing evidence for the norm-based account. However, Rhodes et al. (1998) reported that the stimuli were not created properly, which may have led to an unnecessary distortion of the lateral caricatures, contributing to poor recognition. Therefore, they aimed to replicate whether recognition of lateral caricatures is worse than anti-caricatures. In their experiment, they created line drawings of veridical images, caricatures, anti-caricatures, and four lateral caricatures of familiar celebrities. Lateral caricatures were created by moving the points on the face perpendicular to the direction of the norm-deviation vector (i.e., the identity vector).

Observers were shown these images and were asked to name the celebrity depicted in the image while their accuracy and reaction time were measured. They found that laterals were more accurately recognized than anti-caricatures with no difference in speed of identification, which is not consistent with either of the models.

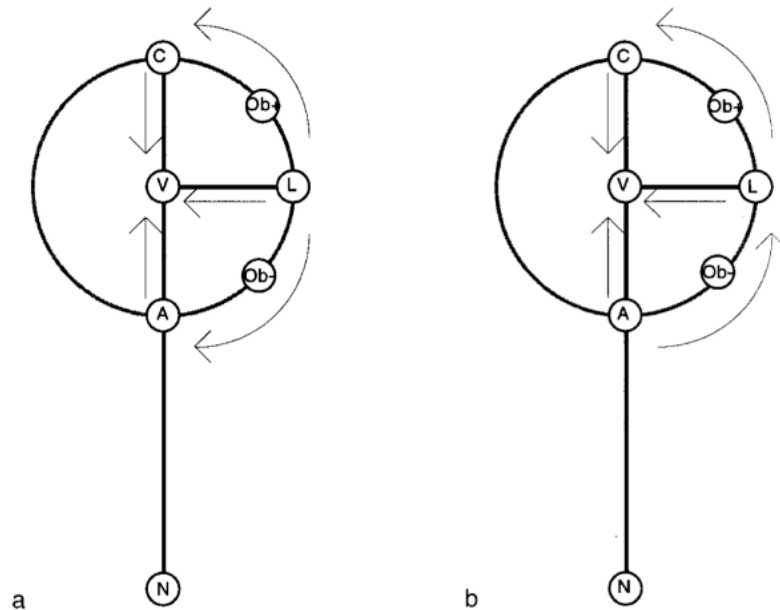
Lewis and Johnston (1998) also aimed to explore the recognition of lateral caricatures by creating photographic caricatures through morphing unfamiliar faces. They created an average (i.e., prototype) face by combining a set of faces which was then morphed with a target face in varying amounts to create veridical images and anti-caricatures (i.e., veridical: $\frac{3}{4}$ face – $\frac{1}{4}$ average, anti-caricature: $\frac{1}{2}$ face – $\frac{1}{2}$ average). The caricature was the original face image as it was the most distinctive due to not being morphed with an average face. The lateral caricature was created through morphing the veridical image with a reference face that was not used in the set of faces that were used in averaging, which distorted the face in an orthogonal direction (i.e., *oblique* caricatures). In their first experiment, to test the processing of these caricatures, they used a method similar to Mauro and Kubovy (1992) where observers were shown the veridical image followed by two test images that were either a caricature, an anti-caricature, or an oblique caricature. Observers were asked to choose the image most similar to the target (veridical) face. They found that caricatures were chosen more often than anti-caricatures and oblique caricatures, with no difference between anti-caricatures and oblique caricatures.

Lewis and Johnston (1998) reasoned that oblique caricatures can be delineated into two distinct types: Positive oblique caricatures are those that are located somewhere between the caricature and the perfect lateral caricature (i.e., equidistant from the veridical face as the caricature and anti-caricature, but maximally dissimilar from both the caricature and anti-

caricature), whereas negative oblique caricatures are located somewhere between the anti-caricature and the perfect lateral caricature (Figure 1.8). According to the norm-based model, caricatures are recognized better because the direction of distortion is the same as the direction of the identity vector, thus caricatures make the identity vector larger while maintaining its direction. For oblique caricatures to create better representations, the angle between the vector associated with the oblique caricature and the identity vector should be smaller (i.e., closer to each other). Therefore, the norm-based model predicts that more distinctive positive oblique caricatures and less distinctive negative obliques will produce representations closer to the veridical (Figure 1.8a). According to the exemplar-based model, recognition depends primarily on the exemplar density in the region where the representation is located. For both positive and negative oblique caricatures, increasing the distinctiveness will move the representation to a less dense region, producing better representations (Figure 1.8b). Therefore, the norm-based model and the exemplar-based model make different predictions for the recognition of negative oblique caricatures.

Figure 1.8

Oblique Caricatures



Note. Different predictions made by (a) the norm-based model and (b) the exemplar-based model. The direction of arrows illustrates the direction where representations that are more similar to the veridical representation. The main difference is the direction of arrow for the negative oblique caricatures. N = Norm (prototype); V = Veridical face; C = Caricature; A = Anti-caricature, L = Lateral caricature; Ob+ = Positive oblique caricature; Ob- = Negative oblique caricature. Taken from Lewis & Johnston (1998). Reproduced with permission from Sage Publications (License Number: 6211090520210).

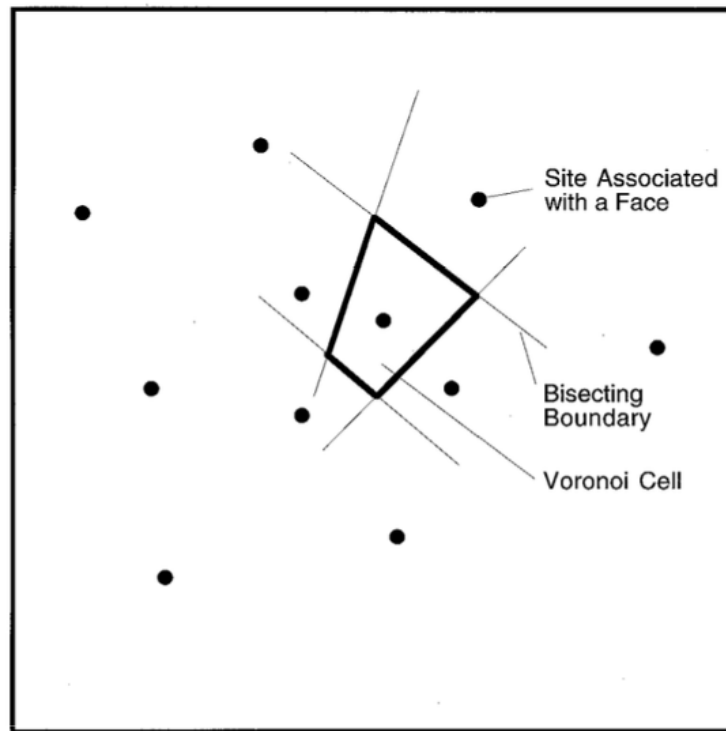
In their second experiment, Lewis and Johnston (1998) aimed to test the perceived similarity between the different types of oblique caricatures. They followed the procedure of the first experiment (i.e., forced-choice similarity task) using the same stimuli while also including the veridical image displayed as the test face. They also collected distinctiveness judgments to determine the direction of the oblique caricatures, which revealed that all oblique caricatures were negative as they were rated less distinctive than the veridical images.

The perceived similarity task showed that when the difference between the distinctiveness of the (negative) oblique caricature and the veridical image is decreased, the probability of choosing the oblique caricature over the veridical image increased, providing evidence for the exemplar-based model (**Error! Reference source not found.b**). In other words, more distinctive negative obliques correspond better with the representation in memory than less distinctive negative obliques.

Lewis and Johnston (1999) proposed the Voronoi-based model, an extension of the exemplar-based model, to provide a unifying account for the caricature effect. Voronoi diagrams are structures that divide a Euclidean space into discrete regions based on a certain number of locations. Each region is referred to as a “cell” that includes locations that are called “sites” (Figure 1.9). According to this model, each face is treated as an identity region (i.e., cell) rather than a single point in the face space as in the traditional exemplar-based model. A site within each region refers to the exact representation of a face with invariant physical properties across the dimensions of the face space. Therefore, although there is a single ideal representation for each face, recognition is achieved when a face corresponds to any representation within the identity region. Since faces are recognized across a range of viewpoints, the Voronoi-based account is particularly compelling since identities are represented as regions.

Figure 1.9

A Voronoi Cell in a Two-dimensional Space



Note. Bisecting boundaries are driven at the halfway-point between two neighbouring sites.

Each site refers to the ideal representation of a face, whereas the cells correspond to the identity region. A Unified Account of the Effects of Caricaturing Faces, Lewis and Johnston, *Visual Cognition*, © copyright #1999, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>

A characteristic feature of the Voronoi model is that it emerges from a normally distributed face space, which creates a predominance of sites that are off-centred¹. The cells with off-centred sites have their sites towards the origin of the face space rather than at the centre of the cell, which means the centre of the cell is often further away from the origin. Recognition is assumed to improve when the representation is closer to the centre of the cell

¹ If the reader wishes to generate a Voronoi diagram, navigate to <https://cfbrasz.github.io/Voronoi.html>.

since the centre of the cell will be more distant from near neighbours. According to the Voronoi-based model, caricaturing does exactly this. The caricature corresponds to a representation that is closer to the centre of the cell, which makes it less similar to the competing representations.

The model can also account for the distinctiveness effect based on exemplar density. Due to the normally-distributed nature of the face space, typical faces that are clustered around the origin have smaller cells. In turn, the noise associated with encoding the input becomes more detrimental to recognition since a small amount of noise is sufficient for the input to fall outside of the cell. Lewis & Johnston (1999) created simple neural network simulations using the Voronoi-based model and were able to replicate the distinctiveness effect and the caricature effect using both two-dimensional and five-dimensional models. Interestingly, they found that more distinctive faces required less caricaturing for optimal recognition (see Benson & Perrett, 1994 above). In line with this model, faces are encoded veridically, and caricatures speed recognition during retrieval; and this blur between encoding and retrieval accounts for the blur regarding whether face representations in long-term memory are veridical or caricaturized. Therefore, according to the Voronoi-based model, face representations are neither veridical nor caricatures, but rather regions that encapsulate both and many other representations.

In conclusion, the face space can provide an account for the finding that caricatures of faces are better recognized than their corresponding anti-caricatures, despite the fact the amount of distortion is the same. It is unclear whether the norm-based model or the exemplar-based model provides a better description of the face space using the caricatures. The norm-based model predicts superior recognition of caricatures because faces are encoded by their

deviations from the norm; exaggerating these deviations enhances distinctiveness. This, in turn, preserves the direction of the identity vector while enlarging its magnitude, which makes them recognized more readily. According to the exemplar-based model, the ease of recognition depends on the exemplar density around the region of the face representation. Caricaturing moves the face representation to a less dense location through enhancing distinctive features, which makes the face easily recognizable. The Voronoi-based model, which is a version of the exemplar-based model, suggests that caricatures are recognized better because caricaturing moves a face representation to a more central point within the identity region, which in turn reduces the competition from the near neighbours.

1.1.4 The Other-Race Effect: “They All Look Alike to Me”

A number of studies have demonstrated an effect of race in recognizing faces – the other-race effect (ORE) is the tendency to recognize faces of one’s own race better than faces of other races (e.g., see Bothwell et al., 1989 for a meta-analysis)². For example, in Shepherd et al. (1974), African and European observers studied African and European faces. They were asked to recognize the study faces among distractors a day later. They found that European observers recognized European faces better than African faces, and African observers recognized African faces better than European faces. Ellis et al. (1975) showed Black and White faces to Black and White observers and asked them to describe the faces to a person who would need to pick them up from a nearby station that did not know them. They assumed that the features that people pay attention to would be reflected in their descriptions of faces,

² The term “race” in this literature is rather loosely defined as major anthropological groups (Valentine et al., 2016), and it is often used interchangeably with ethnicity. Due to the context of the research discussed, race here refers to structural visual information derived from stimuli. Indeed, the *perceived* race of a face is found to influence face processing even if the visual features are left the same, suggesting an overall psychological out-group effect irrespective of physiognomic factors (see Lewis & Hills, 2018). The nomenclature I use to refer to different racial groups is taken verbatim from the papers I cited, which is why the terminology will seem inconsistent.

and they found that White observers were using features that are more useful for discriminating White faces but not necessarily Black faces, such as eye colour, hair colour, and hair texture. Similarly, Black observers used a greater number of useful features to discriminate Black faces compared to White observers, such as hair position, eye size, whiteness of the eyes, eyebrows, and so on.

Although the effect is quite robust (e.g., Bothwell et al., 1989), the theoretical factors contributing to this phenomenon are not fully understood. It is not the case that any one race is inherently more difficult to recognize than a different race as anthropometric analyses provided no evidence for differences in facial heterogeneity based on race (e.g., Goldstein, 1979), and the effect is demonstrated across races as was shown in Shepherd et al. (1974) and Ellis et al. (1975; see also Ng & Lindsay, 1994; Shepherd et al., 1974; Valentine & Endo, 1992). Therefore, it is assumed that there are differences between the memorial representations, or perceptual processing (Megreya et al., 2011) of own- and other-race faces. The most prevalent hypothesis in the literature is the contact hypothesis, suggesting that the ORE occurs due to limited contact with other-race faces. If so, increasing contact should reduce or eliminate the effect.

In Cross et al.'s (1971) study, Black and White observers studied a set of 12 Black and White faces and later were asked to recognize them among distractors. They found that White observers recognized White faces better than they recognized Black faces. Black observers recognized White and Black faces equally well. Interestingly, White observers in segregated neighbourhoods made more false alarms for Black faces compared to White observers who lived in integrated neighbourhoods, suggesting that the ORE can be reduced with increasing contact. Goldstein and Chance (1985) provided evidence that the ORE can be reduced with

training. In their study, White observers were given a paired-associate learning task during a training session to learn Japanese faces over multiple sessions. It was found that training performance improved over subsequent sessions, trained observers performed better than the control group for the recognition task, and the effects of training persisted after five months.

Chiroro and Valentine (1995) provided further evidence for the contact hypothesis. They included Black and White observers with varying self-reported contact with other-race faces. Observers studied faces followed by a recognition test. The A' scores indicated that low-contact observers, irrespective of the race of the observer, demonstrated an ORE. High-contact white observers also demonstrated an ORE, but high-contact Black observers recognized faces of both races equally accurately. Although the contact hypothesis is compelling, there is also evidence that it does not influence recognition performance. For example, Ng and Lindsay (1994) recruited East Asian and White observers from Canada who reported the extent and the intimacy of their relationships with own- and other-race faces. Observers were shown Eastern and White faces along with four comparison images and they were asked to report the similarity between the target face and the comparison faces. Later, they were given a recognition task on those faces. The authors found that both groups of observers demonstrated an ORE in the d' measure that was driven by false alarms. There was no correlation between contact and recognition performance, which was replicated by testing observers from Singapore who had equivalent other-race contact with White observers.

It should be noted that testing for the contact hypothesis is quite challenging because many contemporary societies are multi-racial, there can be social desirability effects at play when reporting contact with other races (Chiroro & Valentine, 1995), and there is increasing influence of visual media where one can be exposed to a diverse range of races. Further,

contact can take many shapes and forms. For example, one can be surrounded by other-race faces but may not feel the need to differentiate among them (e.g., an exchange student only hanging out with a few people of their own ethnicity). Therefore, quality of interactions may be more important than the quantity of interactions, and quality of interactions may be difficult to measure.

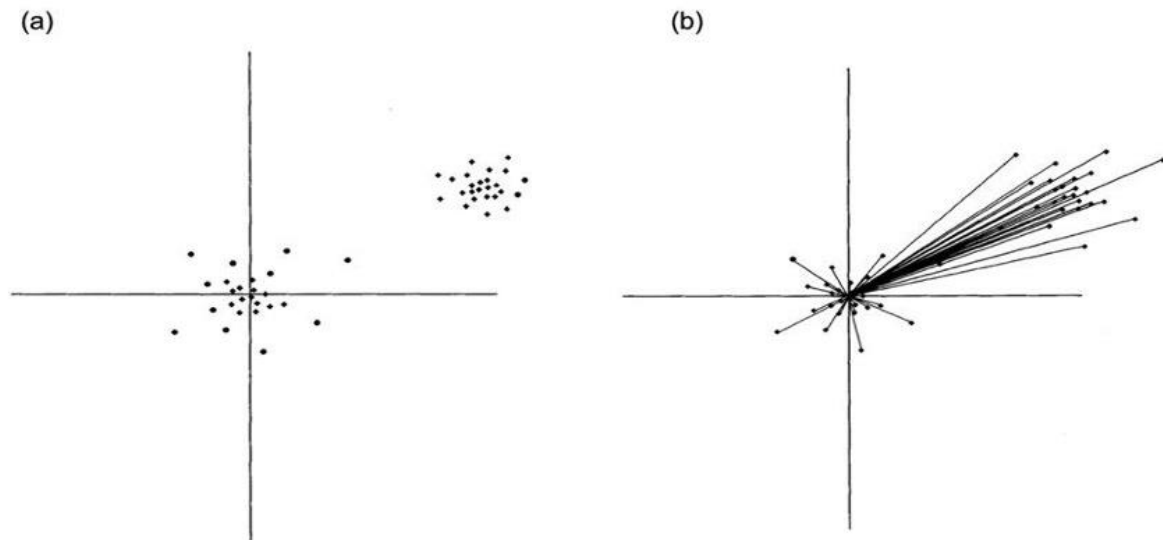
As is found in distinctive faces, there is an asymmetry in the classification and identification performance of other-race faces. While recognition of other-race faces is poor, classifying an other-race face as belonging to that race is better than for own-race faces. For example, Ge et al. (2009) recruited Chinese and Caucasian observers and showed them own- and other-race faces to recognize later. They were also given a race classification task where they categorized the faces as either Chinese or Caucasian. The reaction time results revealed a three-way interaction between the observer race, race of face, and the type of task. Caucasian observers recognized Caucasian faces faster and classified Chinese faces faster, and Chinese observers recognized Chinese faces faster and classified Caucasian faces faster. They also found a negative correlation between recognition and classification performance, suggesting that expertise with a particular race comes at a cost in categorization.

1.1.4.1 Implications for the Face Space. There is some evidence that dimensions used to differentiate other-race faces differ from own-race faces. For example, Ellis et al. (1975) found that Black and White observers differed in their descriptions of facial features of Black and White faces. In Shepherd and Deregowski (1981)'s study, Black and White observers used different attributes to describe the similarity between two Black faces and two White faces. Therefore, according to the face space framework, it is assumed that the face space is constructed most optimally to differentiate among own-race faces, therefore, other-race faces

are encoded more densely than own-race faces (Figure 1.10; Valentine & Endo, 1992; Valentine et al., 2016). Both the exemplar-based and the norm-based model suggest that the ORE is due to the use of inappropriate dimensions when encoding other-race faces since the dimensions of the face space are most appropriate for encoding own-race faces. The two models, however, differ in the encoding processes and how the similarity between two faces is calculated. According to the exemplar-based model, the similarity between two faces is calculated as the Euclidean distance between the two faces, for both own-race and other-race faces. How the similarity is calculated for the norm-based model is not as straightforward. Valentine (1991) and Valentine and Endo (1992) suggest the similarity is determined by the angular separation or the dot product of two vectors. However, these similarity metrics fail to meet the assumptions that collinear representations are not necessarily identical (even though their angular separation is 0°), and collinear representations will be more similar to each other than any other two noncollinear representations that are equidistant (Byatt & Rhodes, 1998). Therefore, according to Byatt and Rhodes (1998), the similarity metric for the norm-based model is the cosine of the angle between two vectors divided by the simple distance between two faces.

Figure 1.10

Own- and Other-race Faces in a Two-dimensional Face Space



Note. (a) A purely exemplar-based model. Around the origin is where own-race faces are encoded. Other-race faces are encoded in a denser cluster as they are encoded using inappropriate dimensions. (b) A norm-based model. Other-race faces are encoded using the own-race norm, which is not optimal for coding other-race faces. Towards an Exemplar Model of Face Processing: The Effects of Race and Distinctiveness, Valentine and Endo, *Quarterly Journal of Experimental Psychology*, © copyright #1992, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>

Although other-race faces are more densely clustered in the face space, there is no *a priori* reason to assume that the *difference* in exemplar density around typical and distinctive faces would be less for other-race faces. Valentine and Endo (1992) aimed to explore whether or how distinctiveness interacts with the other-race effect. They first collected distinctiveness ratings from Japanese and British observers using Japanese and British faces and found that observers of both races agreed on which faces were distinctive. British faces also were rated as more distinctive than Japanese faces for both groups of observers. When observers were

given a recognition task on these faces, they found that both Japanese and British observers demonstrated an ORE in A' scores. Both groups of observers had more hits for British faces. British observers made more false alarms for Japanese faces while Japanese observers' false alarms did not differ as a function of the race of face. Japanese observers correctly rejected Japanese distractors faster than British distractors while British observers' correct rejections did not differ as a function of race of face. In all five measures, an effect of distinctiveness was observed, but there was no interaction among distinctiveness, race of face, and race of observers. In other words, the effect of distinctiveness on recognition was equivalent for both own-race and other-race faces.

In Experiment 3, observers were given a face classification task. Although typical faces were classified faster than distinctive faces consistent with the previous literature, other-race faces were not classified faster than own-race faces, which would be predicted by the exemplar-based model due to the high exemplar density of other-race faces, similar to typical faces. Valentine (1991) found other-race faces to be classified faster; however, that study was not cross-cultural. The authors reasoned that a face classification task may be an inappropriate task to demonstrate an interaction between race of face and race of observer since the face classification task requires a judgment of a superordinate category (face irrespective of race). Therefore, In Experiment 4, observers were given a race classification task with the rationale that a race classification task would require judgment of a subordinate category (race of face). They found that British observers classified Japanese faces faster than British faces, but Japanese observers classified both races of faces equally. Additionally, British observers classified typical British faces faster than distinctive British faces, and Japanese observers classified typical Japanese faces faster than distinctive Japanese faces. Neither group

demonstrated a distinctiveness effect for the other-race faces, demonstrating a greater distinctiveness effect for own- but not other-race faces.

These findings are supported by the exemplar-based model but not the norm-based model. The exemplar-based model suggests that recognition of other-race faces would demonstrate an equivalent distinctiveness effect to own-race faces because the *relative* difference in exemplar density for own- and other-race faces is assumed to be similar. In other words, even though other-race faces are more densely clustered, distinctive other-race faces are still encoded in a relatively more isolated region compared to typical other-race faces. This was demonstrated by the first experiment where there was agreement among observers in both groups on which faces were distinctive irrespective of the race of face, and the lack of interaction of distinctiveness with race effects. The exemplar-based model's prediction for race classification performance is not clearly outlined, however. Since classification relies on a between-category judgment, an effect of distinctiveness in classification of own-race but not other-race faces can be a result of greater exemplar density of other-race faces. Therefore, the exemplar-based model can account for the results obtained from Valentine and Endo (1992). On the other hand, the norm-based model would predict that in a recognition task, the distinctiveness effect would be smaller or absent for other-race faces because the angular separation of other-race faces is smaller than own-race faces, assuming angular separation is an appropriate similarity metric for the norm-based model. This is due to other-race faces being encoded with the own-race norm (see Figure 1.10b).

The norm-based model could account for these findings if other-race faces are encoded with their own norm which reduces parsimony and adds a requirement for the appropriate norm to be selected before deriving individuating information about faces. According to the

face recognition model proposed by Bruce and Young (1986), faces are encoded with semantic and structural codes, with (visually derived) semantic codes corresponding to properties like sex, age, and race, and with structural codes corresponding to identity information. Bruce and Young (1986) argue that these visually derived semantic codes are independent from the recognition of an identity. If an appropriate norm (e.g., someone's race) needs to be selected before accessing structural information, it would mean there is a degree of dependence between visually derived semantics and structural information. Nevertheless, Ge et al. (2009) found a correlation between race classification and recognition of other-race faces so there could be some sort of dependency between visually derived semantics and structural information.

Chiroro and Valentine (1995) aimed to investigate the effect of distinctiveness in recognizing other-race faces while including contact as a factor. They found that level of contact with other-race faces interacted with distinctiveness, such that low-contact observers in both race groups (Black and White) demonstrated an effect of distinctiveness in recognition of own-race faces but not other-race faces. High-contact observers in both groups demonstrated an effect of distinctiveness in recognition of both own-race and other-race faces. The authors provided evidence that the distinctiveness effect is a result of experience with a particular population of faces. This is also in line with the exemplar-based model; if the population of faces encoded in the other-race space is increased, the relative exemplar density can be more accurately established. According to Valentine and Endo (1992), the norm-based model fails to provide an account for the effect of contact; since other-race faces are encoded with an own-race norm, the angular separation among other-race faces will always be limited.

Additional evidence for the exemplar-based model came from Byatt and Rhodes (1998) using caricatures. Since the norm-based model assumes other-race faces to be encoded using the own-race norm, other-race caricatures using an own-race norm (i.e., wrong-norm) should be easier to recognize than caricatures created with the race's own norm (i.e., right-norm). The exemplar-based model makes the opposite prediction, since the caricature advantage stems from reduced exemplar density and other-race faces are encoded with inappropriate dimensions, right-norm caricatures of other-race faces should be easier to recognize than wrong-norm caricatures. To test these predictions, European observers were shown Chinese and European faces and were asked to rate them on distinctiveness and learn their names. Later, they were tested on their recognition of the veridical images, right-norm caricatures, wrong-norm caricatures, right-norm anti-caricatures, and wrong-norm anti-caricatures. They found that own-race veridical faces were recognized as well as own-race caricatures, and that both were recognized better than anti-caricatures. For other-race faces, all images were recognized equally well except for the -50% anti-caricature which was recognized poorly. Importantly, in all conditions, right-norm distortions were more accurately recognized than wrong-norm distortions, providing evidence for the exemplar-based model. The authors argued that the norm-based model could account for these results if other-race faces are encoded with their own norm, but then the norm-based model would fall short in explaining the other-race effect, because the norm-based model's explanation for the other-race effect is that other-race faces are encoded with an inappropriate norm.

This interpretation can be problematic because it assumes that the other-race norm would be optimal for differentiating other-race faces. However, limited exposure to other-race faces can form an other-race norm that is not stable enough to allow for effective

differentiation of other-race faces. Burton et al. (2016) showed that averages of faces stabilize after about 20 images, which could mean that it takes about 20 instances of other-race encounters to form a stable norm of an other-race face. However, this account would suggest that each encounter with a face is included and weighed equally in the formation of the average. How a facial norm emerges, to my knowledge, is not exactly known and is a potential avenue for future research. For now, there does not seem to be clear evidence favoring either of the models, with seemingly more evidence in line with the exemplar-based model (e.g., Byatt & Rhodes, 1998; Valentine & Endo, 1992). The norm-based model seems to be able to account for many findings with the assumption that each category has its own norm, but then the norm-based model becomes indistinguishable from the exemplar-based model.

The face space framework also fits with the mirror effect of recognition and classification of other-race faces (e.g., Chiroro & Valentine, 1995; Ge et al., 2009). In the mirror effect, distinctive faces are recognized faster but classified (as faces) slower. Other-race faces are recognized more slowly but classified (as belonging to which race) faster, which is due to the high exemplar density of other-race faces. Accordingly, recognition of other-race faces is impaired due to the close proximity of near neighbours, whereas classification performance of other-race faces is better because the overall activation of the region is high. An alternative account comes from Levin's (1996; 2000) race-coding hypothesis. Put simply, the race-coding hypothesis suggests that other-race faces are encoded explicitly with their race information at the expense of individuation information. Therefore, the other-race effect is not due to an expert coding of individuation information of own-race faces per se, but rather an expert coding of race information of other-race faces.

Consequently, classifying other-race faces would not be influenced by distinctiveness, because the classification performance is controlled by a unifying race-feature rather than individuating information like typicality. Levin also hypothesized that whereas other-race faces are encoded with a present (positive) race feature, own-race faces are encoded with an absence (negative) of the race feature. Present features are weighed more heavily in decision making (Cherubini et al., 2013), and targets with present features are found faster in visual search tasks than targets with absent features (Treisman & Gormican, 1988). As such, in a visual search task where an other-race face is the target among own-race distractors, the target would be found faster than if it was an own-race face among other-race distractors.

In Levin's (1996) study, observers were given a race classification task on famous and unfamiliar own- and other-race faces where distinctiveness was systematically manipulated through caricaturing using both line drawings (Experiment 1) and photographic stimuli (Experiment 2). The results revealed that observers were faster at classifying other-race faces than own-race faces, but only when the faces were very distinctive (caricatured). There was no classification advantage for typical faces. In another experiment, inverted faces were also used as stimuli to investigate whether disruption in configural processing (associated with expert processing of individuating information; Diamond & Carey, 1986) would account for the classification performance. Similar to Valentine's (1991) finding, inversion did not reduce the classification advantage of other-race faces, indicating that the ORE is not necessarily a result of disruption in configural processing. Lastly, observers were asked to either search for an own-race target among other-race distractors, or for an other-race target among own-race distractors to test for the visual search asymmetry. The results revealed that observers were quicker in locating an other-race face than an own-race face, and that increasing set size made

them disproportionately slower to locate an own-race face than an other-race face. The search asymmetry suggests that other-race faces are encoded with a present feature, whereas own-race faces are encoded with the absence of the race feature. However, when tested cross-culturally, the search asymmetry was replicated for White observers, but Black observers did not demonstrate an asymmetry.

Levin argued that the race-coding hypothesis is an *alternative* to the face space framework. However, Byatt and Rhodes (2004) suggested they do not have to be mutually exclusive, as the dimensions used in encoding race information may receive more weight for encoding other-race faces than own-race faces, which would provide support for both the race-coding hypothesis and the face space theory. To investigate whether the ORE stems from higher exemplar density, Byatt and Rhodes (2004) collected similarity ratings for pairs of own- and other-race faces from one set of observers, and gave a recognition task to another set of observers. The similarity ratings showed that, despite being instructed to focus on individuating information rather than race, own-race faces were rated as more similar to each other than to other-race faces and vice versa. Importantly, other-race faces were more similar to each other than own-race faces were to each other. A multidimensional scaling analysis by Lee et al. (2000) confirmed that own- and other-race faces formed distinct clusters, with other-race faces forming a denser cluster than own-race faces. This clustering was an accurate predictor of the other-race effect.

Rhodes et al. (2009) further explored the race-coding hypothesis as an underlying effect of other-race recognition performance. They reasoned that if coding race information is at the expense of individuating information, then paying attention to race information should be detrimental to identification performance irrespective of race of face. Observers were asked to

either passively view a set of own- and other-race faces, rate the attractiveness of the faces, or rate typicality of the faces based on race. They found that when given a recognition task on these faces, observers in all groups demonstrated an ORE. In fact, observers who rated the faces demonstrated an even better recognition of own-race faces than other-race faces compared to the observers who passively viewed them. The ORE was replicated in a task where observers were asked to classify races of faces, suggesting that directing attention to race information does not eliminate the ORE.

In summary, according to the face space framework, the ORE occurs due to the higher exemplar density of other-race faces compared to own-race faces. This was supported by Byatt and Rhodes (2004) who found that other-race faces form distinct and denser clusters than own-race faces which accurately predicts recognition performance. The face space can also predict how the ORE can interact with phenomena like distinctiveness (Valentine and Endo, 1992), inversion (Valentine, 1991), and caricature (Byatt & Rhodes, 1998). Most findings seem to be in line with the exemplar-based model, but the norm-based model could also account for the results if it is assumed that different face categories have their own norm. However, that would make the norm-based model less parsimonious and theoretically indistinguishable from the exemplar-based model.

1.1.5 The Adaptation Effect: Changing the Norm

Visual adaptation is a process by which the visual system recalibrates systematically after constant viewing of a particular stimulus property (Strobach & Carbon, 2013; Valentine, 1996; Webster & MacLin, 1999). Adapting to a stimulus changes the appearance of a test stimulus in the opposite direction of the adapted stimulus property. For example, adapting to downwards motion makes a stationary object appear like it is going upwards, commonly

known as the waterfall illusion (Addams, 1834). Adaptation effects can manifest in a multitude of visual dimensions, such as orientation (Gibson & Radner, 1937), spatial frequency (Blakemore & Sutton, 1969), and colour (Webster & MacLeod, 2011). More recently, adaptation effects have been demonstrated for face stimuli (e.g., Carbon & Ditye, 2012; Leopold et al., 2001; Strobach & Carbon, 2013; Webster & MacLin, 1999). The multidimensional nature of faces as a visual category means that face adaptation can take many forms depending on the adapting information such as age, gender, configuration, distortion, attractiveness, identity, and so on (Strobach & Carbon, 2013). Face adaptation is also considered a high-level process due to its transference across different sizes and retinal locations (Rhodes & Jeffery, 2006; Valentine et al., 2016).

One of the first demonstrations of face adaptation came from Lewis and Ellis (2000). They showed that repeated presentation of a name or different views of a face lead to semantic satiation (i.e., repeating a semantic concept results in a loss of the concept's meaning). In their experiment, observers were asked to repeat a celebrity's name out loud either three times or 30 times, followed by a face image that either matched or mismatched with the name. They found that observers were slower to indicate whether the face matched the name when they repeated the name 30 times. Similarly, when they were repeatedly shown different images of a celebrity face followed by a test face that was either related (an associate) or unrelated (a non-associate), observers were slower in their judgments of identifying the test face when the images were repeated 30 times compared to three times, suggesting that satiation can occur for non-verbal semantic categories, like facial identity.

Additional evidence in a more visual domain came from Webster and MacLin (1999) who examined the face distortion aftereffects (FDAEs). In their matching task, observers

learned an undistorted face, and later were asked to adjust a test face to match the original face, both before and after adapting to a face. In the ratings task, observers were simply asked to rate whether a displayed image was normal or distorted, also both before and after adapting to a face. They found that when observers adapted to a distorted face, they matched and rated the original face to be distorted in the opposite direction. For example, if the adapting face was expanded, the original face appeared contracted. Additionally, the level of distortion of the adapting image systematically influenced the appearance of the test image. Adapting to a distorted face greatly influenced the appearance of the original face, but adapting to the original face had little effect on the appearance of the distorted face. This suggests that adaptation is not simply a shift in perception but rather a renormalization process. Some faces serve as a “neutral point” (i.e., the norm), and adaptation to these faces does not influence the appearance of distorted images because the adapted stimuli reinforce the existing norm. On the other hand, adapting to a distorted face changes the appearance of a test face in the *opposite* direction, meaning that the norm may have shifted towards the adapted stimulus.

Face adaptation is assumed to be a high-level process since it transcends low-level information like retinal location and size of stimuli (Valentine et al., 2016). Nevertheless, evidence suggests that face adaptation shares dynamic properties similar to low-level adaptation processes like orientation and shape. Leopold et al. (2005) examined these dynamic properties of face identity aftereffects (FIAE). Specifically, they examined the relationship between the temporal presentation of the adapting and test stimuli and the strength of adaptation, which is a relationship previously found for other classes of visual stimuli like orientation and shape. They created faces and anti-faces (i.e., faces that lie on two opposite ends of an identity trajectory passing through the norm, Figure 1.11) assuming a

norm-based model of the face space. After being trained on differentiating the identities used in the study, observers were given an adaptation paradigm where they were first shown a name cue, then adapted to an anti-face in varying durations followed by a test face presented in varying durations. The test face was always the average face. Observers were asked to rate the strength of the apparent identity of the test face that matched the name cue. The results revealed an adaptation effect where observers' ratings of apparent identity were greater and in the opposite direction of the anti-face. Importantly, adaptation was stronger for longer adaptation durations and shorter test durations, mimicking dynamic properties of adaptation to lower-level stimuli. These results indicate that adaptation could be a feature of the visual system in general to adapt to novelties in the environment, and that adaptation could serve as a tool to understand categorical representations of visual stimuli in general (see Webster & MacLeod, 2011 for a review of face adaptation paralleled with colour adaptation).

Carbon and Ditye (2012) examined whether adaptation effects were transferable over different environmental settings, as it could be that being in the same laboratory environment for adaptation and test phases activated episodic representations which in turn activated adapted representations. In their study, all observers went through an adaptation phase in the formal experimental laboratory to either strongly compressed, strongly extended, or veridical celebrity faces. After a 7–10-day delay, half of the observers were tested in the lab, and half were tested in an informal leisure room of the department where the environmental and social properties of the experimental setting were different. The test included a 2AFC task where observers were shown a veridical face paired with either a slightly extended or contracted face. The veridical face could be the same picture, a different picture of the same person, or a novel identity. The task was to choose the veridical face. The authors replicated the adaptation

effect in that adapting to a distorted face made observers more likely to choose the face distorted in the opposite direction (e.g., extended to contracted) as the veridical face.

Adaptation was strongest for the same picture, followed by a different picture of the same identity, and weakest for a novel face. Importantly, the stability of the adapting and the testing environments did not contribute to the adaptation effects, showing that sustainable adaptation is not dependent on episodic representations. Given that adaptation effects are present after a week and in different settings, adaptation may be a continually normalizing mechanism that tunes the entire face space towards newly-encountered visual information.

1.1.5.1 Implications for the Face Space. Face adaptation effects have been used to understand the architecture of face representations, specifically the nature of the face space. Adaptation effects are often used as a support for the norm-based model as they serve as a renormalization process (e.g., Webster & MacLin, 1999), indicating that the norm plays an important role in how faces are encoded. Influential evidence in favor of the norm-based model came from Leopold et al. (2001), who tested FIAEs using realistic faces. In their experiment, a face space was constructed with an average face at the origin. Importantly, they created anti-faces by generating faces through calculating the difference between the face and the average so that anti-faces and their corresponding original faces would lie on the same axis, passing through the average face (Figure 1.11). They found that adapting to an anti-face increased sensitivity in identifying the original face, i.e., adapting to anti-Adam made it easier to identify Adam. Interestingly, adapting to an anti-face made the average face appear as a new identity in accordance with the identity trajectory. In other words, adapting to anti-Adam made the average face look like Adam. This was not true for mismatch trials where the adaptation face and the test face were on different identity trajectories, for example, adapting

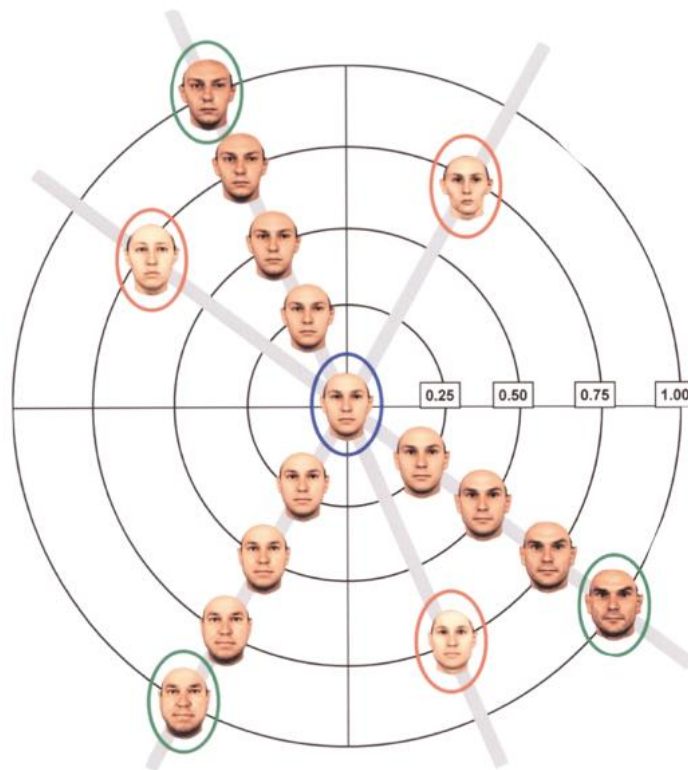
to anti-Adam made it more difficult to identify Jim. These findings highlight the importance of a norm face in differentiating faces since adaptation systematically influenced processing a face on the same identity trajectory but not others.

Rhodes and Jeffery (2006) investigated whether the FIAEs were selective to adapting to opposite faces, or whether an equally dissimilar non-opposite face would induce similar aftereffects. As in Leopold et al., (2001), the authors created faces and anti-faces which lay on opposite ends of the face space while passing through the average face. Non-opposite faces were also created that were rated to be equally dissimilar to the target as the opposite face. Observers were first trained to identify target faces. To collect baseline identification thresholds, observers were asked to identify "weaker" versions of target faces that were morphed with an anti-face or a non-opposite face. Observers were then given an adaptation paradigm where they adapted to either an anti-face or a non-opposite face. The test face was a 50/50 morph of the adapting face with the target face, and observers were asked to identify the test face. They found that adaptation reduced identification thresholds for target faces; in other words, adapting to a face made the morph test face appear more like the target. However, importantly, this effect was significantly larger for anti-faces than non-opposite faces. Interestingly, they also found that identification thresholds were overall lower for faces morphed with their opposite compared to their non-opposite. This was confirmed with a second experiment where observers were asked to rate the similarity between a morph face and one of its constituents. Morphs created with opposite faces were rated as less similar to their constituents than morphs created with non-opposite (but equally dissimilar) faces. These results show that the monotonic distance between two identities in the face space is not

sufficient to explain the adaptation effect. Rather the *direction* of the dissimilarity plays an important role, providing support for the norm-based model.

Figure 1.11

Faces and Their Anti-faces in the Face Space



Note. Taken from Leopold et al. (2001). Faces in green circles correspond to the faces (e.g., Jim, Adam) and the faces marked in red circles that are on the same line as the faces correspond to their anti-faces (e.g., anti-Jim, anti-Adam). The face at the center is the average face. 0.25, 0.50, 0.75, and 1.00 reflect identity strength, manipulated by morphing the average face to the target faces. Reproduced with permission by Springer Nature (License Number: 6211091082939).

Ross et al. (2014), however, provided evidence that exemplar-based models can also account for the FIAEs. They developed computational models using a Gaussian distribution

and PCA, where they constructed a face space assuming either an exemplar-based model, a traditional norm-based model, or a two-pool norm-based model. The difference between the traditional and the two-pool norm-based models is that in the traditional norm-based model, the norm is explicitly represented, and faces are encoded as deviations from the norm; whereas in the two-pool norm-based model the representation of the norm is implicit, and the faces are encoded as opponents based on their perceived characteristics. For example, male and female faces can be encoded as opposites in the gender dimension, or Adam and anti-Adam can be encoded as opposites in some identity dimension (Leopold et al., 2001; Rhodes & Jeffery, 2006). Additionally, the exemplar-based model was created in a way that newly-learned identities were represented by distributions based on their similarity to previously-learned faces as opposed to the more traditional demonstrations of the exemplar-based representations where faces are represented as unique points. Ross et al. simulated the anti-face adaptation in Leopold et al. (2001), adaptation to opposite and non-opposite faces in Rhodes and Jeffery (2006), and the weaker adaptation effects found in adapting to the norm (e.g., Webster & MacLin, 1999). They found that the exemplar-based model and the two-pool norm-based model predicted these findings in the literature, however, the traditional norm-based model fell short in predicting some of the findings.

In summary, adaptation effects have been widely used to support the norm-based model of the face space. Specifically, adaptation to the norm does not create strong adaptation effects (Webster & MacLin, 1999), suggesting that adaptation serves as a renormalization process. Adaptation for face identity is also selective for the opposite faces that have their identity trajectory passing through the norm (Leopold et al., 2001; Leopold et al., 2005; Rhodes & Jeffery, 2006), which highlights the role of the norm in representing faces in the

face space. However, simulations suggest that exemplar-based models make similar predictions about FIAEs as well as norm-based models where the norm is implicitly represented (Ross et al., 2014).

1.2 Telling Faces Together

Valentine's (1991) face space theory provides an account for explaining how the visual system overcomes the homogeneity problem (i.e., all faces share the same basic configuration) when differentiating different individuals from one another. In a multidimensional representational space, each face is encoded in a location relative to other previously encountered faces based on their perceptual similarity to one another. This theory is especially appealing considering the prevalence of such multidimensional spaces to represent and organize other categories of stimuli (e.g., colour; Shepard, 1962). Nevertheless, in the case of face perception and memory, it has some theoretical and experimental limitations, as the more recent literature has demonstrated. Such limitations boil down to the qualitative differences between processing familiar and unfamiliar faces. While the face space theory has greatly focused on differentiating faces from one another, the other side of the coin is the ability to "tell faces together", that is, the visual system's ability to amalgamate different instances of a face into a single representation of that person's identity.

1.2.1 Theoretical Limitations of the Face Space: Beyond a Single Point or Vector

Earlier models of the face space claim that each face is represented as a single point (the exemplar-based model) or a single vector deviating from a norm face (the norm-based model). Such an approach to representing faces does not explain how a familiar face can be recognized across several viewing conditions as familiar face recognition is robust to changes in viewpoint, expression, age, angle, lighting, and even image distortions. For example, Hole

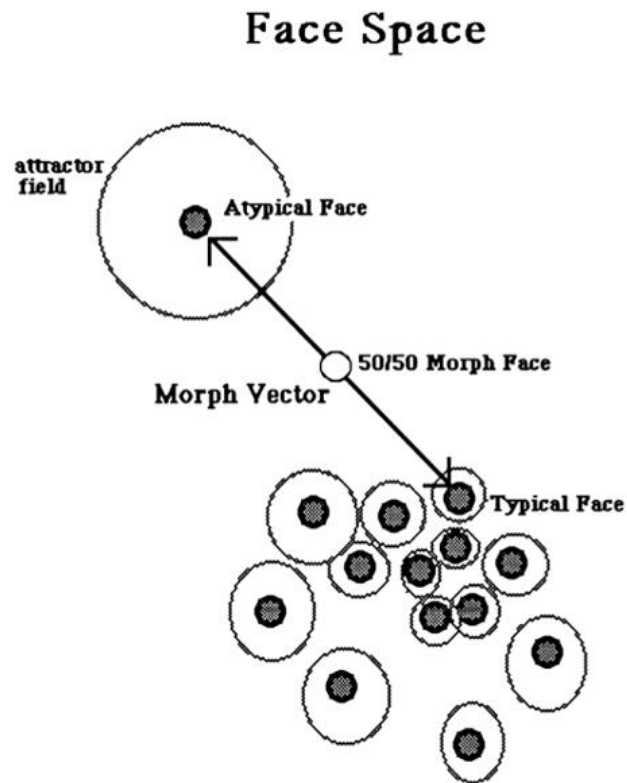
et al. (2002) showed that a familiar face can be successfully recognized even when it is vertically stretched, suggesting that familiar face recognition is robust to even distortions in the facial configuration. Bruce et al. (2001) found that familiar faces can be recognized from low-quality CCTV videos. The ability of the visual system to tolerate such variability for familiar faces is unlikely to stem from a single memorial representation. The assumptions that there is a single “true” representation for a face, and that any change from this true representation is an encoding error or a change in identity, fall short in explaining how we can recognize familiar faces despite the variability in viewing conditions (Burton, 2013).

More recent models of the face space suggested that faces are represented as regions encompassing a range of instances that can be recognized. One such model is the Voronoi model discussed above (Lewis & Johnston, 1999), where a normally distributed Euclidean-based space is divided into discrete regions (i.e., cells; Figure 1.9). Each cell contains a “site” that corresponds to an ideal representation of an identity based on its invariant properties, so the Voronoi model suggests that faces are represented as points but the region surrounding the point allows a face to be recognized across different viewing conditions. The size of the region surrounding a representation is related to how well a face can be recognized, thus, faces with large regions surrounding them can tolerate more variability. Although simulations of this model successfully replicated the distinctiveness and the caricature effects found in the literature, an important limitation of this model is that it assumes unfamiliar faces will be falsely recognized as familiar faces because the entire volume of the space is filled with identity regions. Thus, in contrast to humans who can simultaneously fail to identify a face and not mistake it for some other face, the model cannot give an “unknown” response (Lewis, 2004; Lewis & Johnston, 1999).

Tanaka et al. (1998) suggested that face representations are surrounded by “attractor fields” that can activate a face’s representation through multiple inputs (Figure 1.12). The size of the attractor field of a face is related to the density of other exemplars surrounding the representation in the face space. Since distinctive faces are located in a region with lower exemplar density, it allows their attractor fields to be larger, allowing for more instances to fall within the identity’s region. Evidence for attractor fields came from 50/50 morphs of a typical and a distinctive face. A traditional model of the face space would predict that a 50/50 morph should bear equal resemblance to its parent faces because similarity between two faces is determined by the Euclidean distance of their coordinates in the face space. The attractor field model, on the other hand, would predict that a 50/50 morph would bear more resemblance to the distinctive parent, because it would activate the distinctive face more strongly than the typical face due to the larger size of the distinctive face’s attractor field. In a series of experiments, the authors found that when observers were asked to make a similarity judgment of the 50/50 morph between the typical and the distinctive parent, they chose the distinctive parent, providing evidence for the attractor field model.

Figure 1.12

The Attractor Field Model



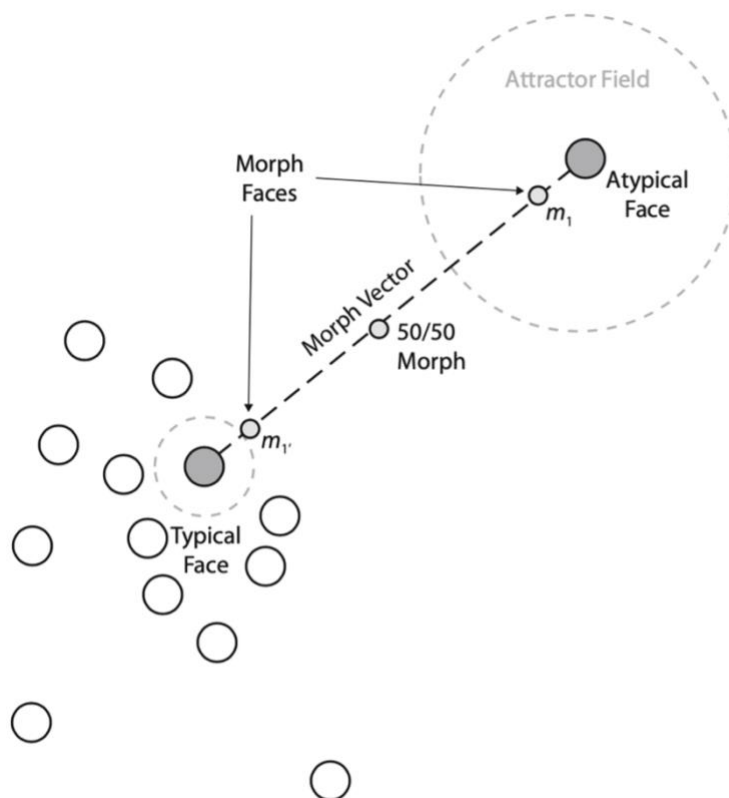
Note. Taken from Tanaka et al. (1998). Atypical faces have larger attractor fields due to low exemplar density, whereas typical faces have smaller attractor fields. A 50/50 morph of a typical and an atypical face is equidistant from its constituents, yet it bears stronger resemblance to the atypical face due to a stronger activation of the atypical face's large attractor field. Reproduced with permission from Elsevier (License Number: 6211091246183).

In another study, Tanaka and Corneille (2007) created morphs of typical and atypical faces with varying ratios to manipulate their Euclidean distance from their atypical and typical parents. Observers were then given a sequential and a simultaneous matching task where they were asked to make a same/different judgment on a morph face paired with its typical or distinctive parent face. In line with the attractor field model, they found that morphs that were

closer to a distinctive face (i.e., distinctive morphs) were less likely to be successfully differentiated from the distinctive face, whereas morphs that were closer to a typical face (i.e., typical morphs) were more likely to be differentiated from the typical face. According to the attractor field model, since distinctive faces have larger attractor fields, a morph that is x units away from a distinctive face will be more likely to fall within the attractor field bounds compared to a morph that is x units away from a typical face (Figure 1.13). This makes the perceived similarity between the distinctive morph and its distinctive parent face larger than the perceived similarity between the typical morph and its typical parent face.

Figure 1.13

Attractor Field Model with Typical and Distinctive



Note. Taken from Tanaka and Corneille (2007). The distinctive (atypical) and typical faces are represented in gray circles, surrounded by their attractor fields that are represented as

dashed lines. Morphs are represented as lighter gray. Although the two m_1 morphs are the same distance away from the typical or the distinctive face, the morph closer to the typical face is more likely to fall outside of the bounds of its attractor field, yielding easier differentiation, whereas the morph closer to the distinctive face is more likely to fall inside of the bounds of its attractor field, yielding more difficult differentiation. Reproduced with permission from Springer Nature (License Number: 6211091353510).

More recent evidence for attractor fields came from Laurence et al. (2016) investigating the ORE. Previous research discussed above framed the ORE as a problem of differentiating other-race faces with the assumption that other-race faces are encoded using dimensions best suited for differentiating own-race faces, which are inappropriate for encoding other-race faces. This leads other-race faces to be encoded in a region with high exemplar density (Figure 1.10), making them more confusable with one another. The greater density of exemplars surrounding other-race faces implies that they will have smaller attractor fields, reducing the tolerable range of within-person variability. In Laurence et al.'s study, observers were given a card sorting task to test for tolerance to within-person variability. The cards included multiple ambient photos, (i.e., viewing conditions like lighting, expression, viewing angle are unsystematically varied) belonging to two unfamiliar own- or other-race identities, and the observers were asked to sort these cards into piles so that each pile corresponded to an identity. The authors found that while the observers created about 7 piles for own-race faces, they created about 9-10 piles for other-race faces, indicating that observers perceived there to be more identities in the other-race set. Importantly, observers made few misidentifications, meaning that they rarely put two different identities in a single pile for both own- and other-race faces, which did not differ significantly. These results confirmed that attractor fields are

smaller for other-race faces, and that it is not necessarily the case that other-race faces “all look alike”, but rather that their individual exemplars “all look different”.

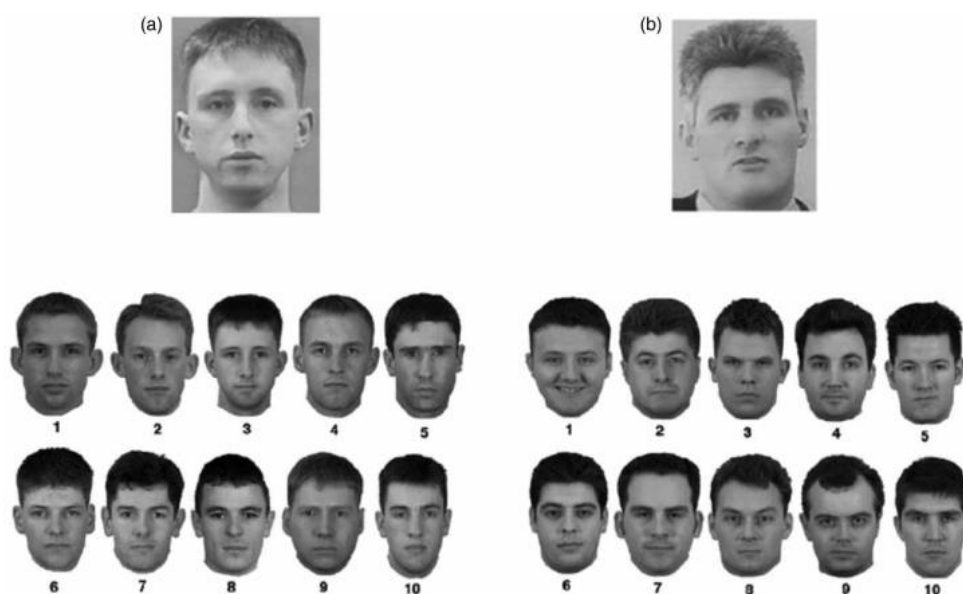
1.2.2 Theoretical Limitations of the Face Space: Differences Between Familiar and Unfamiliar Face Processing

When we assume faces are represented as single points or vectors, it becomes difficult to explain how faces can be recognized across different viewing conditions. Although the Voronoi model (Lewis & Johnston, 1999) or the attractor field model (Tanaka et al., 1998) can provide an account for many views of a face activating the face’s representation, there are certain limitations with the underlying assumptions of these models. Both models assume that the size of a region surrounding a face representation is determined by its distinctiveness. In a normally distributed Voronoi diagram, regions that are clustered at the centre (i.e., typical faces) are necessarily smaller than regions that have their sites on the “edges” of the space (i.e., distinctive faces; see Footnote 1). Similarly, the size of an attractor field is determined by the density of surrounding representations, so distinctive faces have larger attractor fields than typical faces. A larger region surrounding a representation is indicative of how well different instances of a face can be recognized, because a larger region will map more different inputs onto the same identity than a smaller region. However, there are likely factors other than distinctiveness that determine how well within-person variability can be tolerated, and one such factor is familiarity with a face. In fact, there is evidence that processing identity of familiar and unfamiliar faces is qualitatively different (e.g., Megreya & Burton, 2006; 2007); therefore, conflating familiar and unfamiliar faces in a single space fails to account for these differences.

Although unfamiliarity disadvantages have been demonstrated previously for recognition paradigms (e.g., Ellis et al., 1979), more recent research shows that there are also unfamiliarity disadvantages in *perception* tasks such as simultaneous eye-witness lineup paradigms (e.g., Bruce et al., 1999; Megreya & Burton, 2006; 2007). For example, in Bruce et al.'s (1999) study, observers were shown lineups as displayed in Figure 1.14 below where the target could be present or absent in the lineup. The observers were asked to choose the target face (at the top) from the lineup. The observers only reached about 70% accuracy in both target-absent and target-present lineups. This is particularly interesting because the task was untimed, it had no demands on memory (i.e., simultaneous matching), observers knew the target was absent in about half of the trials, and viewpoint and expression were matched across faces. Further, performance declined when viewpoint and expression differed between the target and the lineups, suggesting that unfamiliar face matching is highly image-dependent.

Figure 1.14

Example Lineup Task



Note. Used in Bruce et al. (1999), screenshot obtained from Burton (2013). Can you match the target (on the top) with one of the faces in the lineup? Why has research in face recognition progressed so slowly? The importance of variability, Burton, *Quarterly Journal of Experimental Psychology*, © copyright #2013, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>

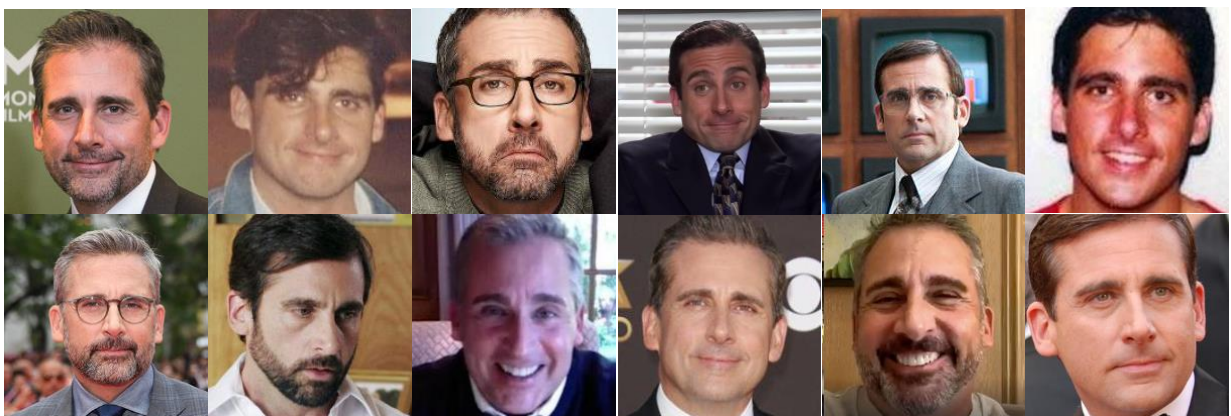
Low accuracy observed in this task is unlikely to be due to task demands. In Megreya and Burton's (2007) study, observers were only shown two images in a same/different judgment task (i.e., a matching task) where two images either depicted the same person or different people. They found the same pattern of accuracy (about 70% correct). Therefore, the issue seems to be not task demands, but rather encoding unfamiliar faces. Familiar face matching, on the other hand, leads to almost perfect performance. In Megreya and Burton's (2006) study, even when observers were briefly familiarized with some faces their accuracy reached above 87%. In Bruce et al.'s (2001) study, observers were able to match familiar faces with above 90% accuracy, even from low quality CCTV videos. This suggests that familiar and unfamiliar face matching rely on qualitatively different processes. In fact, Megreya and Burton (2006), in their paper titled "*Unfamiliar faces are not faces*", showed that unfamiliar face matching accuracy is highly correlated with *inverted* familiar face matching accuracy. Since inversion disrupts configural processing (e.g., Diamond & Carey, 1986), this suggests that configural information cannot be extracted from unfamiliar faces to the same extent. Therefore, while familiar face matching utilizes a stored memory representation, unfamiliar face matching is highly dependent on pictorial cues.

Why do these differences arise when processing familiar and unfamiliar faces? The answer seems to be variability across different views of the same face (Figure 1.15). In a

seminal study by Jenkins et al. (2011), observers were given a card sorting task, including two different identities with 20 photos each. It is important to note that these photos were *ambient*, meaning that viewing conditions were not controlled; lighting, expression, viewing angle, and so on were unsystematically varied. Observers were asked to sort these cards into piles where each pile represents one identity. They found that unfamiliar observers created a median of 7.5 piles, significantly higher than the correct answer which was two piles. Importantly, misidentification errors were quite low - observers almost never placed two different identities in the same pile (fewer than 1% of trials). When the same task was given to a set of observers that were familiar with the identities in the card decks, performance was almost perfect. These results suggest that different views of a face involve high variability, and when a face is unfamiliar, variability is not effectively tolerated. This leads observers to attribute changes in pictures of a face to changes in perceived *identity* of the face, similar to what was found in Laurence et al.'s (2016) study with other-race faces using the same task.

Figure 1.15

Images of Steve Carrell that Depict Within-person Variability



Note. Different images of Steve Carrell across different years, emotional expressions, camera, viewing angle, hairstyles and so on. One can easily observe that photos depicting the same

person can highly vary. Yet, for familiar faces, this variability is readily tolerated, and the face can be successfully recognized. For an unfamiliar viewer, however, it is with great difficulty these photos are judged as depicting the same person.

Familiarity then, can be defined as tolerance to within-person variability. Given that familiarity with a face does not transfer to familiarity to another face (Burton, 2013), the ways in which a face vary are a *unique* property of that face. Computational support for this idiosyncratic variability came from Burton et al.'s (2016) study. In their study, they used a PCA on multiple ambient photos belonging to the same person. A PCA returns components arranged by how much variance is accounted for by each component in a descending order, so that the first component accounts for most of the variance in the data, the second component accounts for the second-most variance, and so on. This allows for observing the variability captured in a set of high-dimensional data, in this case the variability captured by different views of a person. Using PCA, the authors found that while the first three components for each identity commonly captured dimensions like head angle and lighting, starting from the fourth component of the PCA, every identity varied in a unique way. For example, if the fourth component captured a smile for someone, it captured eye movement for another person. Even when there were overlaps in the components between individuals, the *range* over which these components varied was idiosyncratic; for example, a 10° head turn for someone with a longer nose would cast different shadows than for someone with a smaller nose.

It should be noted that within-person variability is not *entirely* idiosyncratic. In a recent paper, Kramer et al. (2018) showed that familiarity with a face can partly benefit unfamiliar

face matching performance³. Additionally, Burton et al. (2016) found that the first three components of the PCA were similar across different faces. Although how a face varied within those components still reflected idiosyncrasy, the most variance captured by the PCA was common across different faces. This is also consistent with Megreya and Burton's (2006) finding that inversion also disrupts unfamiliar face matching, suggesting that some configural information is being extracted from unfamiliar faces. If unfamiliar face matching was entirely feature-based, accuracy matching upright and inverted unfamiliar faces would be the same. Therefore, it is not surprising that some variance would perhaps not be shared across *all* faces to the same extent, but rather across some faces to some extent. This raises an interesting question regarding whether and to what extent the encoding dimensions are shared in the face space. It could be that a layer of the face space represents common variances across different faces that serve to discriminate faces, and that a layer with identity subspaces serves to tolerate within-person variability. A similar mechanism was suggested by Benson and Perrett (1991).

Since there is a large discrepancy between processing familiar and unfamiliar faces due to idiosyncratic variability, some researchers argued that we are not experts at processing faces, only *some* faces, the ones that are familiar to us (Burton, 2013). Whether we are experts at face processing or whether faces are special has been a long debate in the literature (e.g., Diamond & Carey, 1986; Farah et al., 1998; Rossion, 2018). On one hand, faces, despite their homogeneity, can be differentiated fairly well. Importantly, faces are more vulnerable to inversion than other classes of perceptual stimuli (e.g., Yin, 1969), suggesting that faces are

³ In their study, they used PCA combined with Linear Discriminant Analysis (LDA), which combines a more statistical, bottom-up approach (PCA) with a more high-level, top-down approach (LDA) through labeling different "classes" (i.e., identities).

“special” in a sense. Diamond and Carey (1986) argued that expertise with any homogeneous class of perceptual stimulus makes it vulnerable to inversion, suggesting that faces are special only in the sense that we are more likely to become experts in face processing than experts in other classes of stimuli. On the other hand, more recent literature shows that although faces share homogeneity across different identities based on their general configuration, one person’s face involves a high degree of variability that is at least to some extent idiosyncratic (Burton et al., 2016; Jenkins et al., 2011). This variability makes processing unfamiliar faces error-prone, because the unique variability of an unfamiliar face is not learned and thus cannot be tolerated across viewing conditions.

Rossion (2018) argued that face expertise should be considered through comparing typical human observers with populations or species that are not experts at face recognition. For example, patients with prosopagnosia perform worse than typical observers in unfamiliar face matching tasks, suggesting that the ability to recognize faces assists with unfamiliar face matching. This is similar to what was found in Kramer et al. (2018) using PCA and LDA to model face processing. A recent paper by Blauch et al. (2021) using Deep Convolutional Neural Networks (DCNN) to create object- and face-expert models found that face-expert models outperformed object-expert models in unfamiliar face matching tasks, suggesting that familiar and unfamiliar face processing both tap into a shared expertise. However, such benefits to unfamiliar face matching, although useful, are nonetheless a result of learning familiar identities, and not an ability developed to deal with unfamiliar faces (Young & Burton, 2021).

So, are we face experts? The question itself is somewhat problematic. Treating faces as a special case of object recognition, or treating *all* faces as a unified class of perceptual stimulus does not sufficiently capture the differences between familiar and unfamiliar face

processing (Burton, 2013). This has important implications for the face space, which assumes all familiar and unfamiliar faces are encoded in a single space. However, if we assume variability to be idiosyncratic, different identities should be encoded using *idiosyncratic* dimensions. Burton et al. (2016) showed that a PCA model trained with a person's face made fewer errors reconstructing an untrained image of that person's face than reconstructing an untrained identity. In other words, "learning" the variability of someone's face, while useful for recognizing that face, led to making more errors recognizing someone else's face. Therefore, it is not merely the case that faces are represented as points, vectors, or even regions, but rather that faces have their very own subspaces with their own set of dimensions. Indeed, it has been found that computational models that utilise face-specific subspaces outperform traditional models where a single space is shared by all trained faces (Aishwarya & Marcus, 2010; Shan et al., 2003). I provide evidence for this account in Chapter 4.

1.2.3 Methodological Limitations of the Face Space: Exemplar-dependent Recognition

Until the last decade or so, most work done on face processing, including the face space framework, has overlooked the importance of within-person variability. In fact, such variability was purposefully excluded as "noise" (Jenkins et al., 2011). This led to researchers to use images taken under highly controlled settings, with limited variability in the person-dependent (e.g., expression, weight, age), and in the image-dependent information (e.g., viewpoint, lighting, camera; Burton et al., 2011). This is dissimilar to how we process faces in the real world, which involves encountering a range of variability in both person- and image-dependent aspects. Since this variability is a key aspect of face processing, minimizing variability in experimental settings ends up treating face processing as a special case of image processing (Burton, 2013). However, many recognition studies used the same image during

study and test (e.g., Bartlett et al., 1984; Cohen & Carr, 1975; Cross et al., 1971; Ge et al., 2009; Goldstein & Chance, 1985; Hancock & Rhodes, 2008; Light et al., 1979; Ng & Lindsay, 1994), or an image that varies in only a single dimension (expression, e.g., Byatt & Rhodes, 2004; Chiroro & Valentine, 1995; Valentine, 1991, viewpoint, e.g., Hagen & Perkins, 1983; Newell et al., 1999). To have a more complete understanding of face processing, it is imperative to incorporate natural (ambient; Jenkins et al., 2011) images of faces to improve ecological validity, especially since unfamiliar face recognition relies on unsophisticated image-dependent comparisons (e.g., Megreya & Burton, 2006).

Another limitation from previous research comes from the assumption that a single photo is a good representation of a person's face. For example, when collecting ratings to measure facial properties like distinctiveness, only one photo of a person is used. However, when variability is incorporated, different views of a face may elicit different judgments. For example, Jenkins et al. (2011) collected attractiveness ratings for multiple ambient photos of unfamiliar celebrities, and found that variability in attractiveness ratings for one person was greater than variability in attractiveness ratings between people. Similarly, they found that likeness ratings, commonly used as a dependent variable to measure the caricature effect (e.g., Benson & Perrett, 1991; Rhodes et al., 1987), also varied greatly across different photos of a familiar person's face. These results suggest that properties that we may assume are inherent to a face may be captured differently across different photos.

1.3 The current research program and the organization of the thesis

The aim of the current research program is to address the above-mentioned theoretical and methodological limitations to understand how familiarity is represented in the face space. Research on the face space was primarily concerned with how faces are differentiated, which

is a curious problem given that faces share great similarities in their general configuration. Such research was motivated by consideration of faces as a homogeneous category of stimuli while focusing on the nuances among different members of this category. However, while there are similarities across faces, there are great variances within different views of a single face. Therefore, another very curious problem in face recognition is how different views of a face are amalgamated into a robust, stable mental representation that can be used to successfully recognize that face despite such variability. I have argued above that familiarity with a face can be defined as tolerance to its within-person variability.

To investigate how familiarity is represented in the face space, I first report a series of studies investigating how a face's representation in the face space might change as a face becomes familiar through exposure to within-person variability (Chapter 2). Specifically, observers were shown multiple natural images of the same person (i.e., the target face). At regular intervals, the observers adjusted a morph between the target and a similar-looking foil to achieve an equal blend of these two faces (i.e., the point of subjective equality). I expected that if a face's representation grows more distant from others as it becomes familiar, the perceived subjective equality between that face and the foil would change over time. However, this prediction was not supported, suggesting that a face's representation may not move away from others as a function of familiarity.

Then, I report a study investigating how within-person variability is represented in the face space for familiar and unfamiliar faces by constructing a 3-dimensional face space (Chapter 3). Observers were asked to rate the similarity between pairs of faces, which were then analyzed using multidimensional scaling. I found that familiar faces are represented with higher-density clusters in the face space, suggesting that familiarity decreases the distance

between different views of the same person. Supporting the findings I reported in Chapter 2, I found limited evidence in favour of increased distance between identities as a function of familiarity. I also speculated on the dimensions of the face space, which suggested that the same dimensions of the face space that serve to differentiate faces represent within-person variability idiosyncratically.

Lastly, I provided evidence for representing familiar faces as identity-specific subspaces that have a norm-based architecture (Chapter 4). Observers were asked to identify different images of a familiar identity after adapting to an anti-face that is created either by (a) transforming the identity's average face, or (b) by transforming a gender-matched general average. I found that identification thresholds for post-adaptation were further reduced after adapting to an own-face average, suggesting that within-person variability is represented in reference to an identity-specific average.

Together, this research program aims to reconcile the face space theory with newer approaches to face processing that emphasize the importance of familiarity and exposure to within-person variability. In doing so, I hope to explain how the visual system differentiate different faces while successfully tolerating within-person variability for familiar faces.

2 How Does the Face Space Change with Familiarity?

2.1 Introduction

The multidimensional face space theory (Valentine, 1991) provides a model of face recognition that is structurally based in inter-item similarity; that is, faces that are similar to each other are encoded closer together, and faces that are dissimilar are encoded further apart. Accordingly, typical faces look similar to each other and are encoded in a region with high exemplar density, and distinctive faces look dissimilar to other faces and are encoded in a

region with low exemplar density. In turn, while typical faces are easily confusable with similar-looking others and hence are recognized slower, distinctive faces do not have many competing representations surrounding them, and therefore are recognized faster (e.g., Valentine & Bruce, 1986a; 1986b). Although this model provides a unifying approach for many phenomena in the literature, like the distinctiveness, caricature, and the other-race effects, as discussed previously, it is limited in its assumption that each face is represented as a single point or vector in the face space. This raises the question of how the visual system can recognize faces across several viewpoints.

More recent models of the face space have incorporated regions surrounding a face that can allow for multiple inputs to activate the face's representation. One such model is Tanaka et al.'s (1998) attractor field model in which a face's attractor field determines the range of inputs that can be accepted as that face. Therefore, as the size of the attractor field increases, more exemplars can be accepted as that face. In this model, the size of an attractor field is assumed to be determined by the exemplar density of the region in which the face representation is located, so that less exemplar density yields a face representation that is surrounded by a larger attractor field and more exemplar density yields a face representation that is surrounded by a smaller attractor field. Accordingly, distinctive faces have larger attractor fields and typical faces have smaller attractor fields.

Tanaka et al. (1998) provided evidence for the attractor field model by presenting participants with 50-50 morphs of typical and distinctive faces, and found that the 50-50 morph was psychologically more similar to its distinctive constituent. Since a 50-50 morph is equidistant from its constituents, a traditional model of the face space would predict that the morph would bear equal resemblance to the distinctive and the typical constituents. However,

since the distinctive constituent's attractor field is larger, the morph is closer to the attractor field of the distinctive constituent (Figure 1.12). In turn, the 50-50 morph of a typical and a distinctive face appears more similar to the distinctive face.

More recent evidence, using multiple ambient photos for each target, showed that the attractor field model can account for tolerance to within-person variability. In Laurence et al.'s (2016) study, observers were given a card-sorting task of own- and other-race faces and were asked to create a separate pile for each identity. They found that observers incorrectly created more piles for other-race faces than for own-race faces, suggesting that there is less tolerance to within-person variability for other-race faces. According to the attractor field model, since other-race faces are encoded in a region with high exemplar density, they have smaller attractor fields, so two images of the same other-race face are less likely to fall within the same attractor field, and are therefore mistakenly identified as different people. Further, in Koca et al. (2024), we trained observers to become familiar with faces that share a varying degree of similarity with non-target faces to create clusters of identities in the face space in varying densities. We found that faces that shared high similarity with the non-target faces (i.e., higher exemplar density, hence smaller attractor field) were mistakenly identified as "different" when two novel photos of the targets were shown in a same/different matching task. In line with the attractor field model, creating a higher exemplar density surrounding a target face created smaller attractor fields and made it more difficult to tolerate within-person variability.

If the size of an attractor field is associated with the exemplar density of the region where a face is encoded (i.e., distinctiveness), and the range of within-person variability that can be tolerated (i.e., familiarity), then familiar faces should become more distinctive as they

become familiar. In other words, the representations of familiar faces would migrate to a region with lower exemplar density that allows them to be recognized across viewpoints, and be surrounded by a larger attractor field. Support for this prediction came from Faerber et al. (2016) who asked observers to make distinctiveness judgments for familiar and unfamiliar faces, and importantly, their anti-faces. Anti-faces and their corresponding faces are equidistant from the norm, and therefore are located in a region with similar exemplar density. Nevertheless, Faerber et al. (2016) found that familiar faces were rated as more distinctive than their anti-faces, whereas unfamiliar faces were rated as similarly distinctive as their anti-faces. Kramer et al. (2018) found that familiarity results in an increased distance of a face's representation from the nearest neighbour in the face space. Lastly, Kramer et al. (2025) found that similarity ratings between pairs of identities decreased with increasing familiarity. This suggests that familiarity influences how faces are distributed in the face space, where familiar faces likely expand their attractor field as they become familiar, so that more within-person variability can be tolerated.

In another study, Chauhan et al. (2020) provided evidence that familiarity changes the distribution of representations in face space. In their study, they created morph faces by combining either an unfamiliar face with another unfamiliar face (stranger-stranger), an unfamiliar face with a personally familiar face (stranger-friend), an unfamiliar face with the observers' own faces (stranger-self), a personally familiar face with another personally familiar face (friend-friend), and a personally familiar face with the observers' own faces (friend-self). These morphs were created along a continuum with 10% increments. Observers were presented with morphs along this continuum, followed by the two faces used to create the morph, and were asked which of the two identities was more similar to the morph image.

They found that for 50-50 morphs, observers were more likely to indicate that the morph was more similar to the more unfamiliar face. For example, for stranger-friend morphs, observers chose the stranger face, and for friend-self morphs, they chose the friend face. The authors argued that observers become more sensitive to the structural features of familiar faces, hence narrowing the identity-category boundary. In turn, when a 50-50 morph is presented between a familiar and an unfamiliar face, it does not share enough resemblance to the familiar face to be accepted as the familiar face, because the criterion to determine whether an input is similar enough to a familiar face is more conservative for familiar faces. Webster et al. (2004) obtained similar results, finding that observers had narrower category boundaries for own-race and own-gender faces (Webster et al., 2004), indicating greater sensitivity to faces of more familiar categories. Chauhan et al. explained these results in terms of an expansion of identity-specific subspaces in the face space as a function of familiarity.

This finding seems to contradict the predictions of the attractor field model. Since tolerance to within-person variability is almost perfect for familiar faces (e.g., Jenkins et al., 2011), and familiar faces appear more distinctive than they otherwise would be if they were unfamiliar (Faerber et al., 2016), it is reasonable to assume that familiar faces have a large attractor field. Since a 50-50 morph composed of a face with a larger attractor field (e.g., a distinctive face) and a smaller attractor field (e.g., a typical face) bears more resemblance to the face with the larger attractor field (Tanaka et al., 1998), then a 50-50 morph of a familiar and an unfamiliar face should look more like the familiar face. However, Chauhan et al. (2020) found that a 50-50 morph was judged to be more similar to the unfamiliar face.

One possibility is that Chauhan et al.'s (2020) findings could be solely explained by a distinctiveness effect. Even though a face is *relatively* more distinctive in the face space of the

familiar observer than in the face space of the unfamiliar observer (Faerber et al., 2016), this does not necessitate that any familiar face is more distinctive than any unfamiliar face. Therefore, the 50-50 morph between a distinctive unfamiliar face and a more typical familiar face may bear more resemblance to the unfamiliar face due to its larger attractor field. However, if their findings were only due to the initial distinctiveness of the face images, it would mean that the distinctiveness of the face images was coincidentally correlated with their level of familiarity. Therefore, it seems unlikely that their findings can be accounted for exclusively by distinctiveness, but since the a priori distinctiveness of the faces is unknown, it is possible that differences in the a priori distinctiveness of the constituent faces contributed to judgments of 50-50 morphs as well.

Another possibility is that observers were relying on different reference points when making their judgments. Although the 50-50 morph of two faces is assumed to be equally similar to its constituent *photographs*, it does not have to be equally similar to the *identities* that are used to make the morph face. In Chauhan et al.'s (2020) study, observers were asked to judge which of the two identities was more similar to the morph image. For an unfamiliar face, the only reference point available to the observer is the photograph used in the study. For a familiar face, however, observers have an abstract, rich representation of the person's face beyond the photograph used to make the morph. In turn, when judging which *identity* looks more similar to the morph, observers might be more likely to select the unfamiliar face, because the morph looks more similar to their one and only reference point for the unfamiliar face. If observers are using a different representation as a reference point for the familiar face than the test image, that representation may be located further away from the 50-50 morph than the unfamiliar face, especially if the test image is perceived to be poor likeness.

This would also explain why observers were more likely to select “friend” in friend-self morphs. Previous research has shown that observers are poor at judging the likeness of their own faces, and often choose more favorable versions of themselves as better likeness (e.g., Allen et al., 2009; White et al., 2016; 2017). Therefore, a photograph of the observers taken under controlled conditions as was done in Chauhan et al.’s (2020) paper may not be perceived as a good likeness by the observers themselves, making them more conservative in judging the morph as resembling their face. Perhaps different findings would be obtained if observers were asked which of the two *photographs* resemble the morph.

It is also possible that the attractor field model does not sufficiently account for familiarity. Although a larger attractor field allows for more inputs to be recognized as that face, as is the case for familiar faces, a larger attractor field also worsens differentiation performance. Tanaka and Corneille (2007) presented observers with a typical or a distinctive face, followed by either the identical target face image, or a morph face composed of the typical and the distinctive face with varying ratios. Observers were asked whether the faces were the same or different. They found that observers were less likely to report “different” when a distinctive face was paired with a morph that included a higher ratio of the distinctive face compared to a typical face paired with a morph that included a higher ratio of the typical face. In other words, observers were more likely to confuse a morph that was x units away from a distinctive face with the distinctive face than a morph that was x units away from a typical face with the typical face. This is because a face that is x units away from a distinctive face is more likely to fall within the boundary of its attractor field, whereas a face that is x units away from a typical face is less likely to fall within the boundary of its attractor field (Figure 1.13). Therefore, observers were *less* sensitive to slight changes in the appearance of a

face with a large attractor field than a face with a small attractor field. However, Chauhan et al. (2020) found that observers were *more* sensitive to slight changes in the appearance of familiar faces which is contrary to the finding from Tanaka and Corneille (2007).

The assumption that familiar faces have a large attractor field does not hold when it comes to differentiation performance. Indubitably, familiar faces are successfully differentiated from others. In fact, it has been consistently shown that recognition performance for familiar faces is almost always perfect (e.g., Burton et al., 1999; Jenkins et al., 2011). Further, Koca et al. (2024) found that when the exemplar density of faces surrounding a target face was systematically manipulated, in turn manipulating the size of its attractor field, differentiation performance was *better* for faces with smaller attractor fields. Note that in this study, observers were exposed to within-person variability in all conditions, suggesting that the face space, including the attractor field model, should be evaluated in the context of familiarity.

Therefore, it is unclear how the face space changes with familiarity. According to the findings of Faerber et al. (2016), a familiar face is located in a region with less exemplar density than an unfamiliar anti-face, suggesting that it is surrounded by a larger attractor field that allows space for multiple inputs to activate the face's representation (Tanaka et al. 1998; Tanaka & Corneille, 2007). Chauhan et al.'s finding is contrary to the predictions of a larger attractor field surrounding familiar faces – a 50-50 morph of a familiar and an unfamiliar face bears stronger resemblance to the unfamiliar face. Their finding, however, can be explained by the initial distinctiveness of the face images that are used. Further, observers were not asked to make a judgment regarding the constituent *images* of the 50-50 morph, but its constituent *identities*, which likely contaminated the assumption that the 50-50 morph is

equidistant from the familiar and the unfamiliar face. Importantly, in both Faerber et al. (2016) and Chauhan et al.'s (2020) studies, only one photo per identity was used, which does not capture the range of exemplars that correspond to the representation of a familiar face. They also used faces that are initially familiar or unfamiliar, so it is not known how the face space changes over time as a face becomes familiar.

This chapter aims to address this gap by investigating how the face space changes as a face becomes familiar. In a series of studies, observers were trained to learn a newly encountered face while their perception of the appearance of a 50-50 morph between the target faces and unfamiliar foils were measured over different time points. If the attractor field of a face gets larger as the face becomes familiar, which I will refer to as the *expanding attractor field hypothesis*, then the 50-50 morph between the target and an unfamiliar foil should increasingly resemble the target over time. If the category boundary of a face gets narrower as the face becomes familiar, which I will refer to as the *narrowed category boundary hypothesis*, then the 50-50 morph between the target and an unfamiliar foil should increasingly resemble the foil over time. As familiarity increases, in an adjustment task where observers are asked to adjust a morph face to bear equal resemblance to the target and the foil, observers should adjust *away* from the face that they perceive to be more apparent in the actual 50-50 morph to achieve the point of subjective equality.

Both these hypotheses have the assumption that a familiar face moves away from other faces in the face space, expanding the region surrounding them. They differ in terms of whether this expanding region will *pull in* a similar-looking exemplar to improve the likelihood that it would be accepted as the target (i.e., the expanding attractor field

hypothesis), or *push away* a similar-looking exemplar and reject it as the target (i.e., the narrowed category boundary hypothesis).

2.2 Study 1A

2.2.1 Participants

Based on a power analysis conducted using G*Power (Faul et al., 2009), to obtain an effect size of $\eta_p^2 = .02$ (obtained from Koca & Oriet, 2023) with 80% power for a within-subjects ANOVA with 4 levels, 69 participants were required. Fifty-two observers were recruited through the University of Regina Psychology Participant Pool. All observers had normal or corrected-to-normal vision. All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All observers signed a consent form prior to their participation, and received 1% bonus credit toward a psychology course as compensation. Data from four observers who were familiar with the faces used in the study were excluded from the analysis, so the data were analyzed on 48 observers (34 female, 13 male, 1 non-binary, $M_{age} = 22.4$, $SD_{age} = 5.3$).

2.2.2 Stimuli and Apparatus

Sixteen photos each for six targets, one photo each for 24 unique foil identities, and one photo each for 108 unique distractor identities were collected from the Internet. Photos of Turkish celebrities were used to ensure that the faces used were unfamiliar in Canada, and that many high-quality ambient images were available for each face. None of the faces were obstructed (e.g., hands, glasses). Each photo was cropped so that only the face was shown, and photos were delineated (i.e., specific points on the faces landmarked) using Psychomorph 7.1 (Tiddeman et al., 2005) to be used in morphing. Four images from each target were

randomly chosen to be used during testing. For all these test images and their corresponding foils, a morph continuum was created with 5% intervals: 0-100, 5-95, 10-90, 15-85, 20-80, 25-75, 30-70, 35-65, 40-60, 45-55, 50-50, 55-45, 60-40, 65-35, 70-30, 75-25, 80-20, 85-15, 90-10, 95-5, 100-0 target-foil ratio. All test images were converted to black and white, and images that appeared noticeably dark or bright were re-adjusted using Adobe Photoshop (Adobe Inc., 2019) to ensure that the morphs did not appear skewed towards one of their constituents due to image-specific characteristics. The experiment was programmed using jsPsych (de Leeuw, 2015). The data were collected online on pavlovio.org.

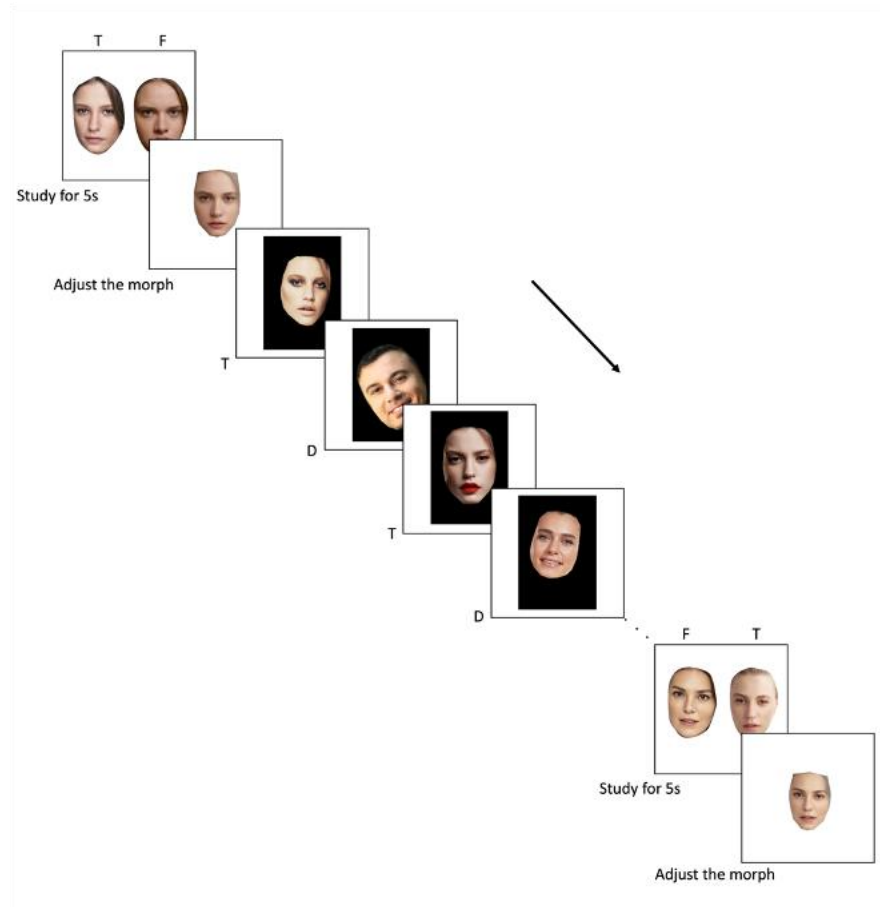
2.2.3 Procedure

First, observers were shown two unfamiliar face images side by side for five seconds and were asked to study these faces. After the faces disappeared, they were asked to adjust a morph face so that it bears equal resemblance to the two face images shown. Observers were able to adjust back and forth between the faces along the morph continuum using arrow keys such that one full cycle brought participants back to their starting point. The morph that was presented initially was randomly selected from the morph continuum, with the exception that it was never the 50-50 morph so that observers had to make at least one toggle to proceed to the next phase of the study. The ratio of the target in the morph that was chosen by the observers in this initial phase was recorded as baseline. After they reported their response, they were shown four ambient images belonging to one of the identities initially shown to the participants to familiarize them with this identity (i.e., the target face). This face was interspersed amongst six unique distractor identities to emulate how we learn faces in the real world (Koca et al., 2024; Koca & Oriet, 2023). Observers were asked to rate the distinctiveness of these ten faces to encourage them to pay attention to the images.

Following the training phase, observers were shown a novel photo of this target identity paired with a novel unfamiliar foil identity for five seconds, and were asked to adjust a morph face again so that it bore equal resemblance to the two face images shown, exactly as they did in the first phase (Figure 2.1). This procedure was repeated a total of three times each for six targets, yielding three training phases and four test phases per target. The trials were blocked by target identity. The order in which the target identities were assigned to the blocks was randomized. The order in which the target and the distractor images were presented during the training phase was also randomized with the constraint that no two consecutive trials depicted the target identity. The order of the target-foil pairs presented during testing was also randomized. In each test trial, target images were paired with a novel unfamiliar identity to eliminate any possible identity- or image-specific effects. This was also done to prevent observers from incidentally becoming familiar with the foil faces during testing. In other words, this was done to ensure that the change in the observers' responses was only influenced by their increasing familiarity with the target. These foil faces matched the target faces demographically (e.g., similar age and gender). Note that, unlike Chauhan et al. (2020), observers were asked to adjust the morph, so it was equally similar to the two face *images*, and not *identities*. This was done to ensure that a 50-50 morph would truly be equidistant from the two test stimuli in the face space and the task would not be contaminated by previous exposure to different target images during the experiment. At the end of the experiment, observers were asked whether they were familiar with any of the faces used in the study prior to the experiment. Data from observers who reported familiarity with the targets ($N = 4$) were removed from the analysis.

Figure 2.1

Procedure Illustration for Study 1



Note. Observers first studied the faces for 5 seconds, followed by adjusting a morph face of the two faces. After a training session, this procedure was repeated for a total of 4 times. T = Target, F = Foil, D = Distractor. Prompts and sliders are not included in the illustration.

2.2.4 Design and Analyses

There was one within-subjects variable with 4 levels: Time Point (T1-T4). The dependent variable was the proportion of the target image present in the morph that was chosen to be equally similar to the two faces (i.e., point of subjective equality). I expected that, at T1, observers would be accurate in selecting the 50-50 morph as equally resembling the two images since both faces are unfamiliar. In other words, the target ratio of the selected

morph at T1 is expected to be around 50%. However, I expected this ratio to change over time as the targets became more familiar.

One possibility is that the target proportion in the morph increased over time, so observers would perceive a morph containing a higher proportion of the target face to bear equal resemblance to the two faces. I call this *the narrowed category boundary hypothesis*. According to Chauhan et al.'s (2020) finding, where a 50-50 morph of a familiar and an unfamiliar face looks more similar to the unfamiliar face, I predicted that observers would perceive the actual 50-50 morph to look more like the unfamiliar face over time. In turn, they would adjust away from the unfamiliar face and choose a morph that has a higher target proportion. If observers were becoming more conservative in perceiving a familiar face as the familiar face so that even slight changes in the appearance of a familiar face would cross the identity boundary, observers would need to add “more” target to the morph mix to be able to perceive the morph to be equally similar to the two faces.

Another possibility is that the target proportion in the morph decreased over time, so that a morph containing a higher proportion of the unfamiliar face would be perceived as the 50-50 morph. I call this the *expanding attractor field hypothesis*. If we assume familiar faces are more distinctive (Faerber et al., 2016), and distinctive faces have larger attractor fields (Tanaka et al., 1998), a face's attractor field should become larger as it becomes familiar. Therefore, a 50-50 morph should look more similar to the familiar face over time, causing observers to adjust away from the familiar face to achieve the point of subjective equality.

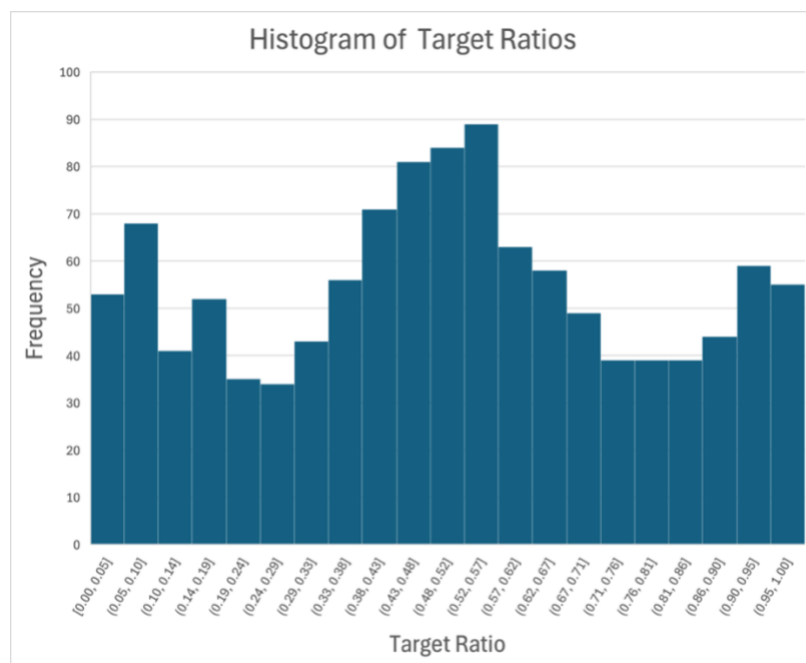
2.2.5 Results

The descriptives revealed that there was no effect of time point on the selected target:distractor ratio, meaning that increasing exposure to a face did not influence the

appearance of the 50-50 morph over time: T1 ($M = 50\%$, $SD = 11\%$), T2 ($M = 50\%$, $SD = 12\%$), T3 ($M = 51\%$, $SD = 13\%$), T4 ($M = 49\%$, $SD = 16\%$). The distribution of responses was examined using a histogram (Figure 2.2). Unexpectedly, responses were distributed in a trimodal fashion, precluding the possibility of conducting an ANOVA. In other words, observers had selected the 0% target or the 100% target image as the midpoint of the target and foil with a frequency comparable to selecting the 50% target image. It could be that unmorphed images appear more "natural", which may have driven observers' preferences. Further, since this is contradictory to what the observers were instructed to do, it could be that the task of memorizing face images was too difficult. This would explain the preference for unmorphed images as they may have facilitated a sense of familiarity while observers were adjusting the morph.

Figure 2.2

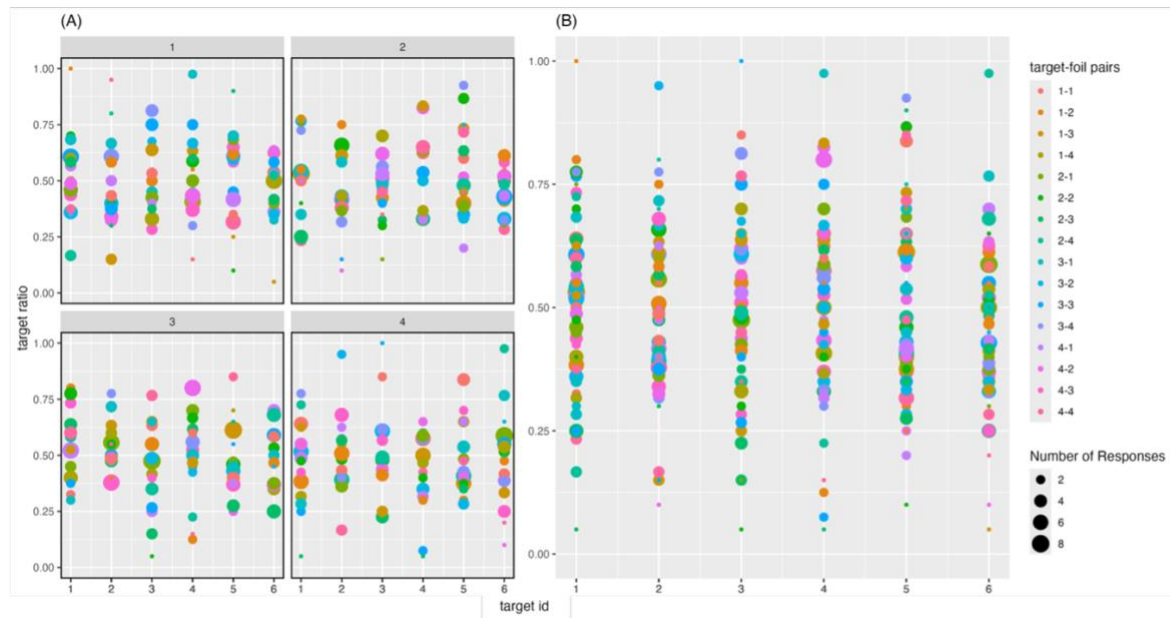
Histogram of Selected Target Ratios in Study 1A



To ensure the results were not due to image- or target-specific effects, the selected target:distractor ratio was plotted for each target-foil pair (Figure 2.3Error! Reference source not found.). Sizes of dots on the graph reflect the number of responses that each target-foil pair received. From the graph, it can be seen that no particular target-foil pair received a response of 0% or 100% target as the mid-point at any time point (separated by panes; Figure 2.3A) except for a few that appear to have only one response. A similar pattern is observed when the data were collapsed across the time point measure (Figure 2.3B). Therefore, there did not seem to be a specific set of target foil pairs that drive the preference for 0% or 100% target.

Figure 2.3

Number of Responses Selected for Each Target-foil Pair



Note. (A) Separate time points and (B) aggregated across time points.

Since the assumption underlying the expanding attractor field hypothesis is that faces become more distinctive as they become familiar, distinctiveness ratings were analyzed for target faces across time point with a one-way ANOVA using JASP (JASP Team, 2025). The

results revealed a main effect of Time Point $F(2, 94) = 5.413$, $MSE = .099$, $p = .006$, $\eta_p^2 = .103$. Posthoc tests using Bonferroni correction revealed a significant *decrease* in distinctiveness ratings between T1 ($M = 3.82$, $SD = .70$) and T2 ($M = 3.62$, $SD = .86$; $p = .008$), with no difference between T3 ($M = 3.66$, $SD = .91$) and other time points. Although significant, the mean difference between T1 and T2 is rather small ($M_{diff} = -.20$) on a 7-point scale. In the following sections, I report a series of pilot studies where I collected data on different variations of this task to test potential methodological limitations contributing to the null findings.

2.3 Pilot Experiment 1: Investigating morph detection and task difficulty

I speculated the preference for selecting the 100% target or foil image may result from (a) a preference for unmorphed images due to the morphed images potentially appearing more unnatural, and (b) difficulty memorizing images. Therefore, in a follow-up pilot study, observers ($N = 37$) were first asked to indicate whether an image is a 50-50 morph of two people or an unmorphed image to investigate whether people can differentiate an unmorphed image from a morphed one. The images were randomly selected from the stimulus set used in Study 1A. Each target was selected once. After they completed the morph detection task, they were given the adjustment task (the testing procedure from the main study). They first completed three practice trials with famous celebrities and received feedback (e.g., “You selected 40% of Ryan Reynolds and 60% of Ryan Gosling.”). Following the practice trials, observers were given a randomly-selected target-foil pair used in the main study (that were not seen in the morph detection task). They saw the target and the foil faces with the adjustment stimulus in the centre. This was done to eliminate the potential difficulty of memorizing images, and to prevent observers from selecting the extreme ends of the morph

continua. At the end of the study, observers were asked to indicate their familiarity with the faces in the study, along with the strategies they used in the task.

2.3.1 Results

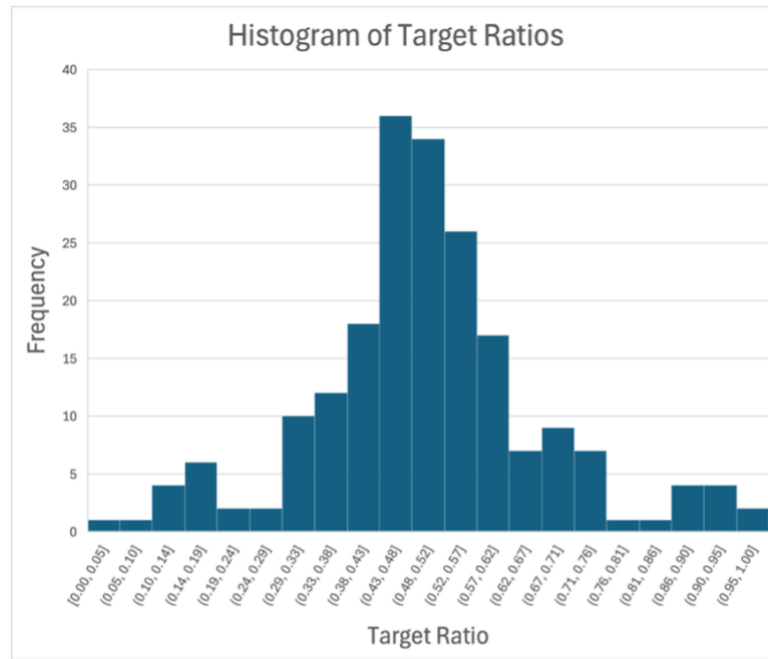
Data from one observer were removed due to not having followed instructions (e.g., attempted to respond without toggling, consistently selected the end points of the morph continua). Data from two observers were removed because they indicated they leveraged the number of toggles to achieve the midpoint. Therefore, the data were analyzed for 34 observers.

The results for the first task showed that observers performed at chance when asked to determine whether an image depicts a morphed image or not $t(33) = 1.504, p = .07, M = 56\%$. Therefore, it is unlikely that the pattern of results observed in the main study is due to a preference for unmorphed images due to their more natural appearance. The p value is approaching significance, but the study is not underpowered, as $N = 27$ is required to detect a medium effect ($d = .5$) with 80% power at $\alpha = .05$.

Results from the second task (excluding practice trials) revealed that the average target ratio selected was 50% with a standard deviation of 8%. Looking at the distribution of responses (Figure 2.4), having the face images remain on screen successfully eliminated selecting the ends of the morph continua as the image equally similar to the two, yielding a normal distribution. This version of the adjustment task was used in a pilot study described below.

Figure 2.4

The Distribution of Selected Target Ratios while Target and Foil Images Remain on Screen



2.4 Pilot Experiment 2: Easier adjustment task

2.4.1 Participants

Twenty-eight observers (22 female, 6 male; $M_{age} = 22.31$, $SD_{age} = 4.43$) were recruited for the experiment. All observers had normal or corrected-to-normal vision. All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All observers signed a consent form prior to their participation, and received 1% bonus credit toward a psychology course as compensation. Data from one observer were removed due to their prior familiarity with the targets used in the study. Data from three observers were removed due to selecting the morph that contains 15% or 85% of the target photo more than three times, which indicates not following the instructions for the

task, especially given that the target and foil images remained on screen for this study. Therefore, the data were analyzed for 24 observers.

2.4.2 Materials and Stimuli

The materials were the same as for Study 1A. Additionally, one photo each for 6 famous celebrities was collected to create practice trials using the same procedure of morphing done in Study 1A.

2.4.3 Procedure

The procedure was identical to Study 1A with the exception that for the adjustment trials, instead of asking observers to view and remember the target and foil images for 5 seconds, both images remained on screen equidistant from their morph at the centre. Therefore, observers were able to adjust the morph by referring to the images on the sides.

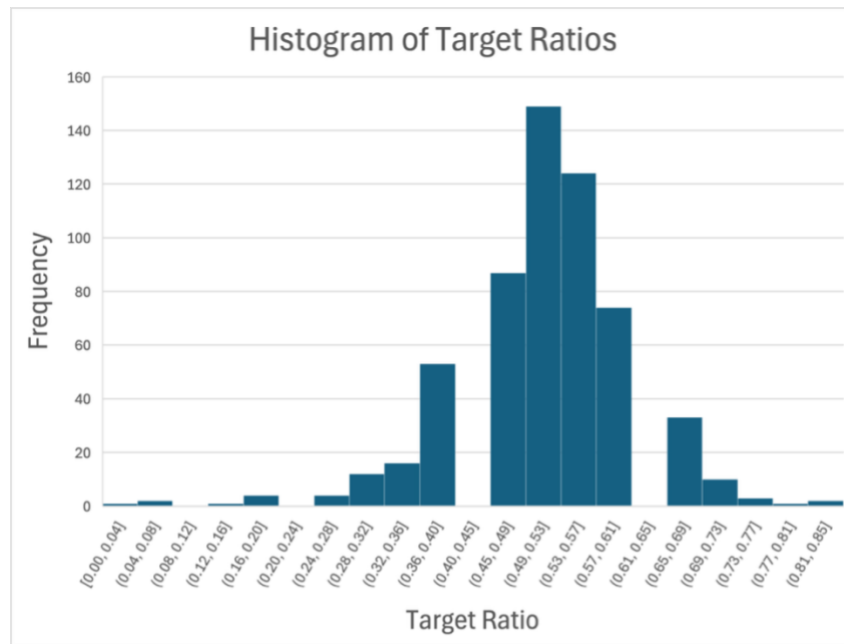
2.4.4 Results

The descriptives indicated there was no effect of time point on the selected target ratio in the perceived 50-50 morph, T1 ($M = 52\%$, $SD = 2\%$), T2 ($M = 49\%$, $SD = 5\%$), T3 ($M = 50\%$, $SD = 5\%$), T4 ($M = 52\%$, $SD = 4\%$), suggesting that increased exposure to a face's within-person variability did not influence the appearance of a 50-50 morph between the face and a similar-looking other.

These findings were similar to Study 1A with the difference that response variability was lower (from 13% average SD to 4% average SD). The histogram of responses is depicted in Figure 2.5. The accuracy of selecting the correct target ratio ($M = 51\%$) and the low variability of responses (especially considering the small number of observers) may suggest that the task was too easy with faces remaining on screen.

Figure 2.5

Histogram of Selected Target Ratios for Pilot Experiment 2



The distinctiveness ratings were also analyzed to see whether distinctiveness ratings given to the targets have changed over time. To improve power, the data were concatenated with Study 1A ($N = 72$), which was not statistically significantly different from this pilot data $F(1, 70) = .872, p = .776$. The data were analyzed using a 3 (Time Point = T1-3) x 6 (Target ID = 1-6) repeated measures ANOVA with the distinctiveness ratings as the dependent variable. Greenhouse-Geisser corrections were applied where necessary.

The results revealed that distinctiveness ratings given to targets *decreased* over time; in other words, targets were rated as more typical with increasing exposure. There was a main effect of Target ID $F(4.375, 311.487) = 6.188, MSE = 1.333, p < .001, \eta_p^2 = .08$. There was a main effect of Time Point, $F(1.720, 122.092) = 7.781, MSE = .752, p = .001, \eta_p^2 = .1$. There was no interaction between Target ID and Time Point.

Post-hoc tests using Bonferroni correction revealed that Target 5 ($M = 4$) was overall more distinctive than Targets 1 ($M = 3.49$), 4 ($M = 3.57$), and 6 ($M = 3.57$). Distinctiveness ratings were the highest at T1 ($M = 3.80$), which were significantly higher than at T2 ($M = 3.62$, $p = .002$, $d = .4$) and T3 ($M = 3.62$, $p = .003$, $d = .4$). However, although significant, .22 is a rather small difference on a 7-point scale.

2.5 Pilot Experiment 3: Improving familiarity

Previous research suggests that a face's representation becomes more distant from others, in other words more distinctive, as it becomes familiar (Faerber et al., 2016; Kramer et al., 2025, White et al., 2022). I found very little change in distinctiveness over time with increased exposure to within-person variability, and the change was in the opposite direction such that the target faces became *less* distinctive. I speculated that perhaps the training paradigm was not sufficiently familiarizing observers with the targets to observe any effects. In previous work (Koca & Oriet, 2023), we found that matching task performance increased for targets trained with the same amount of exposure to within-person variability. However, in that study, we collected attractiveness ratings, which is a conceptual judgment as opposed to the perceptual judgment of distinctiveness. Previous research shows conceptual judgments yield better face recognition than perceptual judgments (Schwartz & Yovel 2019), and learning a name accompanied by a name label improves face recognition more than mere perceptual exposure (Schwartz & Yovel, 2016).

In line with these findings, another pilot study was run with a new training paradigm to better familiarize the observers with the targets by assigning names to targets and collecting attractiveness ratings instead of distinctiveness ratings. Since the task appeared to be too difficult in Study 1A, and too easy in the follow-up pilot experiments, changes were also

made to the adjustment task. First, the target-foil image pair was presented prior to adjustment (as in Study 1A), but observers were allowed to study the images as long as they wished. Additionally, the test images appeared with their names to reinforce learning the target faces. The adjustment procedure itself was also changed. In Studies 1A and the previous pilot experiments, observers were able to adjust the morph in a cyclic manner where they could go back and forth as they wished. In the current study, observers were only allowed to adjust in one direction (unable to go back) and could only adjust until they reached the target or the foil image. This was done to encourage observers to adjust carefully.

2.5.1 Participants

Twenty-seven observers (16 female, 9 male; $M_{age} = 20.78$, $SD_{age} = 2.87$) were recruited for the study. All observers had normal or corrected-to-normal vision. All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All observers signed a consent form prior to their participation, and received 1% bonus credit toward a psychology course as compensation. Data from two observers were removed as they recognized the faces used in the study. Data from six observers were removed due to consistently selecting extreme ends of the morph continua for practice trials, and data from five observers were removed due to consistently selecting extreme ends of the morph continua for the main task. Therefore, the data were analyzed on 14 observers.

2.5.2 Materials and Stimuli

The same materials were used as in Pilot Experiment 2.

2.5.3 Procedure

At the start of the study, observers were given the following instructions:

"Imagine you are a scientist investigating what a face would look like if it was a blend of two people. Your materials for this experiment will be face images with their names. You will need to learn these images and their names, and then you will start mixing these faces. Your task is to adjust this mixture so it looks equally similar to the two images. In between these adjustments, we will ask you learn a person's face (called a target face) while rating the attractiveness of the faces you see. We will give you further instructions as you begin the experiment, which you can do so by pressing space."

Observers were then given practice trials depicting famous faces. They first studied the image pair for as long as they wished along with their name labels. Once they were presented with the morph face, they were able to press the left and right arrow keys to adjust the morph. The arrow keys were labeled with the names of the morph's constituents. Prior to studying the images, they were informed that their laboratory only allows them to add *one* of the two images to the mixture, and that they cannot subtract or add another image. Therefore, if they think the mixture looks more like Person A, they should add more of Person B by pressing the arrow labeled as Person B. Once they start adjusting towards Person B, they could not go backwards. Observers received feedback for practice trials (e.g., "You selected 75% Anne Hathaway and 25% Natalie Portman.").

Upon completing the practice trials, observers were given the same adjustment task using a randomly chosen target face with one of its randomly chosen foils as baseline, as was done in the previous studies. The target proportion in the perceived 50-50 morph image was recorded as the dependent variable. These adjustment trials were interleaved with training sequences, where observers were asked to provide attractiveness ratings. They were also told

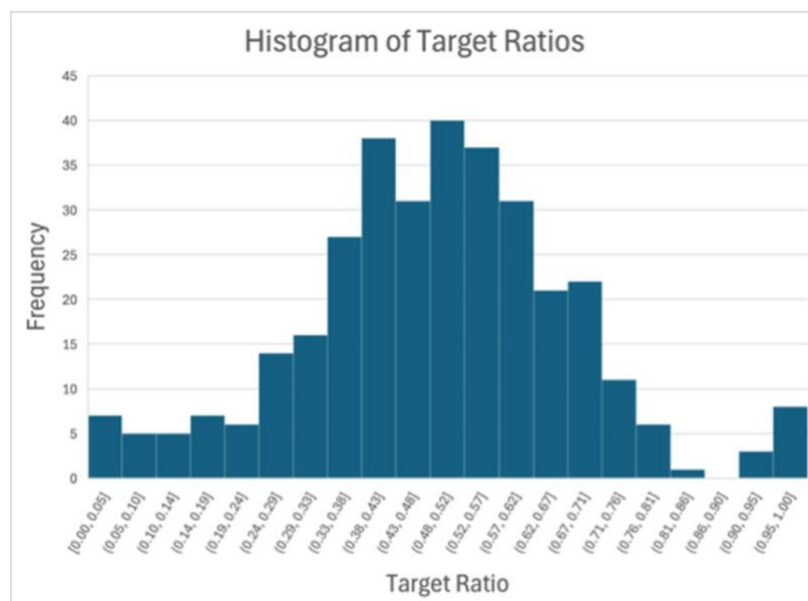
that one of the faces will repeat throughout the sequence, which is the target face. They were given an arbitrarily-assigned name for the target and were asked to pay attention to the target to learn their face. While they were providing attractiveness ratings, all images appeared with a name label. All distractors had a unique name, while the target name repeated with each exposure. The randomization procedure was identical to Studies 1A and the follow-up pilot experiments, as well as the overall structure.

2.5.4 Results

The descriptives indicated there was no effect of time point on the selected target ratio, T1 ($M = 48\%$, $SD = 19\%$), T2 ($M = 49\%$, $SD = 22\%$), T3 ($M = 48\%$, $SD = 21\%$), T4 ($M = 48\%$, $SD = 18\%$). The distribution of responses is depicted in **Error! Reference source not found..**

Figure 2.6

Distribution of Selected Target Ratios in Pilot Experiment 3



2.6 Study 1B

Presenting observers with two face images followed by a morph of the two where they are asked to adjust it to equally resemble the two images is, in essence, asking them to *average*. As previous research has shown, people are rather accurate at representing averages of faces (e.g., de Fockert & Wolfenstein, 2009; Kramer et al., 2015; Koca & Oriet, 2023). Could ensemble coding (i.e., averaging) be responsible for the pattern of results found in the previous studies? It may explain the preference for the extreme ends of the morph (i.e., the exemplars) found in Study 1A, as some studies show equivalent representations of averages and exemplars for averaging the same identity (Kramer et al., 2015), although some (de Fockert & Wolfenstein, 2009) did not find such equivalence averaging different identities. If ensemble coding is responsible for the accuracy with which observers can select the midpoint between the two identities (mean target ratios were always around 50%), an assumption made is that familiar and unfamiliar faces are equally weighed in the averaging process, despite one identity becoming more familiar through exposure to within-person variability. This assumption is supported by previous work. Evidence shows equivalent image-specific representations for familiar and unfamiliar faces for both short-term (Dunn et al., 2019) and long-term (Kramer et al., 2020) memory. Therefore, it is quite possible that observers were averaging the study images, leading them to accurately adjust the mixture.

In Study 1B, I aimed to address the ensemble coding confound by removing the foil image altogether. I asked observers to view only the target image prior to the adjustment trials. Observers were then asked to adjust the morph so that it contained half of the target image. I also decided to collect data in person to ensure that observers understood the instructions.

Prior to data collection in person, I ran a pilot study that replaced the adjustment task with a 2AFC paradigm similar to one used by Chauhan et al. (2020) that encouraged observers to make a target ratio decision for every item in the morph continua. Concurrently, I was running another pilot study with the adjustment task (online) identical to the in-person study described below. I concatenated this data with the in-person version to improve power.

2.6.1 Pilot Experiment 4: Exploring a 2AFC task

2.6.1.1 Participants. Twenty-one observers were recruited for this pilot study. All observers had normal or corrected-to-normal vision. All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All observers signed a consent form prior to their participation, and received 1% bonus credit toward a psychology course as compensation. Data from one observer were removed due to their reported familiarity with the targets used in the study. Data from four observers were removed due to inattentive responding (responded in under 100 ms on more than 30 trials). As such, data were analyzed on 16 observers (14 female, 1 male, 1 unspecified; $M_{\text{age}} = 21.5$, $SD_{\text{age}} = 3.54$).

2.6.1.2 Materials and Stimuli. All materials were identical to Pilot Experiment 3.

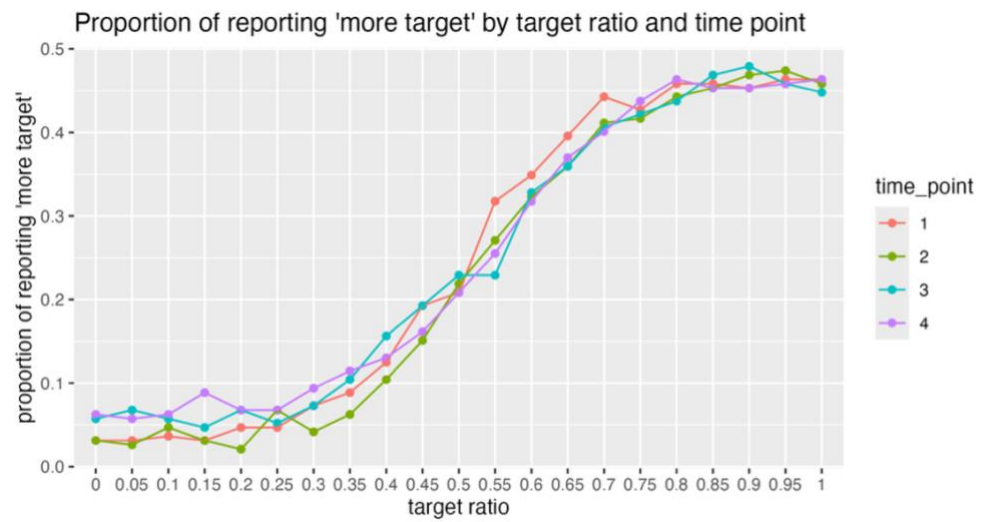
2.6.1.3 Procedure. Observers were given the same scenario about being “face chemists”. This time, however, instead of asking them to make the face mixtures themselves, they were asked to indicate whether a mixture that is presented to them contains more than half of the target’s image or someone else’s image. As was done previously, they first completed practice trials using pairs of three famous celebrities. They first saw one of the randomly-chosen celebrities along with their name (e.g., Ryan Reynolds) and were asked to study this image

with the person's name as long as they wished. Importantly, only one image was shown to prevent observers from averaging. Once they studied Ryan Reynold's image, they were randomly shown one of the 21 images from the target-foil morph continua and were asked whether this mixture contained more than half of Ryan Reynolds' image or someone else's image. They received feedback on whether their response was correct. On trials where the target-foil ratio was 50-50, random feedback was given. Once the practice trials were completed, observers repeated this procedure using the unfamiliar target-foil pairs. In essence, the stimuli were identical to the adjustment task with the difference that instead of a sequential adjustment along the morph continuum (i.e., method of adjustment), observers were randomly presented with an image from the morph continua and were asked to make a judgment at every step (i.e., method of constant stimuli). The proportion of "more target" responses were measured as the dependent variable. The rest of the procedure was identical to Pilot Experiment 3.

2.6.1.4 Results. The proportion of "more target" responses broken down by the actual target ratio and time points are shown in Error! Reference source not found.. Strangely, observers were quite conservative in their "more target" reports, as even the target images themselves (target ratio = 1) are reported to contain more of the target less than 50% of the time. Importantly, there was no shift in the perceived target ratio across time points, especially at the 50-50 target ratio. As a result, I decided to keep the method of adjustment for the main study.

Figure 2.7

Proportion of Reporting “More Target” as a Function of the Target Ratio by Time Point



2.6.2 Main Study: Adjustment task

2.6.2.1 Participants. Based on a power analysis conducted using G*Power (Faul et al., 2009), to obtain an effect size of $\eta_p^2 = .02$ (obtained from Koca & Oriet, 2023) with 80% power for a repeated measures ANOVA with 4 levels, 69 observers were required. 69 observers participated in the study (10 of which participated online as described above). All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All observers signed a consent form prior to their participation, and received 1% bonus credit toward a psychology course as compensation. Data from six observers were discarded due to their prior familiarity with the targets used in the study. Data from five observers were removed due to consistently reporting the ends of the morph continua (over 40% of trials). As such, data were analyzed on 58 observers (42 female, 16 male, $M_{\text{age}} = 21.05$, $SD_{\text{age}} = 5.48$). I did not collect additional data as the data from the

remaining 58 participants yielded a Bayes Factor of 8.440 in favour of the null effect, which is considered moderate evidence (Wagenmakers et al., 2017).

2.6.2.2 Materials and Stimuli. All materials and stimuli were identical to previous studies.

2.6.2.3 Procedure. The procedure was identical to Pilot Experiment 3 with the exception that instead of viewing both constituents of the morph observers were asked to adjust, they only saw the target's image to prevent them from averaging. For data collected in person ($N = 48$), the experimenters walked the observers through the practice trials. After providing verbal instructions, the experimenter completed the first practice trial for the participant by verbally stating their steps. For example, if the morph that was presented looked more like Ryan Reynolds, the experimenter verbally stated that they should be removing Ryan Reynolds from the mixture to achieve a 50-50 blend, and at each step verbally expressed whether they thought the resulting step looked more or less like the target. The experimenter then stayed in the room with the observer as the observer themselves completed the next practice trial. This was all done to ensure that the instructions were understood by the participants. The experiment took about 30 minutes to complete.

2.6.2.4 Results. The target ratio at the point of subjective equality was analyzed as a function of Time Point (T1-T4) using a repeated measures ANOVA in JASP (JASP Team, 2025). The results revealed no main effect of Time Point $F(3, 171) = 1.398$, $MSE = .007$, $\eta_p^2 = .024$, $p > .05$, with moderate evidence in favour of this null effect, $BF_{01} = 8.440^4$. The descriptives show that the mean target ratio selected at T1 was $M = .57$, T2 was $M = .57$, T3 was $M = .56$, and T4 was $M = .58$.

⁴ I also analyzed the data with only the in-person data ($N = 48$) which also revealed no main effect of time point, $F(3, 141) = 2.418$, $MSE = .006$, $p > .05$, $\eta_p^2 = .05$, $BF_{01} = 2.074$.

For exploratory purposes, I re-analyzed these data with target identity added as an interaction term by conducting a 4 (Time Point: T1-T4) x 6 (Target Identity: 1-6) repeated measures ANOVA. The results revealed a main effect of target identity, $F(5, 285) = 4.222$, $MSE = .221$, $p = .001$, $\eta_p^2 = .07$, however, the Bayesian ANOVA provided inconclusive evidence in favour of either hypothesis, $BF_{01} = .993$. Posthoc analyses using Bonferroni correction revealed that the target ratio at the point of subjective equality was lower for Target 6 ($M = .52$) compared to Target 1 ($M = .59$, $p = .015$), Target 2 ($M = .59$, $p = .014$), and Target 5 ($M = .59$, $p = .021$). No other comparisons were significant.

Also for exploratory purposes, I analyzed the attractiveness ratings. In our previous work (Koca & Oriet, 2023) using a similar training paradigm, we found that the standard deviation of attractiveness ratings got lower for targets over time. We speculated that social judgments stabilize with increasing familiarity, which has previously been found to be more stable for familiar than unfamiliar faces (Mileva et al., 2019). Recreating the procedure from Koca and Oriet's (2023) paper, I calculated the standard deviation of attractiveness ratings for each target (Target 1-6) within each time point (T1-T3) and randomly sampled four distractors within each time point. Random sampling was done since each training sequence contained four target and six distractor presentations. Data were analyzed using a 2 (Target: Target, Distractor) x 3 (Time Point: T1-T3) repeated measures ANOVA. The results revealed a main effect of Target, with target identities ($M = .96$) receiving more stable ratings than distractors ($M = 1.56$), $F(1, 57) = 140.202$, $MSE = .299$, $p < .001$, $\eta_p^2 = .71$ with substantial evidence in favour of the alternative, $BF_{10} = 7.064e13$. No other comparisons were significant. To improve power, I also analyzed data concatenated across three studies with the identical training paradigm ($N = 88$; Pilot Experiments 3, 4, and Study 1B adjustment task)

which did not differ significantly $F(2, 85) = .788$, $MSE = .633$, $p > .05$, $\eta_p^2 = .018$. The previous results were replicated, with a main effect of Target $F(1, 87) = 207.251$, $MSE = .206$, $p < .001$, $\eta_p^2 = .704$ with substantial evidence in favour of the alternative $BF_{10} = \infty$. Targets ($M = .95$) received more stable ratings than distractors ($M = 1.52$). No other comparisons were significant. Attractiveness ratings for targets did not stabilize over time.

2.7 Discussion

How does the face space change as a face becomes familiar? Research suggests that familiar faces are located in more isolated regions in the face space than unfamiliar faces (e.g., Faerber et al., 2016; Kramer et al., 2018; Kramer et al., 2025). This leads to the prediction that a face's distance from others in the face space increases with increasing familiarity. This has two possible consequences for the appearance of a morphed face equidistant from an increasingly familiar face and a similar-looking unfamiliar face. According to *the expanding attractor field hypothesis* (Tanaka et al., 1998; Tanaka & Corneille, 2007), the attractor field surrounding a face would expand with increasing familiarity as it moves away from others. In turn, an equidistant morph would be perceived to be more similar to the familiarized target. According to *the narrowing category boundary hypothesis*, an identity-specific subspace for the face would emerge with increasing familiarity. In turn, an equidistant morph would be perceived to be less similar to the familiarized target. Neither of these hypotheses was supported in the current study. In Study 1A, observers were asked to rate the distinctiveness of the faces (perceptual judgment) during training, interleaved with test trials where they were asked to adjust a morph of a target and a foil face along a cycle to achieve 50-50 target:distractor ratio. The point of subjective equality did not change over time, which was not due to image-specific factors. To ensure this was not due to methodological limitations, I

then conducted a series of pilot studies. In Pilot Experiment 1, I showed that observers could not discern 50-50 morphs from unmorphed faces and the adjustment task was easier with morph constituents remaining on screen. As such, in Pilot Experiment 2, observers were given the adjustment task with the faces on screen, but the pattern of findings did not change, and the task seemed too easy as the responses had a small standard deviation (4%). To improve learning the target faces, in Pilot Experiment 3, observers were asked to rate the faces on attractiveness (conceptual judgment; Schwartz & Yovel, 2019) provided with target names, and were able to study the morph images for as long as they wished. To encourage careful adjustments, they were also asked to adjust the morph step-by-step where they could only add or remove the target's image. This manipulation also did not change the point of subjective equality over time. To eliminate ensemble coding as a potential confound, in Study 1B, I removed the foil faces from the pre-adjustment study trial and tested another pilot experiment with a 2AFC task instead of the adjustment task, and there was no effect of time. Overall, exposure to a face's within-person variability did not change the point of subjective equality between that face and similar-looking others. I am confident this was not due to methodological issues. Other potential reasons for the pattern of findings are speculated below.

The targets may have not been learned. Although it is unlikely, this possibility cannot be eliminated entirely because recognition was not directly tested in the current study. Indirect measures of learning showed that while distinctiveness ratings did decrease, the variability of attractiveness ratings, that which we previously demonstrated as an indirect measure of face learning (Koca & Oriet, 2023), remained stable over time. In the same previous work, however, we used the same training paradigm with the same set of stimuli and were able to

improve recognition performance. In the current study, there was a total of 12 exposures to each target depicting high within-person variability (e.g., Corpuz & Oriet, 2022; Ritchie & Burton, 2017), along with top-down support for person recognition from the availability of the names for each face (Schwartz & Yovel, 2016; 2019). Both of these factors have been consistently shown to improve recognition.

Even if the targets were learned, they likely did not become *familiar* in the same way the targets were familiar in Chauhan et al. (2020)'s study. Indeed, Chauhan et al. used personally familiar faces along with the observers' own faces. Research shows differences in personally familiar and experimentally familiar faces (e.g., Herzmann et al., 2004; Natu & O'Toole, 2011), where personally familiar faces are encoded with various perceptual, conceptual, and emotional representations. It is possible that experimentally-built familiarity does not transform the face space like personal, real-life familiarity.

Lastly, it is possible that a face does not become more distinctive (i.e., further from others) over time with exposure to within-person variability, an assumption underlying the rationale of the current study. Indeed, when distinctiveness ratings during training were analyzed (Study 1A and Pilot Experiment 2), distinctiveness ratings *decreased* over time for targets, although the magnitude of the reduction was small (.22 on a 7-point scale). This claim is at variance for some of the previous research cited above. For example, Faerber et al. (2016) found that familiar faces were rated as more distinctive than their anti-faces where the distance from the norm should be equivalent. However, in their study, one of the measures of distinctiveness was the likelihood of spotting a face in the crowd, which is easier for familiar faces not necessarily due to increased distinctiveness, but the presence of the familiarity signal (Tong & Nakayama, 1999). In a PCA + LDA model, Kramer et al. (2018) found that

the distance between a face and its nearest neighbour's centroid increased over time with additional exposures. Kramer et al. (2025) found decreased similarity ratings between identities when they were familiar, indicating increasing distance from others as a function of familiarity. Some research, however, did not find evidence for increased distance with familiarity. In the same work, Kramer et al. (2025) did not find reduced similarity ratings between identities when other-race faces were used. White et al. (2022) did not find evidence for increased distance from the norm as a function of familiarity. It is therefore possible that familiarity contributes little to a face's distinctiveness; rather, distinctiveness might be largely determined by the idiosyncratic variability of a face. The present study provides some support for this claim. Target identity was a significant predictor of target ratio at the point of subjective equality, with some targets mixed less than others in the perceived 50-50 morph. This finding is indicative of distinctiveness varying as a function target identity since observers' point of subjective equality was skewed towards some targets (Tanaka et al., 1998). This specific claim is further explored in Chapter 3.

3 How is Within-person Variability Represented in the Face Space?

3.1 Introduction

In the traditional models of the face space, each face is encoded in a location in the space either as a point or a vector according to its perceived characteristics (Valentine, 1991). Since the dimensions of the face space are unknown, an important factor that provides information about where a face is encoded is the face's distinctiveness. Distinctiveness, by definition, is determined by the inter-item similarity of the faces in the face space. Distinctive faces are assumed to be *dissimilar* to other faces, and typical faces are assumed to be *similar* to other faces and to each other. Indeed, Light et al. (1979) showed that inter-item similarity

ratings for pairs of faces agreed with distinctiveness ratings; that is, faces that were rated as dissimilar when paired with other faces were also rated as more distinctive. Therefore, distinctiveness, which is structurally based on inter-item similarity, is a fundamental property of the face space (Valentine, 2001), especially given that the face space, like other multidimensional perceptual spaces (e.g., colour; Shepard, 1962), is similarity-based.

The distinctiveness of a face can be measured in many ways. For example, Valentine and Bruce (1986a) asked observers to rate how easy it would be to spot the person in a crowd, Light et al. (1979) asked observers to rate the person's similarity to a typical high school male student, and Cohen and Carr (1975) asked observers to simply rank the faces by distinctiveness. Indirect measures of distinctiveness can be collected through (context-free) familiarity and memorability ratings, which have been shown to constitute two orthogonal components of distinctiveness. Specifically, lower familiarity and higher memorability ratings predict higher distinctiveness ratings and vice versa (Vokey & Read, 1992). Another indirect way to measure distinctiveness is to collect inter-item similarity ratings on pairs of faces where more distinctive faces are rated more dissimilar within their pairs (e.g., Johnston et al., 1997; Lee et al., 2000; Light et al., 1979). These studies consistently found superior recognition performance for faces that are more distinctive, indicating that they all measure the same psychological construct of typicality and distance between faces in the face space.

One limitation of these studies, however, is that distinctiveness is measured using a single photo of a person, typically taken under highly controlled conditions. For example, Valentine (1991) collected distinctiveness ratings on photos of faces displaying a neutral expression from a frontal view, and Light et al. (1979) and Vokey and Read (1992) used high school yearbook photos. However, one photo is rarely, if ever, a good representation of a

person, because a person's face varies greatly across different photos (Jenkins et al., 2011). Accordingly, attributes such as attractiveness (Jenkins et al., 2011) or judgments of likeness (Jenkins et al., 2011; Ritchie et al., 2018) also vary across different photos of a person. In fact, Jenkins et al. (2011) found that variability in attractiveness judgments within a person was greater than variability in attractiveness judgments between people, suggesting that attractiveness is as much a photo-centred property as it is a person-centred property (Burton, 2013).

There is no a priori reason to assume that distinctiveness would not be subject to within-person variability. Aspects that make a face distinctive are likely to be present to different extents across different viewing conditions, which is especially intriguing considering that most faces are likely distinctive in at least one dimension. Evidence for this came from Burton and Vokey (1998) who explored the typicality paradox of the face space. Usually, the face space is illustrated as a two-dimensional Cartesian space for ease of interpretation (e.g., see Figure 1.1; Figure 1.4; Figure 1.6; Figure 1.10). In a normally distributed two-dimensional Cartesian space, most faces are clustered near the central tendency, which is described as containing typical faces that resemble most previously encountered faces. However, in studies where typicality ratings are collected, most faces do not end up being rated as “highly typical”, which is counterintuitive considering that most faces are assumed to be typical. Burton and Vokey (1998), however, showed that this is an artefact of conceptualizing the face space in two dimensions. When the normal distribution of exemplars in a space with more than two dimensions is modelled mathematically, it actually appears that the face space is shaped like a “donut”⁵, where highly typical faces are quite rare. This makes intuitive sense

⁵ From my conversations with Dr. R. Kramer.

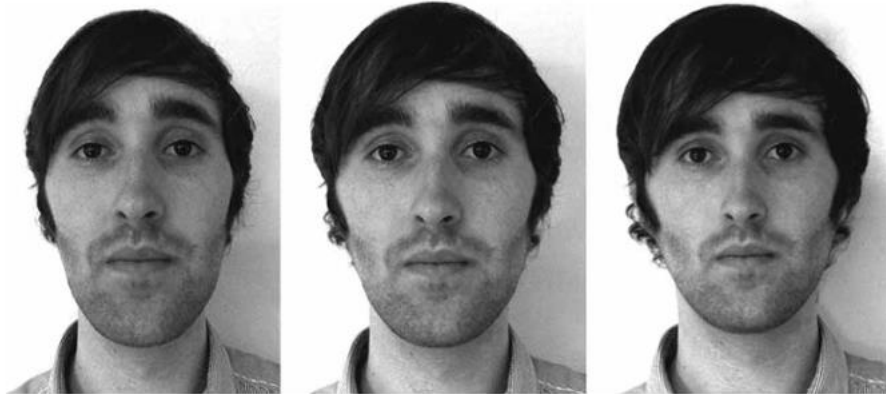
considering the possibly high number of dimensions of the face space (e.g., Lewis speculated there to be 22 dimensions; 2004), where most faces would likely be distinctive on at least one dimension instead of most faces being highly typical on all dimensions.

Since the face space is multidimensional and the dimensions are unknown, it is unclear what exactly makes a face distinctive. Although the idea that the distinctiveness of a face stems from an isolated distinctive feature or a set of features is intuitive, this idea received little support from the literature. For example, Valentine and Bruce (1986b) found that observers are slower when classifying distinctive faces as faces. If a distinctive feature made a face distinctive, distinctive faces would be classified as faces faster, because the salient distinctive feature would signify to the cognitive system that the stimulus is indeed a face. Similarly, Tanaka et al. (1998) found that distinctive faces are vulnerable to inversion, which would not be expected if distinctive faces were encoded by an isolated feature. Assuming inversion disrupts second-order configural processing (i.e., relations between features; Diamond & Carey, 1986), the distinctiveness of a face may be an artefact of its configuration. However, the second-order configuration of a face is vulnerable to within-person variability, considering even a simple change like camera lens distortion can greatly alter a face's configuration (Figure 3.1; Burton, 2013; Sandford & Burton, 2014). Therefore, assuming most faces are somewhat distinctive in at least one aspect (Burton & Vokey, 1998), what constitutes distinctiveness likely varies across different faces. Because faces themselves vary across different views, it becomes highly likely that the perceived distinctiveness would differ across different views of a face. For example, a face with a long nose (i.e., distinctive by an isolated feature) may not appear as distinctive from a frontal view but may appear quite distinctive from a $\frac{3}{4}$ view, or a face that has a long inter-eye distance (i.e., distinctive by the

second-order relational features) may appear distinctive from a frontal view but not as much from a $\frac{3}{4}$ view.

Figure 3.1

Face Images Photographed from Different Distances



Note. As can be seen, the second-order configuration of a face, often expressed as the metric distance between features, changes depending on how much the camera lens distorts the face.

Why has research in face recognition progressed so slowly? The importance of variability,

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The fact that each face varies idiosyncratically leads to the prediction that familiar and unfamiliar faces are processed differently, because the range of idiosyncratic variability across different views of a face is not learned for unfamiliar faces. Therefore, while familiar faces can be successfully recognized across a range of viewing conditions, unfamiliar face recognition is error-prone (e.g., Bruce et al., 1999; 2001; Jenkins et al., 2011; Megreya & Burton, 2006; 2007). This is because familiar face processing relies on an abstract, stable

representation in memory, which makes familiar face processing robust to changes in viewing conditions allowing a novel encounter to be compared to an established memorial representation. On the other hand, unfamiliar face processing is highly image dependent, which makes it vulnerable to such changes. Besides influencing performance in a range of recognition and perception tasks, familiarity also influences social judgments. For example, judgments like attractiveness, trustworthiness, and dominance are more variable across different photos of an unfamiliar face compared to a familiar face (Mileva et al., 2019), and attractiveness judgments become more consistent as familiarity with a face increases (Koca & Oriet, 2023), suggesting a shift from rating “photos” to rating “people”.

Evidence suggests that the perceived distinctiveness of a face also changes with familiarity by moving the representation of the face to a sparser region in the face space. For example, Faerber et al. (2016) asked observers to rate familiar and unfamiliar faces and their corresponding anti-faces, which are equidistant from the norm and therefore assumed to be of equivalent typicality. However, while they found that unfamiliar faces and their anti-faces were rated to be equally typical, familiar faces were perceived to be more distinctive than their anti-faces. Kramer et al. (2025) found that familiar faces were rated as less similar to others than unfamiliar faces, and increasing familiarity with a face reduced similarity ratings between that face and others. Relatedly, Kramer et al. (2018) found that as a computational model became more familiar with a face (i.e., increased number of training images), the distance between the target face and the nearest non-target face increased. These findings suggest that the face space increases the distance of a familiar face from other faces as it becomes familiar, which in turn increases its distinctiveness⁶.

⁶ Valentine and Bruce (1986b) found that familiar faces were classified as faces more slowly than unfamiliar faces, similar to how distinctive faces are classified as faces more slowly than typical faces.

The goal of the current study outlined in this chapter is to investigate how within-person variability is represented in the face space for familiar and unfamiliar faces. To do so, inter-item similarity ratings on pairs of ambient images depicting the same person (familiar or unfamiliar) or different people (both familiar, both unfamiliar, familiar and unfamiliar) were collected. These ratings were then analyzed using multidimensional scaling (MDS), which is a statistical method for mapping out the stimuli as locations in a multidimensional space based on their perceived similarity such that similar items are located near each other, and dissimilar items are located further apart (Hout et al., 2013; Jaworska & Chupetlovska-Anastasova, 2009). For example, if people are given pairs of city names from all around the world and were asked to rate how distant they are from each other, if people's estimates are accurate the resulting MDS solution would appear like a world map. MDS has been previously used to construct face spaces to investigate the distinctiveness and caricature effects (Lee et al. 2000), and the other-race effect (Byatt & Rhodes, 2004). However, to my knowledge, no studies used MDS for mapping out multiple ambient images of familiar and unfamiliar faces. Such a method can allow for a visual construction of the face space that represents within-person variability and by extension, familiarity.

3.2 Methods

3.2.1 Participants

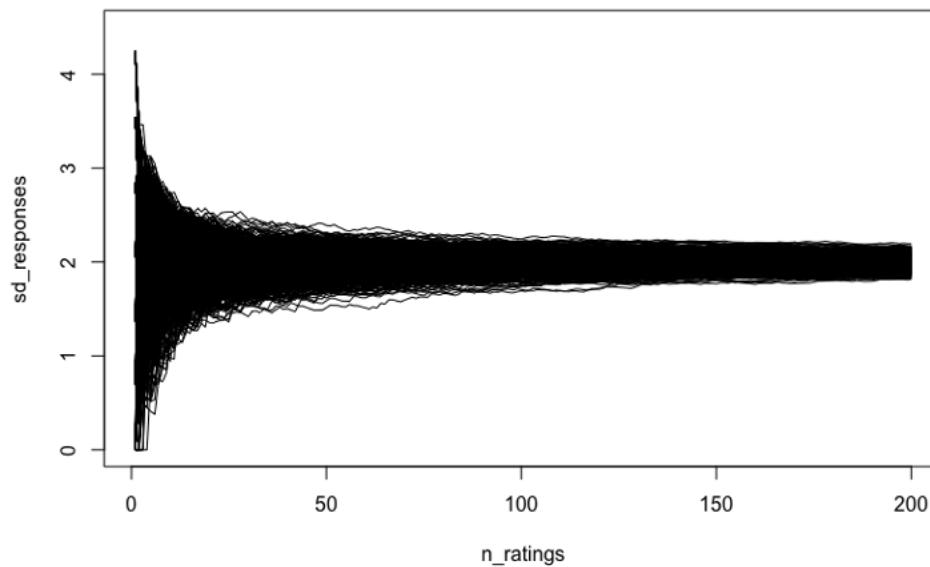
To my knowledge, there is no specific sample size recommendation to carry out an MDS procedure. Therefore, I ran a simple simulation to determine the number of observers to be recruited for the study using R (R Core Team, 2021). The rationale was to obtain the number of ratings required for a given pair in the stimulus set (i.e., a single trial) to achieve a

Speculatively, the main effect of familiarity found in their experiment could be a result of increased distinctiveness of familiar faces.

reasonable consistency across observers (i.e., the point of saturation). The simulation represents a worst-case scenario in which every observer gives a random rating on each trial on a scale of 1-7, which is the rating scale used in the study. After the second simulated observer's rating is determined (randomly), a standard deviation of the two ratings is calculated and stored in a list. After the third observer, a standard deviation of all three ratings is calculated and stored in the same list, and so on for 200 observers. The standard deviation of the accumulated ratings for 200 observers was then plotted, and this simulation was run 1000 times, each representing a single experiment. The resulting plot is shown in Figure 3.2.

Figure 3.2

Simulation Results



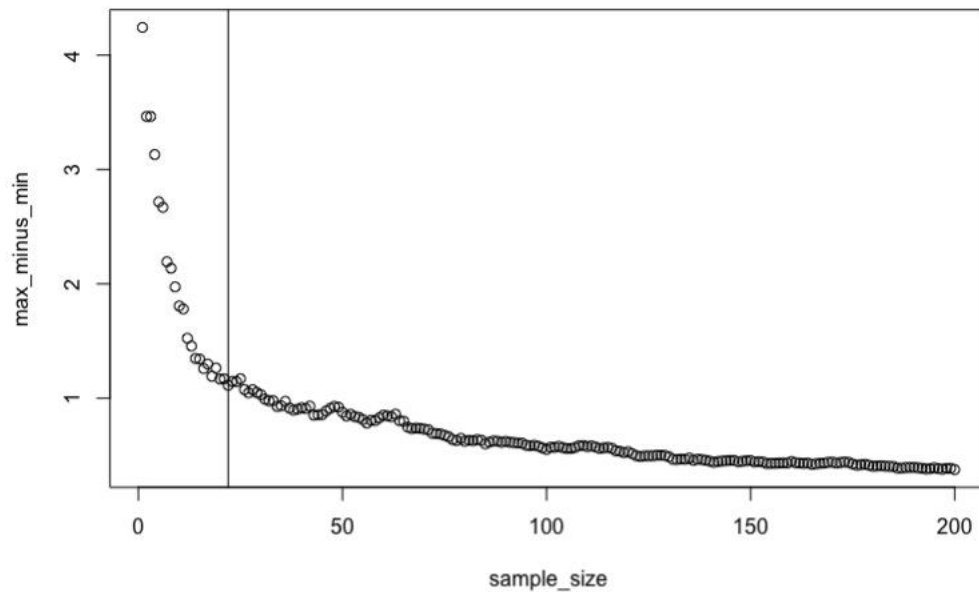
Note. The standard deviation of random responses with each additional rating. Each line on the graph represents one iteration of the simulation.

To find the point of saturation, I then calculated the range of standard deviation (across 1000 experiments) at each number of observers along the x-axis, and then plotted these ranges

(Figure 3.3). Since the goal is to obtain a number where an additional rating would not significantly change the overall consistency of the ratings, I calculated the elbow point of this scree plot using the *smmerc* package (version 1.8.2; French, 2023). For the simulation demonstrated here, the elbow point was at 21, suggesting 21 observers would be sufficient to achieve the point of saturation assuming all observers give a random response. Therefore, I reasoned that 20-25 ratings per pair would be adequate.⁷

Figure 3.3

Simulation Results



Note. The standard deviation ranges (i.e., thickness of lines in Figure 3.2) as a function of each rating that is added. The line vertical line indicates the elbow point of the curve.

Three hundred and sixty-seven observers were recruited through the University of Regina’s Psychology Participant Pool and by word of mouth. Eighteen observers were

⁷ This is also similar to what was used in Lee et al.’s (24 observers; 2000) and Byatt & Rhodes’ (22 observers; 2004) papers that used MDS.

excluded due to failing more than 20% of the attention checks (see below), and 18 observers were excluded due to reporting unfamiliarity with all the targets used in the study. For the remaining observers ($N = 331$; 259 female, 65 male, 6 non-binary, 1 unspecified, $M_{\text{age}} = 21.51$, $SD_{\text{age}} = 4.84$), data were filtered based on reported prior familiarity with the targets. In other words, trials depicting familiar targets that were reported to be unfamiliar and trials depicting unfamiliar targets that were reported to be familiar were excluded for those observers to achieve around 20-25 rating per each possible image pair.

3.2.2 Stimuli and Apparatus

Six photos were collected for each of the 16 target identities (four familiar and female, four familiar and male, four unfamiliar and female, and four unfamiliar and male). All images belonged to celebrities (famous in Canada for familiar faces or abroad for unfamiliar faces) to ensure multiple high-quality ambient photos of the targets were available that depicted high, unsystematic within-person variability. Each target had a near neighbour (i.e., someone that matched their age and hair colour) and a far neighbour. For example, within the female familiar target set, Natalie Portman and Anne Hathaway were near neighbours (in their 40s, brunette) and Margot Robbie and Taylor Swift were near neighbours (in their 30s, blonde). None of the faces were obstructed with objects like hands or glasses, but some faces were partially obstructed with hair (e.g., bangs). All images were cropped and resized to 760 x 1140 pixels. All images were then delineated and masked using PsychoMorph 7.1 (Tiddeman et al., 2005) so that only the head was shown. After delineation, an average was created for each identity. These averages were filtered using wavelet MRF method to have realistic skin texture. This procedure yielded seven photos per identity.

To carry out an MDS procedure, it is recommended that each stimulus to be shown to the observers is paired with every other stimulus in the set without repetition. However, the pairs will only contain male or female faces to prevent observers from making their judgments on the basis of gender. Previous studies that used inter-item similarity ratings (two of which used MDS; Byatt & Rhodes, 2004; Lee et al., 2000; Light et al., 1979) only used male faces, however, in this study both male and female faces were used. Therefore, for each gender, $56 \times 55/2$ pairs were used, which equals 1540 pairs, yielding a total of 3080 pairs. Since presenting observers with 3080 pairs is not feasible in one session, the pairs were randomly divided into 8 subsets, yielding 385 trials per subset. Subsets were selected randomly for each observer to be later concatenated for MDS with the goal of collecting 20-25 ratings per pair. The experiment was programmed using jsPsych (de Leeuw, 2015) and the data were collected online using Pavlovia. All data handling was carried out using R (R Core Team, 2021), including data cleaning, analyses, and plotting.

3.2.3 Procedure

Observers were shown two images at a time, equidistant from the centre of the screen, and were asked to rate the similarity between the two faces on a Likert scale ranging from 1-7 (1- strongly dissimilar; 7- strongly similar; Figure 3.4). The order of image pairs was randomized across observers. Observers were asked to not base their ratings solely on identity, to prevent them from giving really low ratings for pairs that depict different identities, and really high ratings for pairs that depict the same identity. The instructions were as follows:

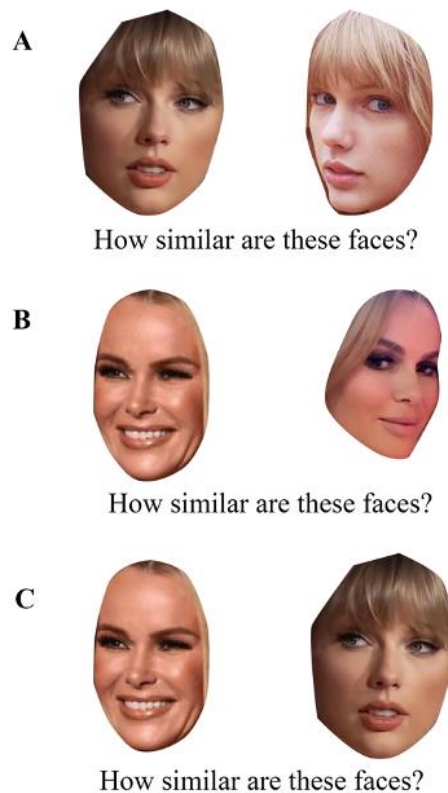
“Please rate the similarity between the faces on the screen. Sometimes the faces will belong to the same person and sometimes they will belong to different people. Keep in

mind that two photos of the same person can look very dissimilar, and two photos of different people can look very similar. Therefore, do not try to base your ratings solely on whether the photos seem to be of the same person, but rather on how similar the photos of the faces look.”

The experiment was carried out in five blocks. Within each block, two trials paired the same exact randomly-chosen image as attention checks. At the end of the experiment, observers were shown an image of each target and were asked whether they were familiar with this face prior to the experiment. If they selected “Yes”, they were given 4 choices that listed names and occupations, one of which correctly described the target. Incorrect choice names were generated using ChatGPT (OpenAI, 2024).

Figure 3.4

Example Image Pairs in Study 2



Note. Example pairs of (A) two familiar faces, (B) two unfamiliar faces, (C) a familiar and an unfamiliar face. The slider is not included in the illustration.

Once 200 observers were collected, trials depicting pairs where observers accurately recognized an unfamiliar face or did not recognize a familiar face (determined post-test) were removed for each observer, which yielded imbalances in the numbers of ratings for each pair. Using the *data.table* package in R (Lim & Qin, 2016), the number of ratings given to each pair was calculated, then the experiment was re-programmed to only depict pairs that received fewer than 25 ratings by binning pairs based on the number of ratings needed to achieve around 20-25.

3.3 Results and Hypotheses

3.3.1 Multi-Dimensional Scaling

Ratings were averaged across participants for each image pair, and then a distance score was calculated by subtracting the mean ratings from 8 (i.e., $8 - M_{\text{pair similarity}}$) to recode the dependent variable as *dissimilarity*. These dissimilarity ratings were then used to create a distance matrix. A distance matrix is a symmetrical matrix that contains the name of each stimulus (i.e., photo) in rows and columns where their intersections indicate their perceived dissimilarity (i.e., distance). It is symmetrical in the sense that rows and columns that contain the same stimulus have a distance of “0” which fall diagonally along the matrix with the same values mirrored on each side. Such matrices were separately created for male and female faces as they were paired separately, and hence separate MDS analyses were conducted on male and female faces using the *smacof* package in R (Rusch et al., 2023).

The primary objective of an MDS solution is to find the best fit using the smallest number of dimensions (Hout et al., 2013; Jaworska & Chupetlovska-Anastasova, 2009). The

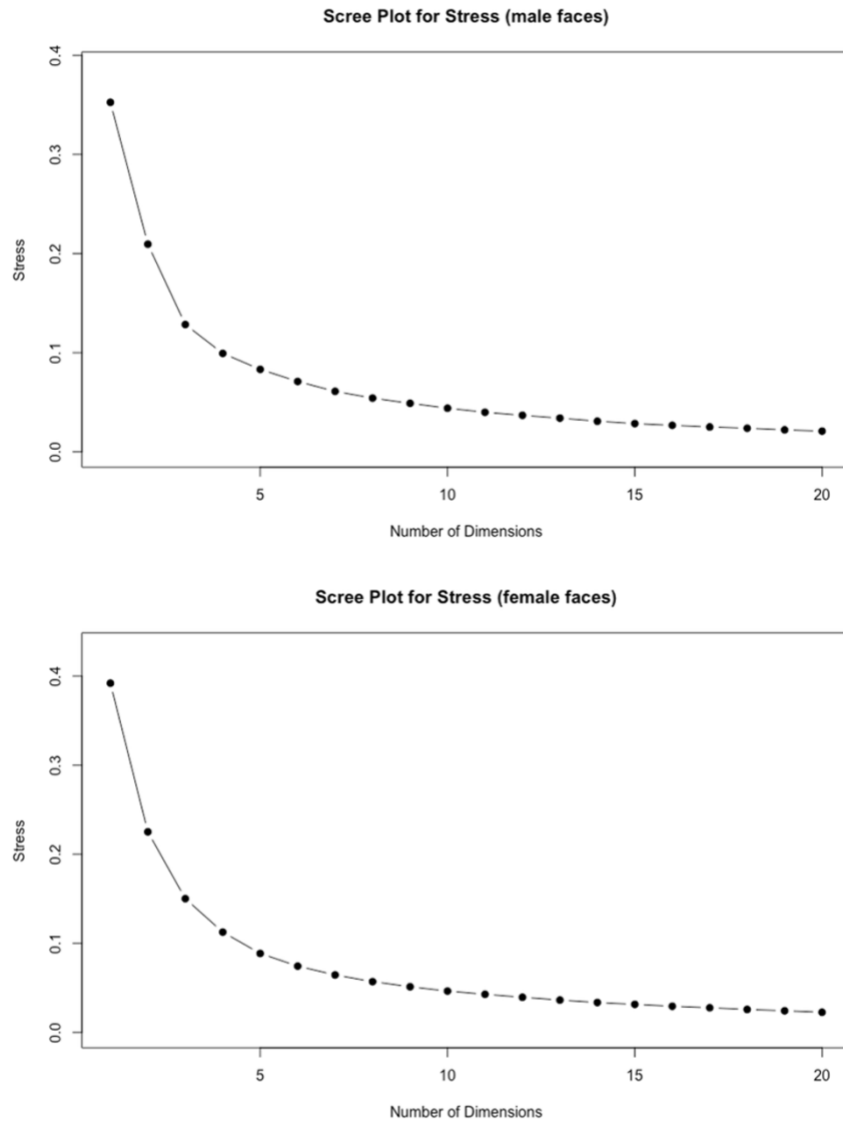
number of dimensions to include in the analysis were selected by creating a scree plot of stress⁸ (σ) values for each possible number of dimensions. Hout et al. (2013) recommend selecting the number of dimensions that correspond to the “elbow” of the plot, which often creates a balance between low stress (i.e., conflict) and the interpretability of the MDS solution. Kruskal and Wish (1978) recommend that stress should not exceed 0.15. For both male and female faces, a 3-dimensional MDS solution, although not corresponding to the elbow point (see

Figure 3.5), had stress values lower than 0.15 ($\sigma = .13$ for male, $\sigma = .14$) and did not alter the interpretability of the solution. Therefore, 3 dimensions were selected as the entire solution can then be interpreted and visually plotted.

Figure 3.5

Kruskal's Stress (σ) as a Function of the Number of Dimensions for the MDS Solution for Male and Female Faces.

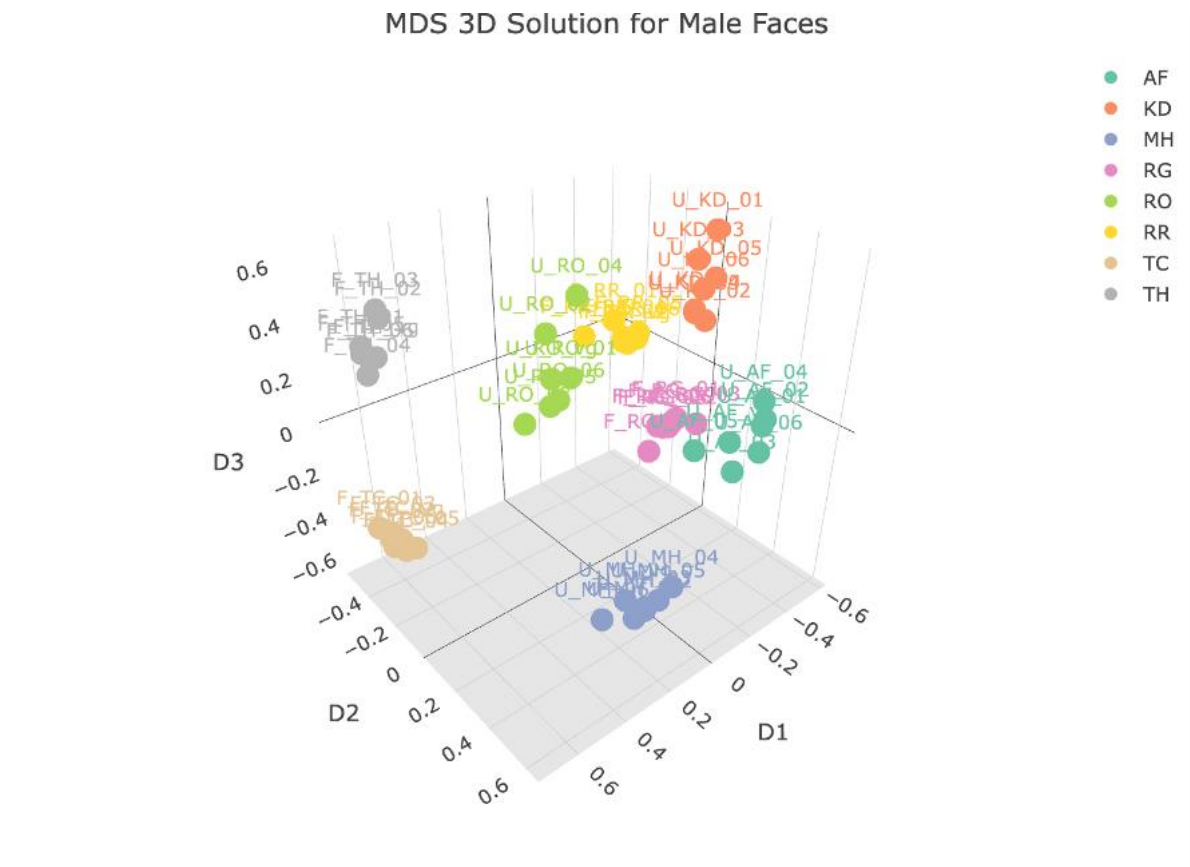
⁸ Collecting inter-item similarity ratings may create certain conflicts. For example, if an observer rates images A and B as highly dissimilar, and images A and C as highly similar, in a two-dimensional space B and C should also be highly dissimilar since B is similar to A and A is dissimilar to C. However, since observers see two images at a time, they may rate B and C as similar as well, which creates a conflict that can be measured with a stress function (Hout et al., 2013).



The 3-Dimensional MDS solution for male faces is presented in Figure 3.6 below. The interactive plot greatly facilitates understanding of the MDS solution and can be viewed via OSF using this link (download required): <https://osf.io/k43wg>

Figure 3.6

The 3D MDS Solution for Male Targets

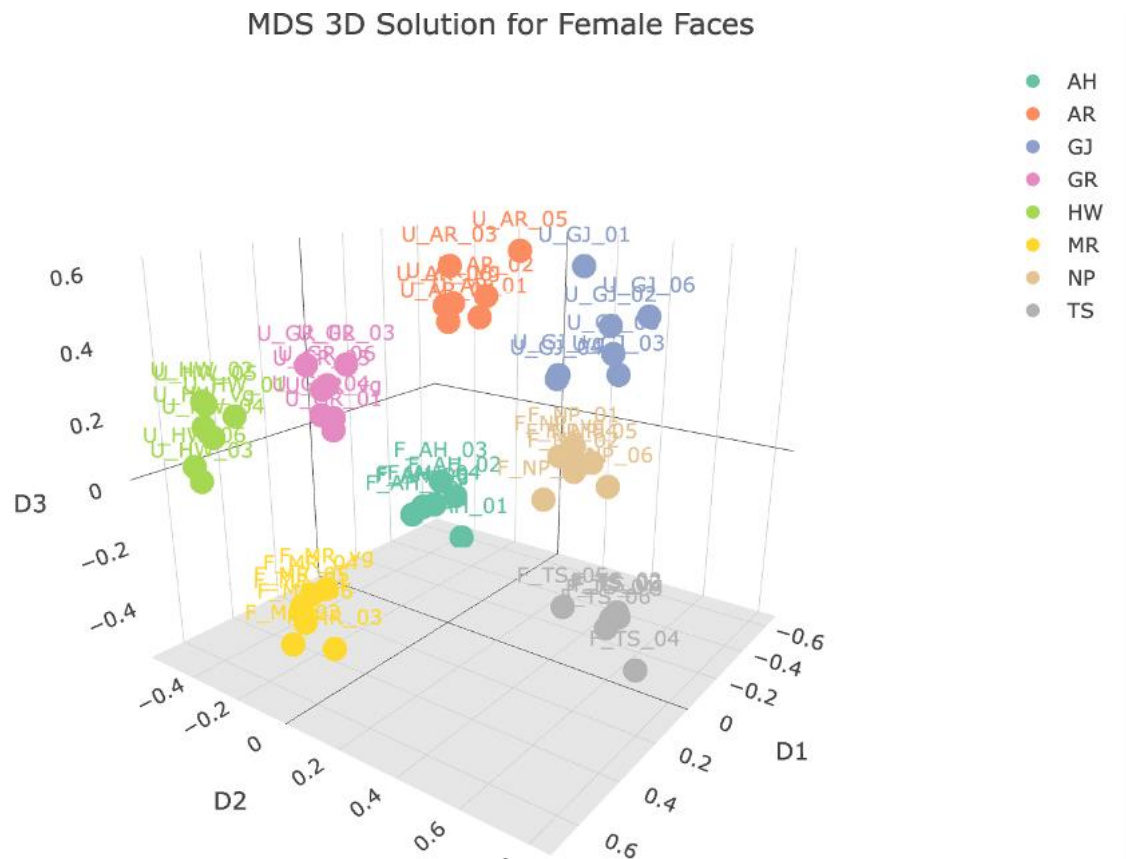


Note. The nodes are labeled as U = unfamiliar, F = familiar, followed by the initials of the target names and the image number. Images numbered as “vg” are average images. Each identity is presented in a unique colour. D1 = Dimension 1, D2 = Dimension 2, D3 = Dimension 3. AF = Alexander Fehling, KD = Ken Duken, MH = Max Hegewald, RG = Ryan Gosling, RO = Rick Okon, RR = Ryan Reynolds, TC = Timothée Chalamet, TH = Tom Holland

I followed the same procedure for female faces. The 3-Dimensional solution for female faces can be seen in Figure 3.7 and can be found on <https://osf.io/pz7ey> (download required):

Figure 3.7

The 3D MDS Solution for Female Targets



Note. The nodes are labeled as U = unfamiliar, F = familiar, followed by the initials of the target names and the image number. Images numbered as “vg” are average images. Each identity is presented in a unique colour. D1 = Dimension 1, D2 = Dimension 2, D3 = Dimension 3. AH = Anne Hathaway, AR = Anna Raads weld, GJ = Gaite Jansen, GR = Gigi Ravelli, HW = Holly Willoughby, MR = Margot Robbie, NP = Natalie Portman, TS = Taylor Swift

Upon visual inspection, distinct identity clusters are readily apparent in the 3D plot for both familiar and unfamiliar faces. This is a rather interesting finding, indicating that facial identity can be represented in the face space with as few as 3 dimensions. In the sections

below, I report my analyses on the coordinates extracted from (a) this 3-dimensional MDS solution as well as auxiliary MDS solutions created by (b) random subsampling of exemplars in the 3-dimensional space, (c) block-based analyses to investigate the change in the space with increased exposure to faces during the experiment, and (d) the 7-dimensional MDS solution, which is the optimal number of dimensions determined by Kruskal's stress analysis.

Differences in distances between identities and exemplars should be interpreted within the coordinate limits. For male faces, the coordinate limits are $-.6$ to $+.6$ for D1, $-.4$ to $+.6$ for D2, and $-.6$ to $+.6$ for D3 (Figure 3.6). For female faces, the coordinate limits are $-.6$ to $+.6$ for D1, $-.4$ to $+.6$ for D2, and $-.4$ to $+.6$ for D3 (Figure 3.7). This means that the volume of the space is 1.44 for male faces and 1.2 for female faces. The maximum possible distance between two points would be approximately 1.97 for male faces and 1.85 for female faces. For reference, a difference of .07 in the exemplar density of two clusters would correspond to about 5% of the entire face space for males and 6% of the entire face space for females. A difference of .07 between two points' distance from all others (between-people distances) would correspond to about 3.56% of the maximum distance for male faces 3.78% of the maximum distance for female faces.

3.3.1.1 Within-person Clusters. Do familiar faces form tighter or more dispersed clusters?

One possibility is that inter-item similarity will be less variable for familiar faces than for unfamiliar faces, creating higher exemplar density clusters for familiar faces. I will call this the *representation anchoring hypothesis*. According to this hypothesis, since familiar face processing relies on an abstracted representation in memory, this representation would serve as an anchor for making inter-item similarity judgments such that identity-specific information would be prioritized over image-specific information. This was consistently

shown for matching tasks (i.e., same/different tasks), where two photos of a familiar face are matched with almost perfect performance, whereas unfamiliar face matching is highly error-prone (e.g., Clutterbuck & Johnston, 2004; Megreya & Burton, 2006; 2007). In turn, any two photos of a familiar face would bear stronger resemblance to each other than any two photos of an unfamiliar face. This was found for social judgments like attractiveness, trustworthiness, and dominance (Mileva et al., 2019), where familiar faces received more consistent ratings than unfamiliar faces. Therefore, the overall inter-item similarity might be higher for familiar faces, creating denser clusters in the MDS space. Further, tolerance to within-person variability is worse for unfamiliar faces (e.g., Jenkins et al., 2011); that is, two photos of an unfamiliar face can look like entirely different people, leading any two photos of an unfamiliar face to be rated as highly dissimilar, creating lower exemplar density clusters for unfamiliar identities. Support for this came from previous research showing that familiarity increased perceived similarity between different images of the same person (Kramer et al., 2025; White et al., 2022).

Another possibility is that familiar faces will have lower exemplar density clusters. I call this the *image anchoring hypothesis*. Since familiar faces rely on an abstract representation in memory, observers would be better able to discern identity-specific and image-specific characteristics of familiar image pairs. Since identity-specific characteristics of familiar faces will appear constant across two photos of the same person, image-specific differences would be more easily discriminable. In turn, two photos of a familiar face may appear more dissimilar than two photos of an unfamiliar face, where image-specific characteristics are not as effectively extracted. This is similar to the notion of “face likeness”, where different photos of a person capture that person’s appearance to varying degrees

(Jenkins et al., 2011), and likeness judgments vary idiosyncratically across viewers (Ritchie et al., 2018). In turn, two photos of a familiar face may be judged as more dissimilar than two photos of an unfamiliar face because they likely do not capture the person’s likeness to the same extent. Naturally, two photos of an unfamiliar face may also capture that face’s likeness to a different extent. However, since face likeness, by definition, relies on an internal representation to be used as a reference (Ritchie et al., 2018), it would not be relevant to unfamiliar observers (i.e., observers cannot tell whether a photo is a good or bad likeness if they are unfamiliar with the person shown).

To calculate the exemplar density of within-person clusters, the average Euclidean distance between each exemplar to other exemplars was calculated using (1):

$$D_i = \frac{1}{N-1} \sum_{j=1, j \neq i}^N d_{ij} \quad (1)$$

D_i is the mean distance of image i to all other images of the same person, N is the total number of images per identity, and d_{ij} is the Euclidean distance between image i and image j . As such, a smaller D_i means higher exemplar density. A linear mixed-effects model was fitted to the data with familiarity as a fixed effect and target identity as a random effect (accounting for within-person variability) using the *lmerTest* package in R (Kuznetsova et al., 2017). R^2 was calculated using the *performance* package in R (Lüdtke et al., 2021). The model estimated fixed effects differences by treating familiar faces as the reference group. As such, the estimates should be interpreted as Unfamiliar - Familiar. The estimates themselves refer to the mean difference in the exemplar density between familiar and unfamiliar faces.

3.3.1.1.1 Main Analysis: 3-Dimensional Space. In the 3-dimensional space with all exemplars included, for male faces, familiarity was a significant predictor of within-person

exemplar density $b = 0.072$, $SE = 0.016$, $t(6) = 4.575$, $p = .004$, with familiar faces ($M = .11$) forming tighter clusters than unfamiliar faces ($M = .19$), supporting the *representation anchoring hypothesis*. Familiarity (marginal R^2) accounted for 36% of the variance. Familiarity and target identity together (conditional R^2) contributed to 41% of the variance. As such, while the effect of familiarity contributed substantially to the variance explained by the model, target identity beyond familiarity contributed weakly (Cohen, 1998). For female faces, familiarity was a significant predictor of within-person exemplar density $b = 0.079$, $SE = 0.025$, $t(6) = 2.797$, $p = .03$, with familiar faces ($M = .14$) forming tighter clusters than unfamiliar faces ($M = .21$), supporting the *representation anchoring hypothesis*. Familiarity (marginal R^2) accounted for 31% of the variance. Familiarity and target identity together (conditional R^2) contributed to 56% of the variance. As such, in contrast to male faces, for female faces both the effect of familiarity and the effect of target identity contributed substantially to the variance explained by the model (Cohen, 1998). These results suggest that familiarity increases the density of identity clusters in the face space.

3.3.1.1.2 Auxiliary Analysis: Subsampling The face space I constructed in this study differs from real psychological face spaces in that it represents an equal number of exemplars for familiar and unfamiliar faces which, by definition is a violation of real-world occurrence since there are necessarily more exemplars contributing to the representation of familiar faces than to unfamiliar faces. To investigate how the face space would represent within-person clusters when fewer unfamiliar exemplars are encoded, I ran five simulations that subsample the data so that all exemplars are selected for familiar faces ($n = 7$) and a random subset of exemplars are selected for each unfamiliar face. The number of exemplars selected ranged from 2 to 6 (hence five simulations). I then repeated the procedure I followed above; I re-fit

the MDS and re-calculated within-person exemplar density in the subsampled face spaces. I ran this subsampling procedure 1000 times per dataset (i.e., male and female faces) and per simulation, and fit a linear mixed effects model to the subsampled data with familiarity as a fixed effect and target identity as a random effect, yielding 1000 models per simulation. I report the mean and the 95% confidence interval of the distribution of the estimates from these models. The estimates refer to the difference in exemplar density between familiar and unfamiliar faces, with positive values indicating higher exemplar density for familiar faces. The difference in exemplar density between familiar and unfamiliar faces is deemed to be statistically significant if the 95% CI excludes 0.

When there were 2 unfamiliar exemplars, both male and female familiar faces formed clusters with equal exemplar density as unfamiliar faces, with the mean estimates of .09, 95% CI [-.02, .28] and .15, 95% CI [-.01, .4], respectively. When there were 3 unfamiliar exemplars, both male and female familiar faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimates of .08, 95% CI [.03, .14], and .11, 95% CI [.01, .26], respectively. When there were 4 unfamiliar exemplars, both male and female faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimates of .07, 95% CI [.04, .11], and .08, 95% CI [.04, .14], respectively. When there were 5 unfamiliar exemplars, both male and female faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimates of .07, 95% CI [.04, .10], and .08, 95% CI [.05, .10], respectively. When there were 6 unfamiliar exemplars, both male and female faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimates of .07, 95% CI [.04, .09], and .07, 95% CI [.05, .08], respectively. These results show that when the face space encodes fewer exemplars for unfamiliar faces, familiar face

clusters are represented with higher exemplar density than unfamiliar face clusters. The addition of unfamiliar exemplars only decreased the confidence intervals of the differences in exemplar density.

It is possible that the subsampling procedure itself, in other words, the mere number of exemplars encoded in the face space, is responsible for the pattern of findings reported above. To test this possibility, I repeated this procedure by subsampling from familiar faces instead. When there were 2 familiar exemplars, both male and female faces formed clusters with equal exemplar density as unfamiliar faces, with the mean estimates of .05, 95% CI [-.05, .1], and .01 95% CI [-.18, .1], respectively. When there were 3 familiar exemplars, familiar male faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimate .06, 95% CI [.02, .1]. Familiar female faces formed clusters with equal exemplar density as unfamiliar faces, with the mean estimate .03, 95% CI [-.11, .09]. When there were 4 familiar exemplars, familiar male faces formed clusters with higher exemplar density than unfamiliar faces, with the mean estimate .07, 95% CI [.04, .09]. Familiar female faces formed clusters with equal exemplar density as unfamiliar faces, with the mean estimate .04, 95% CI [-.07, .09]. When there were 5 familiar exemplars, both familiar male and female faces formed clusters with higher exemplar density than familiar faces, with the mean estimates .07, 95% CI [.05, .09], and .06, 95% CI [.03, .08], respectively. When there were 6 familiar exemplars, both familiar male and female faces formed clusters with higher exemplar density than familiar faces, with the mean estimates .07, 95% CI [.06, .08], and .06, 95% CI [.06, .07], respectively.

These results suggest that the differences in within-person exemplar density between familiar and unfamiliar faces are not merely due to the number of exemplars encoded in the

face space, but the familiarity of the targets. Even when there were fewer exemplars for familiar faces, unfamiliar faces never formed clusters with higher exemplar density. The size of the confidence intervals also remained more or less stable with each addition of familiar exemplars, suggesting that adding familiar exemplars to the face space did not increase the certainty with which the differences between within-person exemplar density is estimated. However, the number of exemplars may have partially contributed to the pattern of findings, at least for female faces, as fewer than 5 exemplars in the female faces face space yielded equivalent exemplars densities between familiar and unfamiliar faces.

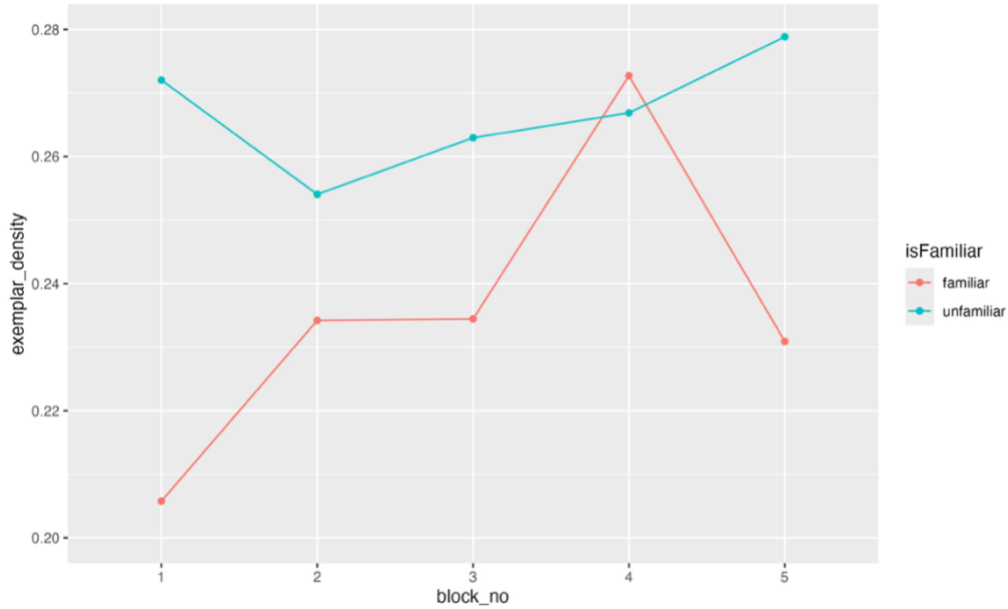
3.3.1.1.3 Auxiliary Analysis: Time. To explore whether or how the face space changes with increased exposure to within-person variability, especially for unfamiliar faces, I also analyzed the data broken down by block number ($n = 5$). Since subsets and the order of pairs were randomized, not all pairs were presented in each block. Specifically, out of 3080 pairs, 18 were missing from the first block, 15 were missing from the second block, 13 were missing from the third block, 12 were missing from the fourth block, and 9 were missing from the fifth block. The MDS cannot be fitted to missing data, so I imputed the missing pairs with the mean dissimilarity rating ($8 - \text{similarity rating}$) given within that block. Following the same procedure, I fit the MDS solution separately for male and female faces within each block, and re-calculated within-person distances using (1). If familiarity organizes within-person variability in the face space by increasing the exemplar density of an identity cluster, the difference between the exemplar density between familiar and unfamiliar faces should be reduced as the experiment progresses. To test this hypothesis, I fit a linear mixed effects model to the data with within-person exemplar density predicted by familiarity and block number (specified as an interaction term) as fixed effects and target identity as a random

effect. I then computed contrasts using the *marginaleffects* package in R (Arel-Bundock et al., 2024) to compare the difference between familiar and unfamiliar faces within each block, and between blocks 1 and 2, 2 and 3, 3 and 4, and 4 and 5.

For male faces (Figure 3.8), in the first block, the difference in exemplar density between familiar ($M = .21$) and unfamiliar ($M = .27$) faces was $b = 0.067$, $SE = 0.019$, $z = 3.533$, $p = .002$. In the second block, the difference between familiar ($M = .23$) and unfamiliar ($M = .25$) faces was $b = 0.020$, $SE = 0.019$, $z = .29$, $p = .29$, which was significantly less than the difference in the first block, $b = 0.046$, $SE = 0.021$, $z = 2.25$, $p = .02$. In the third block, the difference between familiar ($M = .23$) and unfamiliar ($M = .26$) faces was $b = 0.029$, $SE = 0.019$, $z = 1.52$, $p = .12$, which did not differ from the difference in the second block, $b = -0.009$, $SE = 0.021$, $z = -.42$, $p = .67$. In the fourth block, the difference between familiar ($M = .27$) and unfamiliar ($M = .27$) faces was $b = -0.006$, $SE = 0.019$, $z = -.312$, $p = .75$, which was significantly less than the difference in the third block, $b = .034$, $SE = 0.019$, $z = 1.67$, $p = .048$. In the fifth block, the difference between familiar ($M = .23$) and unfamiliar ($M = .29$) faces was $b = .045$, $SE = 2.56$, $z = 1.67$, $p = .01$, which was greater than the difference in the fourth block, $b = .054$, $SE = 0.021$, $z = 2.61$, $p = .009$, and did not differ from the difference in the first block, $b = .02$, $SE = 0.021$, $z = .89$, $p = .37$. Overall, for male faces, the evidence for convergence is mixed – while the difference between familiar and unfamiliar faces was reduced from the first block, it was “restored” in the last block.

Figure 3.8

The Difference in Exemplar Density Between Familiar and Unfamiliar Male Faces by Block Number

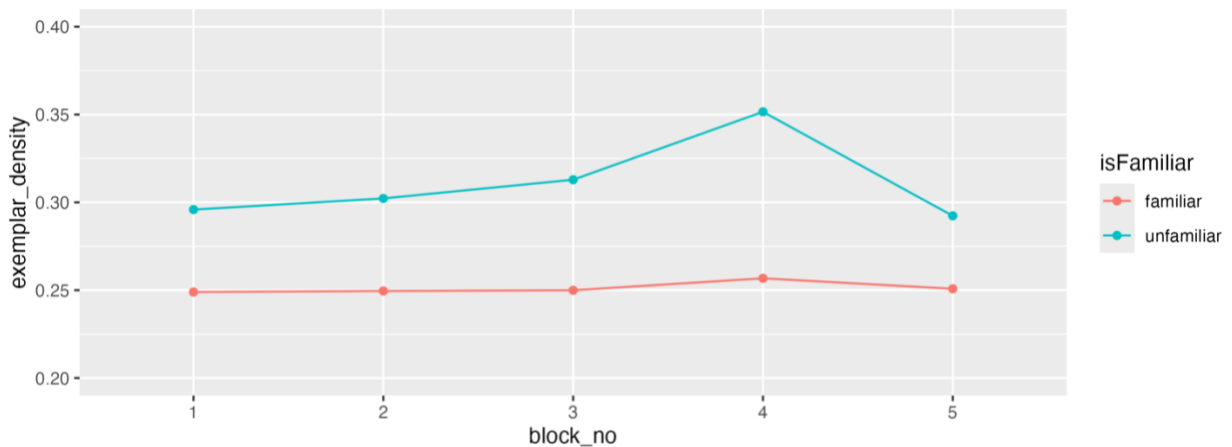


For female faces (Figure 3.9), in the first block, the difference in exemplar density between familiar ($M = .25$) and unfamiliar ($M = .30$) faces was $b = 0.047$, $SE = 0.024$, $z = 1.95$, $p = .05$. In the second block, the difference between familiar ($M = .25$) and unfamiliar ($M = .30$) faces was $b = 0.053$, $SE = 0.024$, $z = 2.18$, $p = .03$, which did not differ from the first block, $b = -0.006$, $SE = 0.023$, $z = -.247$, $p = .81$. In the third block, the difference between familiar ($M = .25$) and unfamiliar ($M = .31$) faces was $b = 0.063$, $SE = 0.024$, $z = 2.61$, $p = .009$, which did not differ from the second block, $b = -.01$, $SE = 0.023$, $z = -.44$, $p = .66$. In the fourth block, the difference between familiar ($M = .26$) and unfamiliar ($M = .35$) was $b = 0.095$, $SE = 0.024$, $z = 3.93$, $p < .001$, which did not differ from the third block, $b = -.03$, $SE = 0.023$, $z = -1.36$, $p = .17$. In the fifth block, the difference between familiar ($M = .25$) and unfamiliar ($M = .29$) was $b = 0.042$, $SE = 0.024$, $z = 1.72$, $p = .08$, which was significantly less than the difference in the fourth block, $b = -0.053$, $SE = 0.023$, $z = -2.27$, $p =$

.02, but not different than the difference in the first block, $b = 0.005$, $SE = 0.023$, $z = .23$, $p = .82$. Overall, for female faces, there was no compelling evidence for a convergence between familiar and unfamiliar faces over time.

Figure 3.9

The Difference in Exemplar Density Between Familiar and Unfamiliar Female Faces by Block Number



3.3.1.1.4 Auxiliary Analysis: The 7-Dimensional Solution. Lastly, I also constructed a 7-dimensional MDS solution and recalculated within-person exemplar density using (1) since 7 dimensions was the optimal number of dimensions according to Kruskal’s stress analysis. In a PCA model representing within-person variability Burton et al. (2016) found that the first 3 dimensions captured more rigid changes like head rotation, and after the fourth dimension the space represented non-rigid changes such as expression and speech. After the fourth dimension, the authors observed idiosyncratic changes across identities. Although PCA spaces are constructed somewhat differently than MDS spaces, it is still possible that the effects of familiarity are captured differently when considering a more than 3-dimensional solution. The pattern of findings, however, was replicated from the 3-dimensional space. Specifically, for male faces, familiarity was a significant predictor of exemplar density $b =$

.049, $SE = .013$, $t(54) = 3.709$, $p < .001$. Familiar faces ($M = .30$) formed tighter clusters than unfamiliar faces ($M = .35$). Familiarity explained 20% of the variance in the model (moderate; Cohen, 1998), while target identity explained none of the variance. For female faces, familiarity was a significant predictor of exemplar density $b = .078$, $SE = .021$, $t(6) = 3.719$, $p = .01$. Familiar faces ($M = .30$) formed tighter clusters than unfamiliar faces ($M = .38$). Familiarity contributed substantially to the model (marginal $R^2 = .35$), and familiarity + target identity (conditional R^2) explained 47.5% of the model. Target identity contributed moderately to the model.

3.3.1.2 Between-people Clusters. Does familiarity reduce the similarity between the familiar face and others? In other words, are familiar face clusters located further away from other faces compared to unfamiliar face clusters? I expected that familiar faces would form more distinct clusters than unfamiliar faces. In other words, familiar faces would be clustered further away from each other and other unfamiliar faces. This would be in line with the findings of Faerber et al. (2016) who found that familiar faces were rated as more distinctive than their corresponding unfamiliar anti-faces. Since faces and anti-faces located in regions with similar exemplar density have similar distinctiveness, this finding shows that familiarity shapes the face space so that familiar faces move further away from their near neighbours. A similar finding came from Kramer et al. (2018) in their PCA + LDA model for face recognition, where the distance between a target face and the nearest non-target centroid increased as a function of familiarity.

Recently, Kramer et al. (2025) found that similarity ratings between similar-looking identities were lower when one or both identities were familiar. This prediction is also in accordance with Tanaka et al. (1998)'s attractor field model. In this model of the face space,

an attractor field of a face increases with decreasing exemplar density surrounding a face's representation. A larger attractor field indicates that more exemplars can be accepted as that face, allowing for more tolerance to within-person variability (Laurence et al., 2016).

Assuming faces become more distinctive as they become familiar (Faerber et al., 2016), their attractor fields would also get larger, suggesting that they would be located in a region with lower exemplar density. However, there is also evidence that shows no change in between-people distance with familiarity. For example, White et al. (2022) found that familiar and unfamiliar identity averages were equally similar to a gender-matched average (i.e., the prototype face), even though they predicted familiar face averages to be more distinctive. In the series of studies reported in Chapter 2, I did not find evidence for increasing distance from others as a function of exposure to within-person variability. Therefore, it is unclear how familiarity would change the distance between the familiar face and others.

There are two ways in which familiarity can change the distribution of identities in the face space. If the face space expands globally, then the distance between an identity and all others would increase (Faerber et al., 2016; White et al., 2022). If the face space expands locally, then only the distance between near neighbours would increase (Kramer et al., 2018; Kramer et al., 2025). To investigate global expansion, I calculated the mean distance of each exemplar of an identity to all exemplars of other identities using Equation (2). To investigate local expansion, I identified the average distance between each exemplar in an identity cluster and each exemplar in their nearest neighbour's identity cluster derived from distances calculated in Equation (2)⁹.

⁹ Another way to investigate distances between clusters is to calculate the distances between their centroids. I chose to calculate point-based distances as different values can be obtained for each item (i.e., exemplar), allowing greater variability for statistical testing. Centroid-based distance calculations are also less robust when there are outliers or when the clusters are not uniformly distributed. Since within-person variability

$$D_i = \frac{1}{N} \sum_{j=1}^N d_{ij} \quad (2)$$

D_i is the mean distance of image i to all other exemplars of different identities, N is the total number of images of identities except the target, and d_{ij} is the Euclidean distance between target's image i to non-target images j . A larger D_i means further distance from others. A linear mixed-effects model was fitted to the data with familiarity as a fixed effect and target identity as a random effect (accounting for within-person variability) using the *lmerTest* package in R. R^2 was calculated using the *performance* package in R. The model estimated fixed effects differences by treating familiar faces as the reference group. As such, the estimates should be interpreted as Unfamiliar minus Familiar. The estimates themselves refer to the mean difference in the exemplar density between familiar and unfamiliar faces, with positive estimates indicating greater distance from others for unfamiliar faces, and negative estimates indicating greater distance from others for familiar faces.

3.3.1.2.1 Main Analysis: 3-Dimensional Space. For male faces, familiarity was not a significant predictor of distance between identities, neither globally, $b = -0.08$, $SE = 0.037$, $t(6) = -2.012$, $p = .09$, nor between nearest neighbours $b = 0.02$, $SE = 0.035$, $t(6) = 0.643$, $p = .54$. For the global expansion model, familiarity (marginal R^2) accounted for 26% of the variance. Familiarity and target identity together (conditional R^2) contributed to 73% of the variance. This suggests that the distance between a face and others (i.e., distinctiveness) is better explained by target identity, since target identity explained almost half of the variance in the model (~47%). For the local expansion model, familiarity (marginal R^2) accounted for 4% of the variance. Familiarity and target identity together (conditional R^2) contributed to

is idiosyncratic, I assume that clusters of identity representations in the face space would not be uniformly distributed.

66% of the variance. Again, the distance between a face and their nearest neighbour is mostly explained by target identity (~62%).

For female faces, familiarity was not a significant predictor of distance between identities, neither globally, $b = -0.01$, $SE = 0.045$, $t(6) = -0.183$, $p = .86$, nor between nearest neighbours $b = 0.03$, $SE = 0.110$, $t(6) = 0.232$, $p = .82$. For the global expansion model, familiarity (marginal R^2) accounted for .3% of the variance. Familiarity and target identity together (conditional R^2) contributed to 55% of the variance. This suggests that the distance between a face and others (i.e., distinctiveness) is better explained by target identity, since target identity explained almost half of the variance in the model (~54.7%). For the local expansion model, familiarity (marginal R^2) accounted for .6% of the variance. Familiarity and target identity together (conditional R^2) contributed to 88% of the variance, suggesting that the distance between a face and their nearest neighbour is mostly explained by target identity (~87.4%).

3.3.1.2.2 Auxiliary Analysis: Subsampling. I repeated the subsampling procedure I conducted in the within-person distance calculations to test whether the above-reported findings are a result of the equal number of exemplars included for familiar and unfamiliar faces because, as noted above, familiar faces should (by definition) have more exemplars associated with them than unfamiliar faces. The estimates again refer to the difference in distance from others between familiar and unfamiliar faces. Negative estimates mean larger distance for familiar faces than unfamiliar faces.

When there were 2 unfamiliar exemplars, both male and female familiar face clusters were located further from all other faces in the face space (i.e., the global expansion model) than unfamiliar faces with the mean estimates of $-.06$, 95% CI $[-.08 \text{ } -.04]$ and $-.04$, 95% CI $[-$

.08 -.02], respectively. Male and female familiar faces both had similar distance from their nearest neighbours as unfamiliar faces (i.e., the local expansion model), with the mean estimates of .01, 95% CI [-.09 .10], and .02, 95% CI [-.05 .10], respectively.

When there were 3 unfamiliar exemplars, both male and female familiar face clusters were located further from all other faces in the face space than unfamiliar faces with the mean estimates of -.06, 95% CI [-.08 -.05], and -.04, 95% CI [-.06, -.01], respectively. Male and female familiar faces both had similar distance from their nearest neighbours as unfamiliar faces, with the mean estimates of .03, 95% CI [-.06 .08], and .01, 95% CI [-.06 .08], respectively.

When there were 4 unfamiliar exemplars, both male and female familiar face clusters were located further from all other faces in the face space than unfamiliar faces with the mean estimates of -.07, 95% CI [-.09 -.05], and -.03, 95% CI [-.04, -.01], respectively. Male and female familiar faces both had similar distance from their nearest neighbours as unfamiliar faces, with the mean estimates of -.01, 95% CI [-.08 .04], and .01, 95% CI [-.08 .06], respectively.

When there were 5 unfamiliar exemplars, both male and female familiar face clusters were located further from all other faces in the face space than unfamiliar faces with the mean estimates -.07, 95% CI [-.08 -.06], and -.02, 95% CI [-.03, -.01], respectively. Male and female familiar faces both had similar distance from their nearest neighbours as unfamiliar faces, with the mean estimates of -.01, 95% CI [-.05 .02], and .01, 95% CI [-.07 .07], respectively.

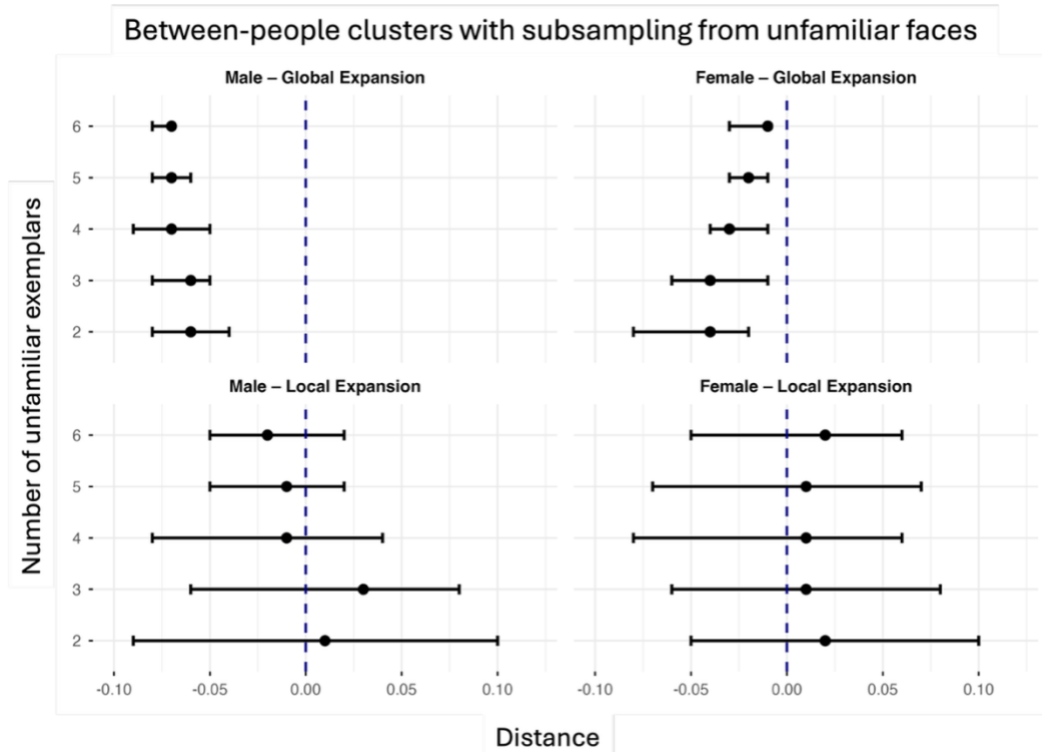
Lastly, when there were 6 unfamiliar exemplars, both male and female familiar face clusters were located further from all other faces in the face space than unfamiliar faces with

the mean estimates $-.07$, 95% CI $[-.08 \text{ } -.07]$, and $-.01$, 95% CI $[-.03, -.01]$, respectively. Male and female familiar faces both had similar distance from their nearest neighbours as unfamiliar faces, with the mean estimates of $-.02$, 95% CI $[-.05 \text{ } .02]$, and $.02$, 95% CI $[-.05 \text{ } .06]$, respectively. These results suggest that the lack of observed between-people distances between familiar and unfamiliar faces may be a result of the equal number of exemplars encoded in the face spaces created in this study. The local expansion model was not supported in any of the subsampling analyses. A summary of these CIs can be seen in

Figure 3.10.

Figure 3.10

The Confidence Intervals of the Estimates of the Difference in Between-people Distances Between Familiar and Unfamiliar Faces



Note. For the global expansion (distance from all other identities) and local expansion (distance between nearest neighbours) models. If the confidence interval includes 0, there is no statistically meaningful difference between familiar and unfamiliar faces.

To test whether this pattern of findings reported above is merely due to the number of exemplars encoded in the face space irrespective of observers' familiarity with the faces, I repeated the subsampling procedure by subsampling from familiar faces. Overall, while all subsampling of male faces resulted in familiar faces located further from all other faces (i.e., the global expansion model) and their nearest neighbours (i.e., the local expansion model) than unfamiliar faces, subsampling of female faces resulted in equivalent distance from all other faces and their nearest neighbours between familiar and unfamiliar faces. When there were 2 familiar exemplars, familiar male faces were located further from all others and their nearest neighbours than unfamiliar faces, with the mean estimates $-.10$, 95% CI $[-.11, -.08]$ and $-.20$, 95% CI $[-.24, -.08]$, respectively. Female familiar faces had equivalent distance from

all others and their nearest neighbours with unfamiliar faces, with the mean estimates .02, 95% CI [-.01, .05] and -.06, 95% CI [-.11, .07], respectively.

When there were 3 familiar exemplars, familiar male faces were located further from all others and their nearest neighbours than unfamiliar faces, with the mean estimates -.09, 95% CI [-.1, -.08] and -.16, 95% CI [-.18, -.06], respectively. Female familiar faces had equivalent distance from all others and their nearest neighbours with unfamiliar faces, with the mean estimates .02, 95% CI [-.01, .04] and -.04, 95% CI [-.06, -.01], respectively.

When there were 4 familiar exemplars, familiar male faces were located further from all others and their nearest neighbours than unfamiliar faces, with the mean estimates -.08, 95% CI [-.09, -.08] and -.11, 95% CI [-.15, -.05], respectively. Female familiar faces had equivalent distance from all others and their nearest neighbours with unfamiliar faces, with the mean estimates .01, 95% CI [-.01, .03] and -.04, 95% CI [-.07, .01], respectively.

When there were 5 familiar exemplars, familiar male faces were located further from all others and their nearest neighbours than unfamiliar faces, with the mean estimates -.08, 95% CI [-.09, -.08] and -.10, 95% CI [-.12, -.07], respectively. Female familiar faces had equivalent distance from all others and their nearest neighbours with unfamiliar faces, with the mean estimates 0, 95% CI [-.01, .01] and -.02, 95% CI [-.05, .01], respectively.

When there were 6 familiar exemplars, familiar male faces were located further from all others and their nearest neighbours than unfamiliar faces, with the mean estimates -.08, 95% CI [-.08, -.07] and -.07, 95% CI [-.10, -.04], respectively. Female familiar faces had equivalent distance from all others and their nearest neighbours with unfamiliar faces, with the mean estimates 0, 95% CI [-.01, .01] and -.01, 95% CI [-.04, .03], respectively. These results suggest that the difference between familiar and unfamiliar faces observed during

subsampling unfamiliar faces is not merely due to fewer exemplars encoded in the face space for unfamiliar faces. Even when there were fewer familiar faces represented in the face space, unfamiliar faces were never located further from others than familiar faces. These spaces supported both the global and local expansion models, but only for male faces. Since it is theoretically implausible that more exemplars for unfamiliar faces are represented in the face space than for familiar faces, I will not further speculate on these findings.

3.3.1.2.3 Auxiliary Analysis: Time. To explore whether and how familiarity changes the distance between identities, I have repeated the block-based analyses I reported for the within-person clusters for the between-people clusters, for both the global and local expansion models.

For male faces, familiar and unfamiliar faces did not differ across any blocks globally. The difference between familiar and unfamiliar faces did not change across blocks. For the local expansion model, familiar ($M = .55$) faces were significantly closer to their nearest neighbour than unfamiliar ($M = .73$) faces in the fourth block, $b = .172$, $SE = .060$, $z = 7.9$, $p = .004$. This difference was significantly greater than the difference between familiar and unfamiliar faces in both the third block, $b = -.085$, $SE = .007$, $z = -12$, $p < .001$ and the fifth block, $b = -.098$, $SE = .007$, $z = -13.8$, $p < .001$. All other comparisons across blocks were nonsignificant.

For female faces, familiar and unfamiliar faces did not differ across any blocks globally. The difference between familiar and unfamiliar faces did not change across blocks. Similarly, for the local expansion model, familiar and unfamiliar faces did not differ across blocks. The difference between familiar ($M = .87$) and unfamiliar ($M = .77$) faces in the third block was greater than the difference in the second block, $b = .062$, $SE = .020$, $z = 3.15$, $p = .002$, and

the fourth block, $b = .070$, $SE = .020$, $z = 3.56$, $p < .001$. Overall, there was no meaningful pattern of difference between familiar and unfamiliar faces across blocks, neither for the global nor the local expansion model.

3.3.1.2.4 Auxiliary Analysis: 7-Dimensional Space. In the 7-dimensional space where between-people distances were calculated using (2), the results were replicated from the 3-dimensional space. For male faces, familiarity was not a significant predictor of distance between other identities neither globally $b = -.057$, $SE = .028$, $t(6) = -1.992$, $p > .05$ nor locally, $b = .029$, $SE = .058$, $t(6) = .51$, $p > .05$. For the global expansion model, familiarity explained .25% of the variance, and target identity explained 45% of the variance. For the local expansion model, familiarity explained 2.9% of the variance, while target identity explained 84% of the variance. For female faces, familiarity was not a significant predictor of distance between other identities neither globally $b = -.008$, $SE = .029$, $t(6) = -0.276$, $p > .05$ nor locally, $b = -.056$, $SE = .090$, $t(6) = -0.622$, $p > .05$. For the global expansion model, familiarity explained .5% of the variance, and target identity explained 46% of the variance. For the local expansion model, familiarity explained 4.3% of the variance, while the target identity explained 86% of the variance.

In summary, to test whether familiar face representations are located in more isolated regions in the face space, I calculated the average distance between each exemplar of an identity cluster and each exemplar of all other identity clusters (global expansion model), or the identity cluster of the nearest neighbour (local expansion model). I then compared these distances for familiar and unfamiliar faces using the 3-dimensional space, using the 3-dimensional space while subsampling from unfamiliar and familiar exemplars, and using the 7-dimensional space. Overall, there was some evidence in support of the global expansion

model, specifically when there were fewer unfamiliar exemplars in the face space, but all other comparisons were nonsignificant. I found no support for the local expansion model. If familiarity increases the distance between identities in the face space, it is done so by increasing the distance between the familiar face and all others, and not necessarily nearest neighbours.

3.3.1.3 Where is the average face? The location of the average was explored in previous studies that used inter-item similarity ratings (Byatt & Rhodes, 2004; Lee et al. 2000) to investigate the norm-based model of the face space, and it was found that the average face was not at the origin. However, the procedure used in these studies created averages that were smoother in texture than exemplar images, possibly contaminating the inter-item similarity judgments. In this study, an average face was created using a software that can average faces without sacrificing skin-like texture (see Materials and Stimuli). Additionally, the averages were created by averaging within-person images, and their relationship with the exemplars of the same identity was investigated to explore a norm-based model of identity-specific subspaces. Previous research showed that an average representation is extracted from different views of a face (e.g., Burton et al., 2005; Kramer et al., 2015), and refined with increased familiarity (Koca & Oriet, 2023), suggesting averaging can be an underlying mechanism for face familiarity as it can yield a stable representation that can tolerate variability across different views of a face (Bruce, 1994). This raises the question of whether an average face would be located at the origin of an identity-specific subspace, perhaps as a reference point for exemplars to be encoded, a question I further explore in Chapter 4. In that case, I would expect the average-centroid distance to be smaller for familiar faces than for unfamiliar faces. This expectation receives support from White et al. (2022)’s work in which observers were

asked to rate the similarity between familiar and unfamiliar identity-specific averages and their constituent exemplars: Familiar face exemplars were closer to their identity-specific averages.

Since the absolute distance between individual exemplars and the average would naturally be smaller for faces that are more densely clustered, I calculated the distance between each exemplar and the centroid of the identity cluster proportional to the average distance among the individual exemplars within an identity cluster. Four target identities per familiarity level precludes null hypothesis significance testing, as such, only descriptives are reported. Contrary to my prediction, familiar face averages were not closer to the centroid than unfamiliar face averages. For male faces, the distance between the average and the centroid did not differ between familiar ($M = .35$, $SD = 1.5$) and unfamiliar faces ($M = .36$, $SD = .96$). For female faces, the average was closer to the centroid for unfamiliar faces ($M = .48$, $SD = 1.25$) than familiar faces ($M = .54$, $SD = 1.41$). It is worth noting that the data are quite noisy (standard deviations are highly large relative to the means), so these comparisons should be regarded as preliminary.

3.3.1.4 What do the dimensions represent? To investigate what the dimensions of the face space might represent, I plotted the 2-dimensional MDS solutions with each node overlaid on its corresponding image (Figure 3.11; Figure 3.12; Figure 3.13) to visually speculate on the dimensions that I plan to formally test in future work. Akin to a PCA solution, the dimensions in the MDS solution are ordered from the most variance captured to the least. It is apparent that the first dimension represents a blonde/brunette spectrum for both male and female faces (Figure 3.11; Figure 3.13), suggesting that colour information is greatly used to differentiate individuals. However, each identity varies little by colour, yet identity clusters do disperse

along this dimension, suggesting that the same dimensions that encode between-people variability encode within-person variability differently. For example, Holly Willoughby's exemplars (female faces, bottom right corner cluster, Figure 3.11) seem to vary in head rotation for the first dimension. Ryan Reynolds' exemplars (male faces, bottom left corner cluster, Figure 3.11) may vary in age for this dimension, as his youngest photo is located further from others on the D1 axis.

What is represented in the second and third dimension is less clear (Figure 3.12). These dimensions may represent some identity-specific information, as identity clusters disperse little along the D2 and D3 axes. For male faces, identities seem to be separated by familiarity in D2 and for female faces, identities seem to be separated by familiarity in D3, as familiar faces are encoded along the negative values and unfamiliar faces are encoded along the positive values. Interestingly, it is apparent that as in D1, how different identities are organized along D2 and D3 is different from how within-person variability is organized along the same dimensions. For example, many targets vary by head rotation along the second dimension and facial expression along the third dimension, but the overall space does not reflect such an organization across identities. For some targets, positive expressions are encoded with a more positive value within their identity cluster in D3, whereas for others, positive expressions are encoded with a more negative value within their identity cluster in D3. This is consistent with Burton et al. (2016)'s argument that the very dimensionality with which each identity is encoded in the face space is idiosyncratic. For example, while Taylor Swift's iconic lip colour or bangs are useful for recognizing her, lip colour or hairstyle are unlikely useful identifying aspects of Ryan Reynolds. Otherwise, all faces would be organized based on a particular factor or a set of factors (e.g., facial expression) in the face space

irrespective of identity. Identity-specific clustering seems to contribute more to the organization of faces in the face space compared to between-people variability, a speculation that is consistent with the above-reported findings regarding familiarity increasing the density of identity clusters with inconsistent evidence of its contribution to between-people variability. I provide further support for this conclusion in the section below.

Figure 3.11

Dimensions 1 and 2 with Images

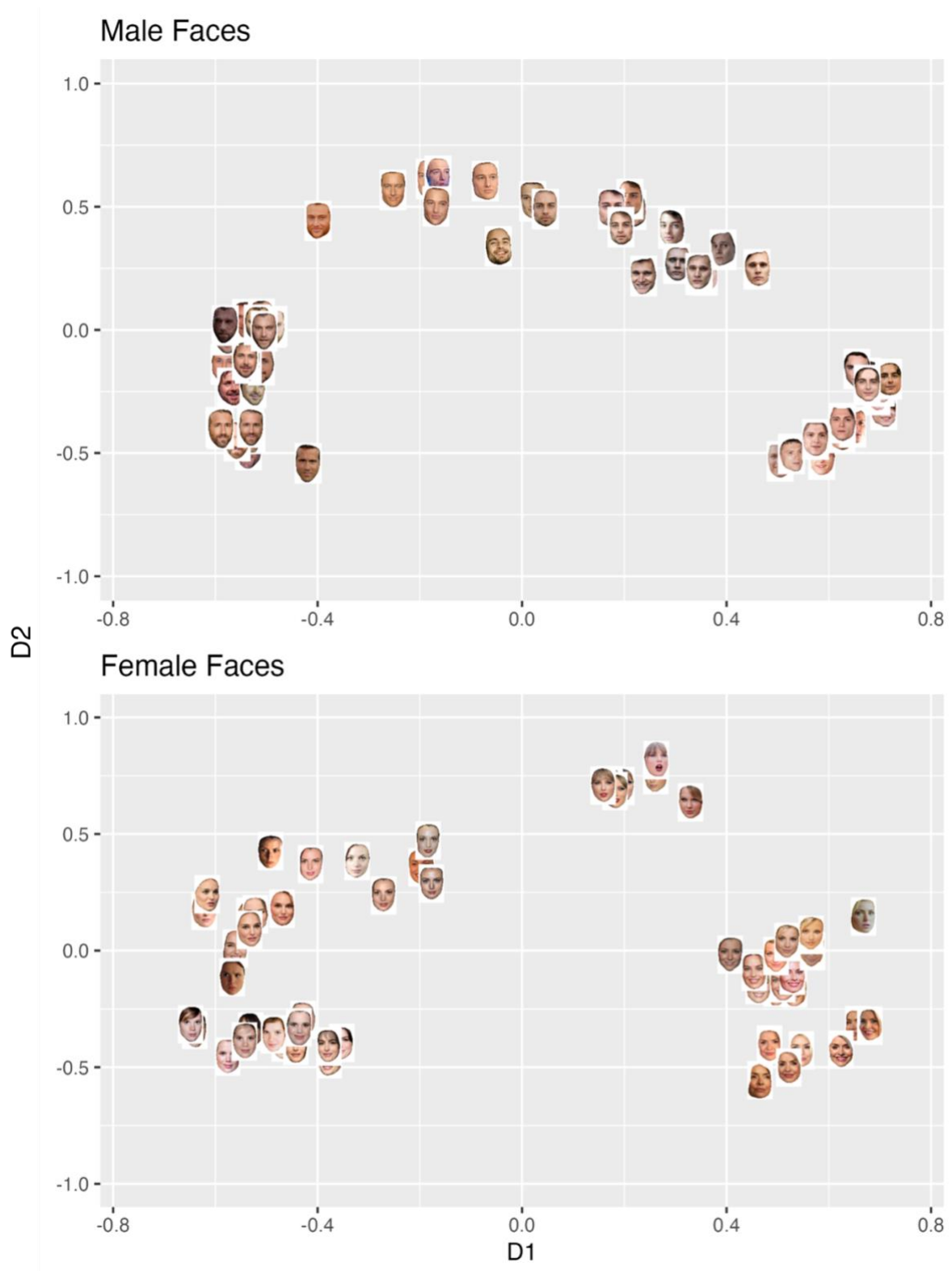


Figure 3.12

Dimensions 2 and 3 with Images

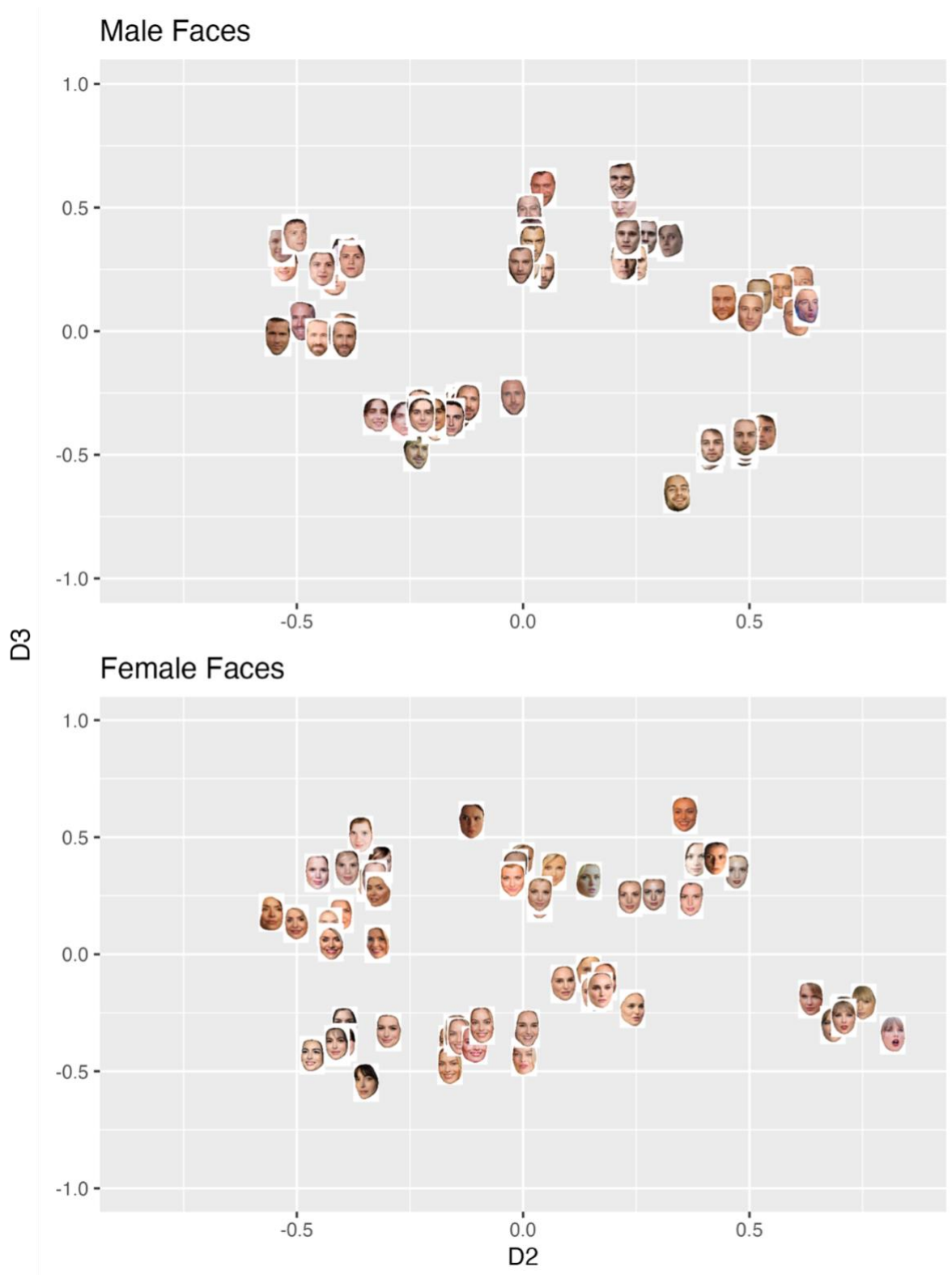
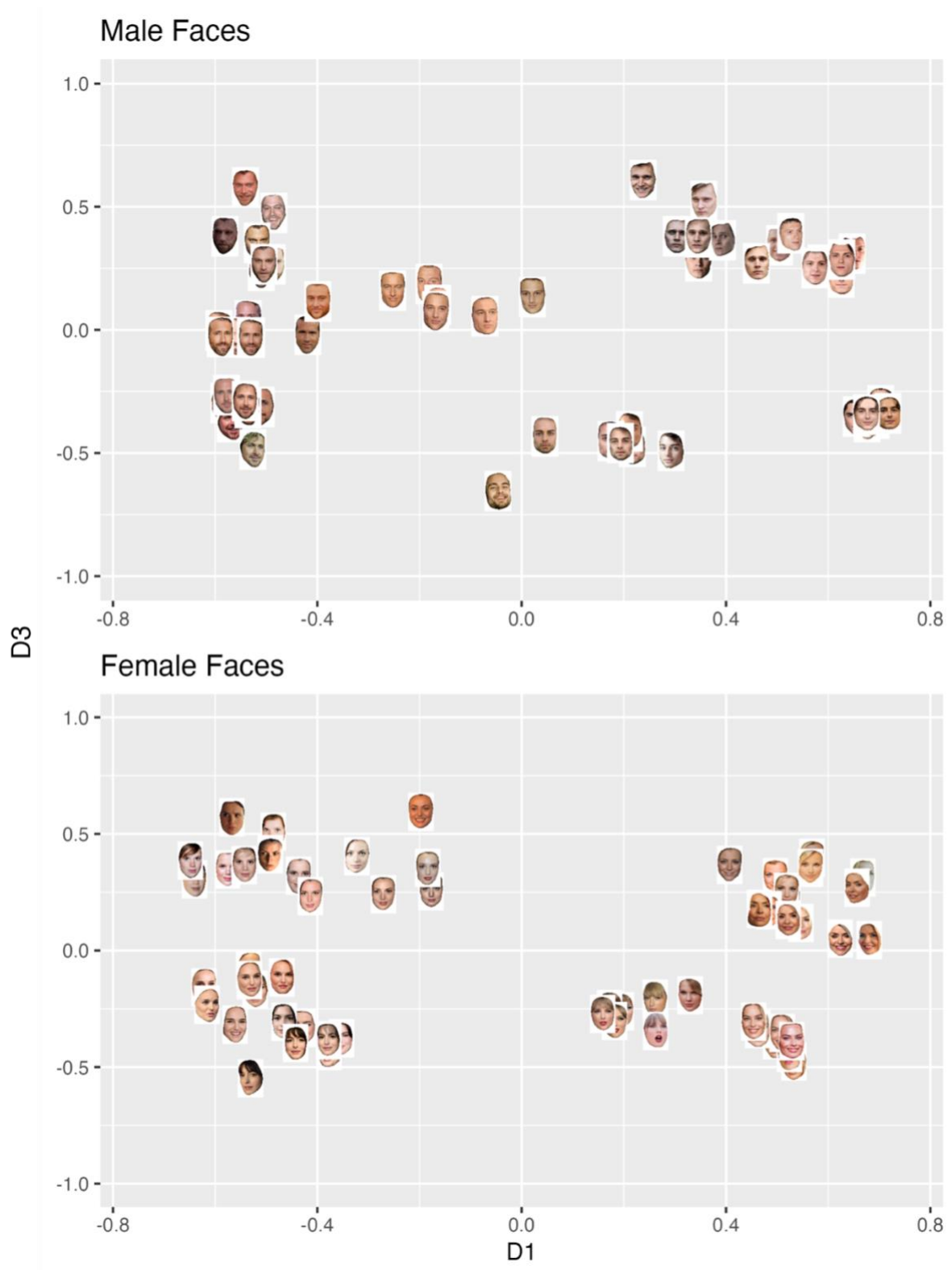


Figure 3.13

Dimensions 1 and 3 with Images



3.3.2 Rank-Image and Rank-Identity Correlation

Given that people's faces vary highly and idiosyncratically across different viewpoints (e.g., Jenkins et al., 2011), and that most faces are somewhat distinctive on at least one dimension (e.g., Burton & Vokey, 1998), I expected that perceived distinctiveness will vary across different photos of a person as different photos would capture the distinctiveness of a face differently. Inter-item similarity ratings have been shown to agree with distinctiveness ratings (Light et al., 1979), such that items that are consistently rated as similar to other items are also rated as more typical, and items that are consistently rated as dissimilar to other items are rated as more distinctive. An average "image-distinctiveness score" was calculated for each image by subtracting each rating from 8, and then averaging them for each image, so that images that were consistently rated as dissimilar to the images they were paired with received a higher image-distinctiveness score, and images that were consistently rated as similar to the images they were paired with received a lower image-distinctiveness score. For each identity, an identity-distinctiveness score was obtained by averaging over the distinctiveness scores of each identity. These scores were used to explore whether the variability in the image-distinctiveness scores contributes to the variability in the identity-distinctiveness scores. To do so, I followed the procedure of Jenkins et al. (2011), which was used for likeness and attractiveness ratings.

The procedure involved comparing two correlations: Rank-Identity correlation and Rank-Image correlation. To calculate the Rank-Identity correlation, the identity-distinctiveness scores (ranging from 1-7) were arranged in an ascending order (for male and female faces separately), and each identity received an identity-distinctiveness rank (from 1-8). The Rank-Identity correlation was a Pearson correlation between the identity-

distinctiveness scores and the identity-distinctiveness ranks, which naturally yielded a very high correlation of $r = .97, p < .001$ for male faces, and $r = .93, p < .001$ for female faces. To calculate the Rank-Image correlation, the image-distinctiveness scores (ranging from 1-7) were correlated with the identity-distinctiveness rank (each image of the person will have the same rank, ranging from 1-7 as above), which yielded $r = .86, p < .001$ for male faces, and $r = .67, p < .001$ for female faces. These two correlations were then compared using Fisher's z test, which revealed that the two correlations were significantly different, $z = 4.83, p < .001$ for male faces, and $z = 5.70, p < .001$ for female faces. These results suggest that the variability in the image-distinctiveness scores is not accounted for by changes in identity. In other words, a more typical face overall can sometimes look distinctive, or a more distinctive face overall can look typical. Different photos of a person capture their distinctiveness differently.

3.4 Discussion

The main findings of this study can be summarized as follows:

1. Familiar faces formed higher density clusters in the face space than unfamiliar faces.
2. Familiarity did not reliably change the distance between identities in the face space neither globally nor locally.
3. Averages were not closer to the centroids of familiar identity clusters.
4. The same dimensions that encode between-people variability encode within-person variability idiosyncratically.
5. Distinctiveness of a face is subject to within-person variability.

Familiarity organizes within-person variability in the face space by decreasing the psychological distance between different exemplars of the same person, yielding higher

exemplar densities for familiar faces. This finding provides support for *the representation anchoring hypothesis*. When we see a familiar face, image-specific variations (e.g., lighting, angle) are readily tolerated by anchoring perception to the person depicted in the image. This finding corroborates previous evidence showing higher similarity ratings between two images of the same person as a function of familiarity (e.g., Kramer et al., 2025) while accounting for the relative distance among multiple familiar and unfamiliar identities in the face space. This “compressing” effect of familiarity was evident when there were fewer unfamiliar exemplars included in the current face space, showing that the psychological construct of familiarity is the result of increased density of within-person clusters. In fact, even when there were fewer familiar face images selected, the effect of familiarity persisted.

Familiarity, however, may not organize face representations in memory by increasing the psychological distance between identities. This is interesting because the compression of exemplars through familiarity would naturally move those exemplars away from others. Nevertheless, in all the models testing the distance between identity clusters, familiarity was not a significant predictor, and contributed little to the model’s explained variance whereas target identity contributed substantially to the models. There was, however, some support for the global expansion model when there were fewer unfamiliar exemplars included in the MDS space, with no reliable evidence in support of the *local expansion hypothesis*. It is unclear why a face’s distance from all others would increase without reliably increasing the distance between near neighbours. If increasing the psychological distance between faces in the face space would serve to differentiate faces, it would be most beneficial for faces that are most similar to each other.

The observed effects of familiarity are not a result of experimental familiarity built through exposure to unfamiliar faces. When I re-analyzed the data broken down by block number, I found that the density of exemplars increased over time for unfamiliar faces, but the difference in between-people distances between familiar and unfamiliar faces did not change as a function of time. This may explain the findings in the series of studies in Chapter 2 in which I did not find evidence that exposure to a face's within-person variability influenced the point of subjective equality between that face and similar-looking others. The observed effects of familiarity are also not due to the limited number of dimensions in the face space, as the optimal number of dimensions for the MDS solution was 7, but the 3-dimensional solution was used to plot and calculate distances between exemplars. Repeating the analyses for the 7-dimensional space replicated the findings of the 3-dimensional space. This suggests that facial identity can be represented with as few as 3 dimensions in the face space.

The findings from this study support the notion that within-person variability is idiosyncratic. First, the dispersion of exemplars within identity clusters does not appear to be uniform. Second, the same dimensions that encode between-people variability encode within-person variability differently. This is also supported by the finding that target identity contributes greatly to between-people distances – faces are organized in the face space as distinct identity clusters with their own set of dimensions. Third, the distinctiveness of a face is also subject to within-person variability. A face that is overall more distinctive can have views that look more typical and vice versa. As such, I recommend that studies investigating facial distinctiveness incorporate within-person variability instead of measuring distinctiveness using one or two images per identity.

A surprising finding was that the averages were not closer to the centroids of familiar face clusters than unfamiliar face clusters. This could be a result of using few images per identity ($N = 7$). Familiar face averages are likely created by averaging a greater number of encounters and may be more stable than the averages used in the current study, as averages stabilize at around 20 images (Burton et al., 2016). Unfamiliar face averages, on the other hand, would be constituted entirely by the stimuli viewed in the study, which matches the average that was shown to the observers. Indeed, when 20 images are incorporated into an average, familiar faces are rated to be more similar to their averages than unfamiliar faces (White et al., 2022).

It would be interesting to replicate this work using a greater number of identities, as the current study included only four identities per cell. Reconstructing a more realistic face space would also include fewer exemplars for each unfamiliar identity; indeed, when data were reanalyzed using fewer unfamiliar exemplars, some familiarity effects were observed for between-people distances that were otherwise absent. The current study is also limited in its approach to familiarity. Familiarity varies across individuals based on prior experience, as such, treating familiarity as a binary variable may not be sufficiently sensitive. Further, familiarity was not manipulated in a crossed design. Since within-person variability is idiosyncratic, and there were target-specific effects across different analyses, the possibility that the results can be explained by target-specific differences cannot be fully eliminated. Nevertheless, the models that were fitted to the data included target identity as a random effect to mitigate this limitation.

The present study provided evidence that a psychological face space that has its structural basis in inter-item similarity can account for the organization of facial identity in

memory. Familiarity influences this organization of faces by decreasing the distance between different exemplars of the same identity with inconclusive evidence regarding its effect on between-people distances. The present work also showed that a 3-dimensional similarity-based face space can represent distinct identity clusters. These clusters are not uniformly distributed and reflect the idiosyncratic nature of within-person variability. In the following chapter, I explore the architecture of such within-person clusters.

4 Exploring Identity-Specific Subspaces in the Face Space

4.1 Introduction

Familiarity yields remarkable benefits for face processing. While observers have great difficulty recognizing novel instances of unfamiliar faces, and perceptually matching two photos of an unfamiliar face, such tasks become trivial for familiar faces, even under poor viewing conditions (e.g., Bruce et al., 1999; 2001; Hancock et al., 2000; Megreya & Burton, 2006; 2007). This led some researchers to argue that familiar and unfamiliar faces are processed in qualitatively different ways such that unfamiliar face processing recruits unsophisticated strategies that rely on image-based cues, while familiar face processing matching relies on abstract representations in memory.

For example, Megreya and Burton (2006) showed that inverted familiar face processing was strongly associated with upright unfamiliar face processing, suggesting that configural information is not as effectively extracted from unfamiliar faces. Further, in card sorting tasks, where observers are asked to sort multiple ambient unfamiliar face photos in different piles each corresponding to an identity, they make many more piles than they should given the number of identities in the set while rarely placing two different identities in the same pile (e.g., Jenkins et al., 2011; Laurence et al., 2016; but see Andrews et al., 2015). In other words,

while unfamiliar observers have great difficulty judging two separate exemplars of the same person as depicting the same person, it is easier to accurately judge that two photos belonging to different people are different people. It seems, then, that the problem of face recognition is not only solved by learning to tell faces apart (i.e., discriminate), but also by learning to tell faces “together” (i.e., amalgamate). This is because faces vary in tremendous ways – one person can look drastically different across different viewing conditions (e.g., angle, lighting, age, weight; Figure 1.15). When a face is familiar, this variability is readily tolerated and the face can be recognized. When a face is unfamiliar, the visual system cannot tolerate the variability across different instances of the face; hence, it mistakenly categorizes these separate instances as different identities.

Importantly, this variability is unique to each face because being familiar with one identity does not transfer that familiarity to another. Burton et al. (2016) provided computational evidence that faces vary in idiosyncratic ways – when multiple ambient photos of different celebrities were used in training a PCA model, the model explained the variability in photos of different people in different ways. For example, the same component captured facial expression for one identity but gaze direction in another identity. This means that for a face to be familiar, observers need to be exposed to a sufficient range of variability in that face so that a novel instance is more likely to be successfully recognized. Accordingly, many studies have shown that exposure to high, unsystematic within-person variability captured in sets of ambient images yields superior recognition performance (e.g., Andrews et al., 2015; Burton et al., 2016; Corpuz & Oriet, 2022; Ritchie & Burton, 2017).

Robust and seemingly qualitative differences between processing familiar and unfamiliar faces give rise to the idea that familiar faces are represented as identity-specific

subspaces in the face space. This idea is particularly compelling considering that faces vary in idiosyncratic ways. Therefore, it is unlikely that each face loads differently onto a shared set of dimensions as is often conceptualized in the face space, because the very dimensionality with which each face is encoded is idiosyncratic (Burton et al., 2016), a notion that is supported by the findings in the previous chapter. For example, suppose there is a dimension of the face space that encodes age. How would this dimension accommodate different exemplars of a familiar face that vary in age? If we assume that each face is encoded using a common set of dimensions, we need to assume that the age dimension of this space hosts multiple representations of a familiar face in varying age groups. However, recent evidence suggests that this is not the case. Laurence et al. (2022) investigated whether a single representation of a familiar face can incorporate different ages, or whether multiple representations are needed to represent a familiar face across different ages (i.e., younger adult, older adult) by presenting older adults with the faces of older famous adults who had aged alongside them (e.g., Paul McCartney).

In their first experiment, they used a long-lag repetition priming task where observers were primed with images of famous faces followed by a test phase. The test image could be the same image shown during the priming phase, a different image of the same person from the same age, or an image of the person depicting a different age than the priming phase. During testing, observers' task was to indicate whether the test image was a famous or a non-famous person, and their reaction times were recorded. They found that the effect of priming was similar across the three test conditions, indicating that the priming effect can transcend different ages of the same face. In their second experiment, an adaptation paradigm was used. Observers viewed an adaptor image that depicted either an older or a younger photo of a

target face, followed by a test image that included the target face morphed with a non-target identity by 50%. The age of the test image was either congruent or incongruent with the age of the adaptor image (e.g., older-older or younger-older, respectively). Observers were asked to indicate whether the test image better resembled the target or the non-target identity. They found that the magnitude of the aftereffect was similar across the congruent and incongruent conditions, suggesting that identity aftereffects are not contingent on the person's age. Together, these findings suggest that a single familiar face representation can accommodate changes in age.

This is especially fascinating considering that aftereffects are found to be category-contingent, meaning that they do not transfer over different age, race, or species categories (Little et al., 2008). When a face is familiar, however, changes in such categories¹⁰ do not impact the magnitude of adaptation, suggesting that familiar identity representations can encompass a wide range of variability across different viewing conditions. Support for this conclusion came from Hole (2011), who found that identity-specific aftereffects transfer over a range of viewing conditions for familiar faces like inversion, changes in viewpoint, and even vertically stretching the face three times the normal height. Transference of aftereffects within different images of a familiar identity despite drastic changes in low-level image properties (Hole, 2011), and despite distinct changes in categories like age (Laurence et al., 2022), suggest that familiar faces may be represented as subspaces in the face space.

Further evidence for identity-specific subspaces comes from computer vision, where PCA models that utilize identity-specific subspace methods outperform the traditional Eigenface methods (Aishwarya & Marcus, 2010; Shan et al., 2003). Put simply, in the

¹⁰ Certainly, a familiar identity is unlikely to switch between particular categories like race, unless perhaps in rare cases like Michael Jackson.

traditional Eigenface methods, the model is trained with multiple standardized images each belonging to a different individual whereas in the subspace method multiple models are created per individual, each trained with multiple images of that individual. The model's recognition performance is often measured by projecting a novel image in the face space and calculating its distance from previously trained exemplars, which is referred to as reconstruction error. Evidence suggests that reconstruction error of a novel image of a previously trained identity is reduced for identity-specific subspace models (Aishwarya & Marcus, 2010; Shan et al., 2003) relative to traditional Eigenface models.

The identity-specific subspace account can also explain how refining a familiar face's representation can facilitate *both* amalgamation and differentiation. In the attractor field model (Tanaka et al., 1998), a face with a larger attractor field improves the likelihood that more exemplars can be recognized as that face. There is evidence in the literature suggesting that the size of an attractor field is associated with tolerance to within-person variability (Laurence et al., 2016), and there is also evidence suggesting that the region surrounding a familiar face's representation in the face space is larger (as indexed by distinctiveness) than the region around an unfamiliar anti-face located in a region with comparable exemplar density (Faerber et al., 2016). This indicates that familiar faces should have a larger attractor field, allowing for multiple exemplars to be accepted as that face (i.e., amalgamation).

However, a larger attractor field also increases the likelihood of false alarms. Tanaka and Corneille (2007) found that discriminating between a distinctive target (i.e., a face with a large attractor field) and a similar-looking distractor was more difficult than discriminating between a typical target (i.e., a face with a small attractor field) and an equally similar-looking distractor. However, they only used unfamiliar faces in their study. A more recent

study by Chauhan et al. (2020) found that observers were less likely to judge a 50/50 morph composed of a familiar and unfamiliar face to resemble the familiar face, suggesting that when a face is familiar, observers are *more* conservative in accepting a similar-looking exemplar as the familiar face. Further, Koca et al. (2024) found that manipulating the size of the attractor field surrounding a face's representation in a familiarization paradigm did not influence differentiation performance in a matching task. Contrary to the predictions of the attractor field model, the series of studies reported in Chapter 2 did not find increased distance between a face and a near neighbour as a function of increased exposure to within-person variability.

The assumption that the size of the attractor field surrounding a face's representation increases with familiarity can explain the process of amalgamation, but it does not necessarily explain differentiation performance for familiar faces. Perhaps a limitation of most of the face space models (e.g., the attractor field model, Tanaka et al., 1998; the Voronoi model, Lewis & Johnston; 1999) is the assumption that *all* faces belong to a particular social category are represented in a similar portion of face space. Because familiar and unfamiliar faces are represented very differently, it is difficult to make unifying predictions with a model of face recognition that possesses the same set of functions and dimensions for processing both familiar and unfamiliar faces. However, this can be reconciled if it is assumed that the same faces are represented in different *layers* of face space. One layer of the face space would represent all faces and would be responsible for discriminating faces from each other. Another layer of the face space represents familiar faces as distinct categories with their own set of idiosyncratic dimensions, and would be responsible for amalgamation. Such a mechanism was previously proposed by Benson and Perrett (1991).

Qualitative differences between processing familiar and unfamiliar faces, the phenomenon that within-person variability is idiosyncratic, evidence from computational models trained for face recognition, and current limitations within the models of the face space all raise the possibility that there are identity-specific face spaces for familiar faces that can successfully allow observers to recognize novel instances of that face, while achieving near-perfect performance in discriminating the face from similar others. The goal of the study in this chapter is to take first steps into exploring the properties of an identity-specific subspace.

An intriguing question is whether norm-based coding is a property of identity-specific subspaces as hypothesized for the population face space. Previous research showed that averaging faces can be an underlying mechanism of face learning (Burton et al., 2005; Koca & Oriet, 2023; Kramer et al., 2015). This suggests that an identity-specific norm is represented for familiar faces, which is a compelling account since different exemplars of a face vary greatly, and the process of averaging would wash out this within-person variability allowing identity-invariant aspects of a face to be represented in memory. Recently, however, Davis et al. (2024) argued that ensemble coding (i.e., set averaging) may not be an underlying mechanism of face learning. In their study, observers were presented with an array of four images depicting the same person. The array contained either four different ambient images (the different-images condition), the average of these four images presented four times (the average-only condition), or a single image presented four times (the same-image condition). Recognition was tested using a novel image of the person. The authors found that presenting observers with the average set yielded no recognition advantage. In fact, recognition performance was similar in the same-image and the average-only condition, both worse than

the different-images condition. Since presenting observers with an average did not generalize over recognition of a novel image, the authors reasoned that extracting an average representation is not sufficient and that exposure to within-person variability is imperative to learning a new face.

However, presenting observers with an average image may not be an appropriate operationalization of ensemble coding. To an unfamiliar eye, an average image is just another exemplar of that face, which would explain why Davis et al. (2024) found comparable recognition performance between the average-only and the same-image condition. If ensemble coding is assumed to be an underlying mechanism of face learning, it is important to consider not only the average representation that is created as an end product, but also the *process* of extracting this average representation. Given that ensemble coding serves to reduce redundancies and noise in the perceptual domain (perhaps even the social domain, see Whitney & Yamanashi-Leib, 2018), arguably, ensemble coding is only meaningful when the ensemble has some sort of underlying variability. If there is no variability in the set, then any one item is as good a representation as the average. Therefore, I argue that exposure to variability, in the case of faces, within-person variability, is an imperative aspect of the ensemble coding process and should be considered in investigations of ensemble coding as an underlying mechanism of face learning.

Nevertheless, as Davis et al. (2024) argued, it is unlikely that *only* an average is represented for familiar faces. The idiosyncratic nature of within-person variability suggests that not only the central tendency, but also the distribution of exemplars is critical for face processing (Burton et al., 2016). Importantly, many identity-invariant aspects of a face that one can conceptualize, like metric distances between features, or as found in Chapter 3,

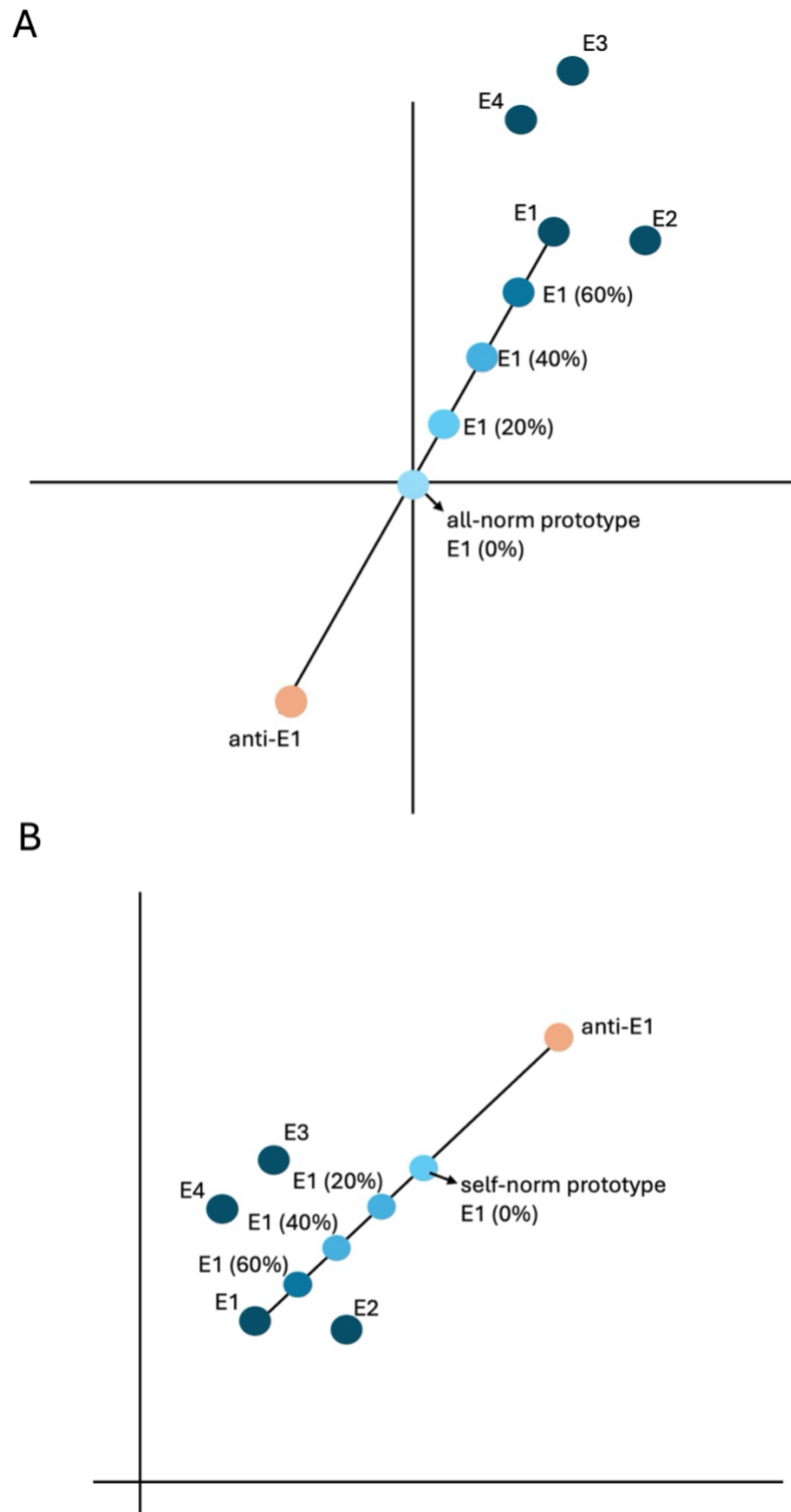
distinctiveness, are subject to within-person variability. If representing varying exemplars is important for recognizing a face, then what is the purpose of representing the average in addition to exemplars? Since an average representation is refined with increasing familiarity (Koca & Oriet, 2023), I propose that this identity-specific norm serves as a reference point to allow within-person variability to be encoded.

In this study, I investigated whether different exemplars of a face are encoded relative to a norm by using an adaptation paradigm similar to Leopold et al.'s study (2001). In their experiment, Leopold et al. created anti-faces by exaggerating an average face in the opposite direction of an identity's vector (Figure 1.11). They found that adapting to an anti-face, such as "anti-Adam", made the average face appear more like Adam, decreased the identification threshold for recognizing Adam, and increased the identification threshold for recognizing a non-Adam face. This suggests that adaptation to an anti-face shifts perception in a *systematic* way - that is, along the identity trajectory of the anti-face - providing evidence for norm-based coding of the face space. If norm-based coding is also a property of the hypothesized identity-specific subspaces, then adapting to an anti-exemplar of a familiar face should produce similar results for different exemplars of a familiar face (i.e., adapting to "anti-Adam-Exemplar-1" should lower the recognition threshold for "Adam-Exemplar-1"). Further, if familiar faces are represented with an identity-specific average that is activated when a familiar face is seen, adaptation effects should be greater for identity-specific averages than gender-specific averages. A simplified diagram of this idea is depicted in **Error! Reference source not found..** Assume that for both panels, the baseline identification threshold (the minimum amount of exemplar needed in the morph for that image to be classified as that specific image) is 60% E1 (shorthand for "Exemplar 1"). If familiar faces are represented in

reference to an identity-specific norm, adapting to an anti-face created through distorting the self-norm prototype in the opposite direction of E1 (Figure 4.1B) would reduce the identification threshold to, say, 20% E1. Adapting to an anti-face created through distorting the all-norm prototype in the opposite direction of E1 (Figure 4.1A) would reduce the identification threshold to, say, 40% E1. A larger adaptation effect in the self-norm condition would be taken as evidence for norm-based identity-specific subspaces. Further, when the self-norm average is shown post adaptation, the appearance of the average image would be biased towards E1, suggesting that adapting to an anti-exemplar would increase sensitivity to the image on its exemplar trajectory. The opposite pattern or an equivalent adaptation effect would be observed if familiar faces are represented at the same layer of the face space as all others and would activate an all-norm prototype.

Figure 4.1

All-norm and Self-norm Face Spaces



Note. A simplified diagram depicting the (A) gender-matched norm face space and (B) self-norm identity-specific subspace in the current adaptation paradigm. E1-4 refer to specific exemplars (images) depicting the same person. The Prototypes are depicted in light blue. Anti-exemplars are depicted in orange. E1 trajectory is depicted as a black line crossing through the respective prototype with a morph continuum between E1 and the prototype. Numbers and distances are completely arbitrary.

An interesting question is to ask what would happen to the identification threshold for the nonmatching-exemplar images (E2-4, Figure 4.1). One possibility is that the identification threshold will increase for nonmatching exemplars, suggesting that increased sensitivity to the matching-exemplar (E1) will make it more difficult to identify nonmatching exemplars. This is similar to what was found in Leopold et al. (2001)'s and Rhodes and Jeffery (2006)'s studies using nonmatching-*identities*. The authors found that while adapting to anti-Adam made it easier to identify Adam (matching identity), it made it more difficult to identify Jim (nonmatching identity). This is because adapting to anti-Adam increased sensitivity to Adam's facial features which are not likely present in Jim. A nonmatching *exemplar*, however, is still the same identity (i.e., it is a nonmatching exemplar for Adam) and hence has the same identity-specific characteristics as the matching-exemplar, so it is likely that the features to which the observers have increased sensitivity after adaptation to an anti-exemplar will be present in the mismatching-exemplar as well. Previous research showed that adaptation effects transfer within different views of the same identity (Carbon & Ditye, 2012; Hole, 2011; Laurence et al., 2022). Therefore, adaptation to an anti-exemplar can decrease the identification threshold for the mismatching-exemplars. Nevertheless, I predict that even in

that case, the magnitude of identification threshold decrease will be larger for the matching-exemplar than for the nonmatching exemplar.

4.2 Methods

4.2.1 Participants

Typically, adaptation studies are conducted with very few participants (less than 10) with thousands of trials (e.g., Leopold et al., 2001). In this study, I aimed to have less trials to prevent participant fatigue, so I aimed to collect data from 20 observers for two conditions. Fifty-one participants were recruited through the University of Regina Psychology Participant Pool and by word of mouth. All participants had normal or corrected-to-normal vision. All procedures were carried out in accordance with the Canadian Tri-Council Policy Statement on the ethical treatment of research participants and were approved by the University of Regina Research Ethics Board (#874). All participants signed a consent form prior to their participation, and participants who were recruited through the participant pool received 1% bonus credit toward a psychology course as compensation. Participants who had less than 80% accuracy in the practice trials ($N = 8$) did not complete the main task. Data were thus analyzed for 42 participants (31 female, 11 male; $M_{age} = 22.69$ $SD_{age} = 6.23$).

4.2.2 Stimuli and Apparatus

The stimuli construction followed Leopold et al.'s (2001) approach, but instead of using images of different people, stimuli were created by using different images of a single identity. To do so, 12 images each for six famous celebrities (three female) were collected from the internet as targets ensuring that none of the faces were obstructed (e.g., glasses, hand, hair), both pupils were visible, and none of the images depicted a big smile with teeth. To create gender-matched all-norm averages, averages from familiar and unfamiliar faces (used in the

previous studies in this research program) were used, which consists of 6 famous (including the targets) and 7 unfamiliar faces for each gender.

Using PsychoMorph 7.1 (Tiddeman et al., 2005), all images were delineated by marking points that correspond to facial features to create averages and anti-faces¹¹. Self-norm averages were created by averaging all 12 images of the same person, and all-norm averages were created by averaging gender-matched identity averages. Anti-faces were created by transforming each exemplar image away from the self-norm and all-norm by 80% (-0.8 strength). This was done by creating a vector between the exemplar images and the average, and then transforming the average image along that vector in the opposite direction of the exemplars. Only shape transformations were applied, following the procedure in Leopold et al. (2001) and Rhodes and Jeffery's (2006) work. More recent research (Rogers et al., 2022) emphasizes the importance of both shape and texture in recognizing faces. The choice of using only shape transformations was a practical one, as factors like image sharpness or quality influenced the appearance of texture-based anti-faces (e.g., if the image is grainy, the anti-face image yields an extra smooth texture to the point that contours of the face become invisible). Once anti-faces were created for all exemplars of a face, four out of 12 anti-faces and their exemplars were then selected to be used as adapting and test stimuli, respectively, based on the following criteria:

¹¹ It was not mentioned in previous studies using this paradigm (Leopold et al., 2001; Rhodes & Jeffery, 2006) whether the test stimuli were incorporated into the average. In this study, the averages that were used to create the anti-faces did include the test stimuli, as norm-based spaces would represent exemplars as deviations from the norm that includes those exemplars.

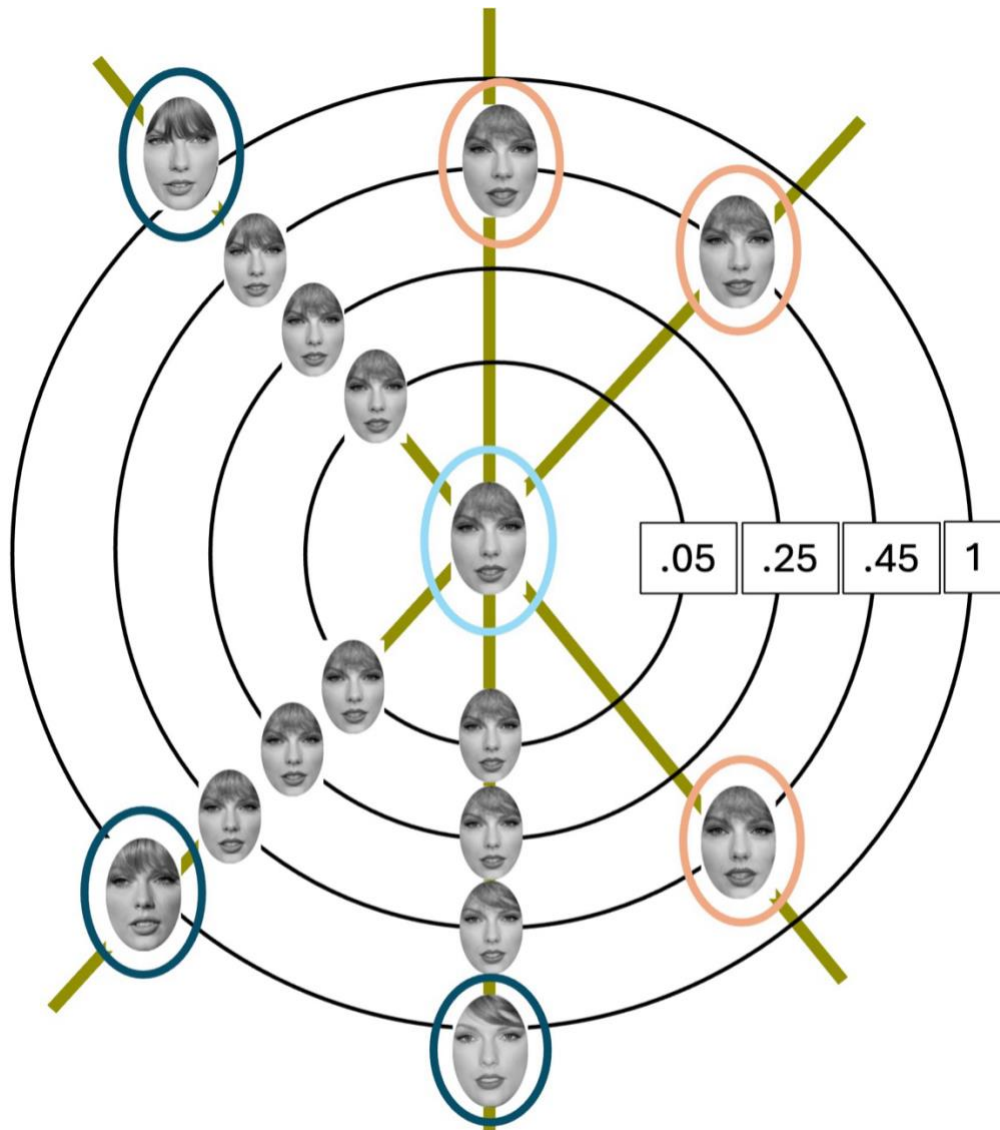
(a) The exemplars and anti-faces are sufficiently distinct from one another as the task is to distinguish images of the same person. This was accomplished by selecting images that differed in viewing angle and facial expression.

(b) The anti-faces were not too distorted. Some anti-faces, especially those created from exemplars that had a large head rotation were very distorted as the anti-face would have a head rotation in the opposite direction.

Once the test stimuli were selected, the average face and each exemplar were morphed in different ratios, from 0% (average face) to 45% exemplar strength in steps of 5%. This was done to create a set of “weaker” versions of each exemplar that gradually look more similar to the average face in order to measure identification thresholds. Once all stimuli were created, each stimulus was put in an oval mask (2:3 aspect ratio) with eyes aligned and was converted to black and white. An example stimulus structure can be seen in Figure 4.2.

Figure 4.2

Norm-based Identity-specific Subspace Structure



Note. An example stimulus structure used in the study depicting exemplars (dark blue), the average (light blue), anti-faces (orange), and some of the faces on the morph trajectory. Interestingly, anti-exemplars still appear recognizable as Taylor Swift, providing anecdotal evidence for norm-based identity-specific subspaces where within-person variability is encoded in reference to a norm.

The experiment was programmed using PsychoPy3 (Peirce, 2007) and data were collected in-person. Data were collected using monitors with a 1920x1080 resolution. Since face identity adaptation effects do not depend on retinal size (Rhodes & Jeffery, 2006; Valentine et al., 2016), observers' distance from the screen was not restricted.

4.2.3 Procedure

The procedure closely followed Leopold et al.'s (2001) study, with the difference that anti-faces were created by transforming either a self-norm average or a gender-matched all-norm average, which was manipulated between-subjects. The order of completing the task with self-norm and all-norm anti-faces was counterbalanced across participants. Before the experiment, observers were given a list of 6 targets and were asked with which target they were most familiar. The person they selected was shown to them throughout the study. If they indicated that they were not familiar with any of the faces, they did not participate in the main task. The study consisted of a practice phase which was always completed first, a baseline phase, and an adaptation phase. The order of the baseline and adaptation phase were counterbalanced across participants to mitigate practice effects.

4.2.3.1 Practice. Following Leopold et al. (2001)'s procedure, observers first completed a practice phase to familiarize them with the test image. During the practice phase, they were shown four different images (i.e., exemplars) of the target face, and they were asked to press labeled buttons to indicate which particular image they saw. For example, if the exemplar is an image of Keanu Reeves in the Matrix, they would be asked to press "1", if it is an image of Keanu Reeves in John Wick, they will be asked to press "2", and so on. This procedure was carried out for 80 randomized trials, and participants received feedback. Participants who had below 80% accuracy in the practice phase ($N = 8$) were not eligible to participate in the

main task. Before proceeding to the test phases, the participants were informed that they will repeat the same task but will now see weaker versions of the images, and sometimes they will be asked to look at another face image in between trials. They were also informed that they would not receive feedback in these tasks.

4.2.3.2 Baseline Phase. Baseline identification thresholds of the exemplar images were measured using the method of constant stimuli. To do so, participants were presented with the exemplars and their weaker versions (created by morphing the exemplars with the average of the target in varying ratios) and were asked to identify the exemplar as they did in the training phase. There were 20 trials per exemplar-strength level, yielding 200 randomized baseline trials.

4.2.3.3 Adaptation Phase. In the adaptation phase, participants were shown anti-exemplars presented on the screen for 5 seconds. After viewing the adapting stimulus, participants were shown a test stimulus that varied in exemplar strength. The test stimulus was either an exemplar image along the same vector of the anti-exemplar (matching exemplar), or an exemplar image of the same person with a different vector direction (nonmatching exemplar). As was done in the practice and baseline phases, participants were asked to press the key corresponding to the specific image they saw. There were 20 trials per exemplar-strength level per matching condition, yielding a total of 400 randomized trials. The trials were separated into 10 blocks. The study took about an hour to complete.

4.3 Results

Responses were recorded as correct for trials where an exemplar was correctly identified, and when the average test image (0% exemplar strength) is identified as the corresponding exemplar of the adapted anti-exemplar. Accuracy was calculated for each

observer within each exemplar strength level (0% - 45% in steps of 5%) within each adaptation condition (baseline, matching exemplar, nonmatching exemplar). Using the *drc* package in R (Ritz et al., 2015), a four-parameter logistic function (slope, lower asymptote, upper asymptote, threshold) was fitted to the data separately for each adaptation condition within each norm condition (self-norm and all-norm), following the procedure of Leopold et al. (2001) and Rhodes and Jeffery (2006). Curves were fitted to concatenated observer data to reach model convergence. The model estimates the identification threshold as the midpoint of the lower and upper asymptotes. The magnitude of the adaptation effect is the decrease in identification threshold in the matching-exemplar condition from baseline, indicating that it is easier to identify an exemplar after adapting to an anti-exemplar.

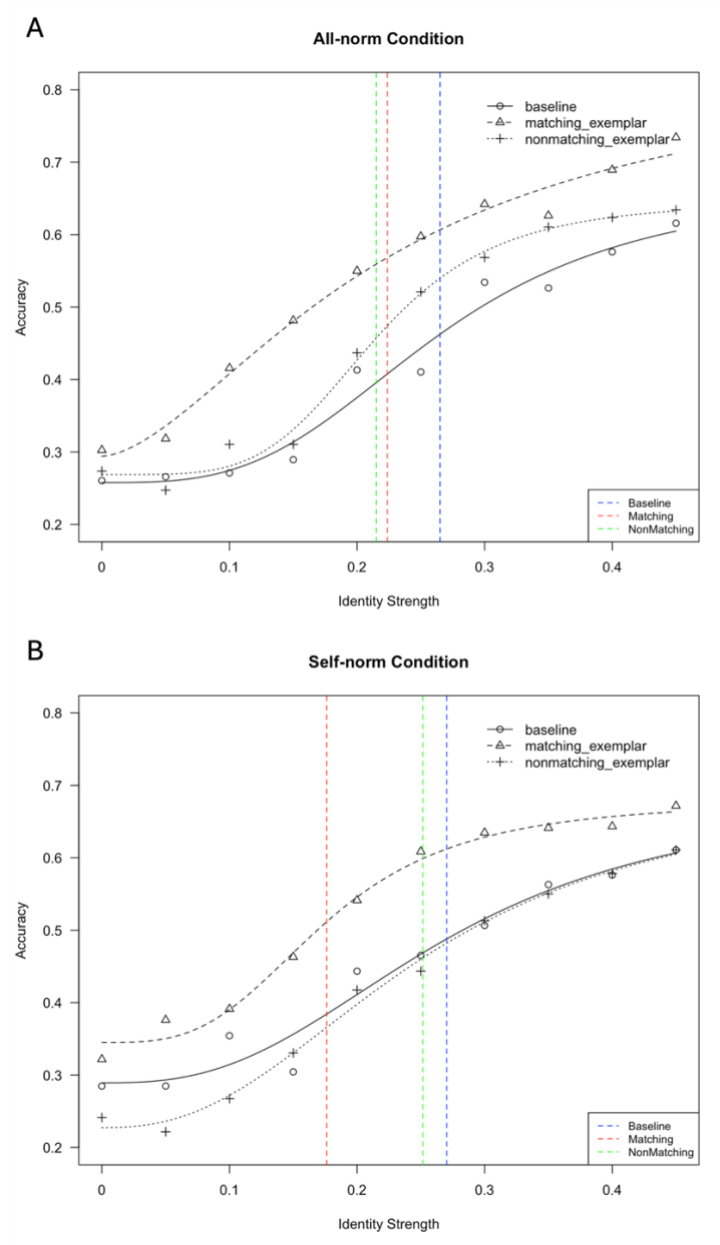
For the all-norm condition, the baseline threshold was estimated to be 26.5%, the matching exemplar threshold was estimated to be 22.4%, and the nonmatching exemplar threshold was estimated to be 21.5% (Figure 4.3A). The identification threshold was reduced from baseline by 4.1% in the matching exemplar condition, and by 5% in the nonmatching exemplar condition. The identification accuracy for the average image was at chance (i.e., 25%) for the baseline and nonmatching exemplar conditions, and above chance in the matching exemplar condition. Specifically, baseline accuracy was 26.1%, 95% CI [22.6%, 29%] and nonmatching exemplar accuracy was 27.4%, 95% CI [23.8%, 30%], and matching exemplar accuracy was 30.3%, 95% CI [25.1%, 33.7%]. These results suggest that adapting to an anti-face made it easier to identify both matching and nonmatching exemplar images, and biased the perception of an average face in the direction of the matching exemplar image.

For the self-norm condition, the baseline threshold was estimated to be 27%, the matching exemplar threshold was estimated to be 17.6%, and the nonmatching exemplar

threshold was estimated to be 25.2% (Figure 4.3B). The identification threshold was reduced from baseline by 9.4% in the matching exemplar condition, and by 1.8% in the nonmatching exemplar condition. The identification accuracy for the average image was at chance for the nonmatching exemplar, and above chance for the baseline and matching exemplar conditions. Specifically, nonmatching exemplar accuracy was 24.1%, 95% CI [19.4%, 26.1%], baseline accuracy was 28.5%, 95% CI [25.4%, 32.4%], and matching exemplar accuracy was 32.2%, 95% CI [30.1%, 38.1%]. These results suggest that adapting to an anti-face made it easier to identify a matching exemplar image. Adaptation to an anti-face biased the perception of an average face in the direction of the matching exemplar image.

Figure 4.3

Study 3 Results



Note. The four-parameter logistic curves for the baseline, matching exemplar, and nonmatching exemplar conditions for (A) the all-norm and the (B) self-norm conditions. The identification thresholds are indicated as dashed vertical lines.

4.4 Discussion

In this study, I investigated whether the architecture of identity-specific subspaces for familiar faces is organized in a norm-based fashion with their average serving as the reference for within-person variability to be encoded. If identity-specific subspaces have norm-based encoding, adapting to an anti-face, which is created by transforming an identity-specific average face away from specific exemplars (i.e., individual images), should make it easier to identify the specific exemplars. I also used gender-matched averages as a control to compare the magnitude of the adaptation effects for self-norm and all-norm anti-faces. I hypothesized that if viewing a familiar face activates the identity-specific average as a reference point, the adaptation effect (i.e., reduction in identification threshold from baseline) should be larger in the self-norm condition than in the all-norm condition. Indeed, the results revealed that the magnitude of the adaptation effect was more than twice as large in the self-norm condition than in the all-norm condition, providing evidence for a norm-based identity-specific subspace architecture. If familiar faces were represented in the same layer of the face space as others, adapting to gender-matched all-norm anti-faces would yield a similar or larger magnitude of adaptation.

In both the all-norm and self-norm conditions, adaptation to an anti-face shifted the appearance of the average face in the direction of the matching exemplar, providing further support for norm-based encoding. Identification thresholds were also reduced for nonmatching exemplars in both conditions, which were test stimuli viewed after adapting to an anti-face that was created by transforming a different exemplar. Interestingly, the magnitude of the adaptation effect for matching and nonmatching exemplars was similar in the all-norm condition, suggesting that adaptation did not increase sensitivity to exemplars systematically. It is possible that adapting to a gender-matched anti-face increased sensitivity

to the target identities generally, but it is unclear why it made it easier to identify specific images of target identities. In contrast, in the self-norm condition, the magnitude of the adaptation effect was trivial for nonmatching exemplars (less than 2%). This means that adapting to an anti-face increased sensitivity to specific exemplars systematically, suggesting that within-person variability is encoded in reference to an identity-specific norm.

Could the data be explained by an exemplar-based model of identity-specific subspaces? Since exemplar-based accounts of the face space assume that a face representation is activated based on its similarity (Euclidean distance) to the test face, adaptation effects would be stronger for test faces that are similar to the anti-faces, irrespective of vector trajectory. Therefore, identification thresholds would increase for exemplars that are perceptually similar to the anti-faces. This leads to the prediction that adaptation effects would be larger in the self-norm condition, as anti-faces created with identity-specific averages bear more resemblance to the exemplars (in fact, self-norm anti-faces appear identifiable; Figure 4.2). This prediction is supported by the current data. However, exemplar-based accounts would also predict larger adaptation effects for the nonmatching exemplars in the self-norm condition than in the all-norm condition because nonmatching exemplars also bear more resemblance to self-norm anti-faces than all-norm anti-faces. This prediction is not supported by the current data – identification thresholds are lower for nonmatching exemplars in the all-norm condition and have larger adaptation effects than the self-norm condition. However, similarity of exemplars and anti-faces is not explicitly tested in current work and is a potential avenue for future research.

A limitation of the present study is that the task was to differentiate different images of the same person, which may be unusual for face recognition in the real world. Indeed, familiar

face recognition relies on amalgamation of exemplars as opposed to differentiation. It would be interesting to replicate this study with a paradigm that requires amalgamation of exemplars – perhaps by incorporating different identities as test stimuli so that identification of a matching exemplar is done by identifying the target identity. It would also be interesting to include a condition that incorporates unfamiliar faces. I avoided doing so for the present study as exposure to multiple ambient images of an unfamiliar face would yield an average representation in memory (Kramer et al., 2015) which may contaminate the adaptation effects.

Averaging has been proposed as an underlying mechanism of face learning (e.g., Kramer et al., 2015), including our own previous work which provided direct evidence for averaging over time with increased exposure to within-person variability (Koca & Oriet, 2023). The averaging account is compelling as averages retain identity-invariant information while discarding image-specific noise. However, what constitutes identity-invariant information is unclear, as many aspects of faces are subject to within-person variability (Burton et al., 2016). It appears that face variability itself is identity-specific (e.g., Jenkins et al., 2011), which suggests that representing averages in the absence of within-person variability is unlikely (Davis et al., 2024). If within-person variability is represented in memory, why do we average? The present work suggests that averaging serves as a reference point for within-person variability to be encoded. Averaging, then, provides a system of organizing within-person variability in memory.

5 General Discussion

Face recognition is a perceptually challenging problem. While different individuals share common patterns (i.e., two eyes above a nose above a mouth), different views of one individual have great variability from one instance to the next (e.g., age, make-up, weight,

lighting, viewing angle). This means that the visual system has to both differentiate similar-looking people, and amalgamate different views of the same person. How does the visual system organize faces in memory to achieve face recognition? In this work, I argue that the face space theory can serve as a unified account for solving both the problems of homogeneity and variability. My use of the adjective “unified” is an intentional one, as it is an homage to the title of Valentine’s paper introducing the face space theory in 1991. This unification involved common phenomena in the face recognition literature that I outlined in the Introduction of this paper. Around the time Valentine’s influential work was published, the problem of face recognition was conceptualized primarily as a problem of homogeneity. In turn, it was assumed that face recognition was achieved by individuation of the members of this homogenous class of visual stimuli. This assumption informed the methods of face recognition research through creating homogeneous conditions by incorporating face stimuli taken under highly controlled settings and by repeating images during study and test to measure recognition. This approach, although incomplete in its assumptions, lent itself to the conceptualization of the face space theory which became fundamental in our (or at least my) understanding of human cognition.

How the problem of face recognition is conceptualized in the literature has recently shifted. What I humorously refer to as the “variability revolution” I believe was marked by the seminal work published by Jenkins and colleagues in 2011¹². When multiple naturalistic photos depicting the same individual were used as stimuli, face perception was greatly challenged by not homogeneity but by *variability*. In other words, observers were able to

¹² I would like to humbly note that this poorly-described brief history of this field is not through my witnessing these shifts but rather inferences I made based on my readings. I was not even an undergraduate student when Jenkins’ paper was published, and was not even a born human when Valentine’s work was published. I apologize for any oversight or incorrect presentation of information.

differentiate similar-looking individuals but could not amalgamate different photos of the same person. This was only true for unfamiliar faces as performance for differentiating and amalgamating was near-perfect when familiar faces were used. This work and the subsequent work of others had important implications for fundamental assumptions made in the field of face recognition research. First, different photos of the same person have immense and idiosyncratic variability; in fact, there is likely as much within-person variability as there is between-people variability. Second, the problem of face recognition may be primarily of variability more so than of homogeneity. Third, familiar and unfamiliar faces are processed differently by the cognitive system, and familiarity can be operationalized as tolerance to within-person variability.

The visual system, then, deals with homogeneity between people which is solved through differentiation, and variability within a person which is solved through amalgamation. These two problems and their respective solutions are seemingly opposite but share the common denominator of similarity: Different people look *similar* to each other, and one person can look *dissimilar* from one encounter to the next. Therefore, the face space theory, which has its structural basis in inter-item similarity, can provide a parsimonious account for solving the problem of face recognition. A similarity-based system is intuitively appealing; similarity is a fundamental Gestalt principle that the visual system uses to organize many domains of visual information¹³. As such, the primary goal of the work I present here is to understand how face familiarity, and by extension, within-person variability, is represented in

¹³ I am curious whether other Gestalt principles are used to organize faces in memory in a multidimensional space. The first principle that comes to mind is the proximity principle. For example, we may encode people that we see together as near neighbours in the face space. Dr. C. Oriet and I have actually discussed this as a group-facilitation of face recognition, like recognizing a member of a group may assist recognition of other members.

the face space. Although Valentine conceptualized the dimensions of the face space as serving to differentiate faces, I argue that the same organizational system can serve to amalgamate different views of the same person.

To understand how familiarity organizes representations of faces in the face space, I conducted three studies that aimed to investigate (a) how a face's location changes in the face space with increased exposure to within-person variability, (b) how within-person variability is represented in the face space for familiar and unfamiliar faces, and (c) the architecture of identity-specific subspaces.

I found that familiarity may not organize faces in the face space by increasing the distance between similar-looking individuals. In a series of studies reported in Chapter 2, I did not find evidence for an increased distance between two similar-looking individuals as a function of increased exposure to within-person variability. Specifically, I predicted that as a target face becomes familiar, a 50-50 morph the target face and a similar-looking foil would be biased either towards the target face (Tanaka et al., 1998) or towards the foil face (Chauhan et al., 2020). Neither of these predictions were supported, and the distinctiveness ratings for familiarized targets did not increase over time. In fact, there was evidence to suggest that they decreased but only by a small amount. Further evidence for this came from the work I reported in Chapter 3. I constructed a multidimensional face space through collecting similarity ratings from pairs of faces depicting multiple naturalistic images of familiar and unfamiliar identities. I found limited evidence supporting increased distance between identities as a function of familiarity. Only when there were fewer exemplars included in the model for unfamiliar faces was there support for larger between-people distances for familiar faces than for unfamiliar faces globally in the face space, but no

increased distance between near neighbours. I do think this is compelling evidence since real psychological face spaces would include fewer representations of exemplars for unfamiliar faces than for familiar faces. However, I did not find consistent evidence in favour of increased between-people distances as a function of familiarity.

Is the problem of homogeneity solved by familiarity? Overall, the evidence from current work suggests that familiarity may contribute little to the distance between identities in face space. This conclusion is at variance with some previous findings suggesting greater distance between individuals as a function of familiarity (Faerber et al., 2016; Kramer et al., 2025; Kramer et al., 2018), but supports other findings suggesting no difference in between-people distances between familiar and unfamiliar faces (Kramer et al., 2025; White et al., 2022). It also aligns with evidence from card sorting tasks where misidentification errors are low even for unfamiliar faces (Jenkins et al., 2011; Laurence et al., 2016), and from training paradigms where training benefits matching task performance through increasing accuracy in match trials (telling people together) but not mismatch trials (telling people apart; Koca & Oriet, 2023; Sandford & Ritchie, 2021; Ritchie et al., 2018). It is therefore possible that the problem of homogeneity is not primarily solved by familiarity. Indeed, many relate to the embarrassing experience of waving at a stranger due to mistaking them as a friend, an error that arises despite high levels of familiarity. If perceptual properties of facial identity contribute more to a face's distance from others than familiarity, as evidenced by the current work, the inconsistency of findings in the literature may partly be a result of identity-specific effects. Perhaps increasing between-people distances with familiarity is most useful for faces that are more typical since distinctive faces are already discriminable. How the face space expands as a function of familiarity is inconclusive from the current work.

Importantly, the findings in the current work revealed that familiarity organizes within-person variability in the face space through decreasing the distance between representations of different images of the same person. In the multidimensional space I constructed in Chapter 3, there was consistent evidence for familiar faces being represented with higher exemplar density, even when the models were re-run with fewer familiar exemplars encoded in the face space. I also found that familiarity organizes within-person variability in reference to an identity-specific prototype (Chapter 4). When adapting to an anti-exemplar of a familiar person's image, the sensitivity for identifying the image increased, and the perception of an average image shifted in the direction of the matching exemplar. The magnitude of this adaptation effect was larger when the adapting anti-exemplar was made from an identity-specific average compared to a gender-matched average. This suggests that familiar faces are represented in distinct subspaces, and viewing a familiar face activates a person-specific prototype as opposed to a general prototype. I did not find that identity-specific averages, however, were more distant from the centroids of familiar face clusters than unfamiliar face clusters in the study reported in Chapter 3. This could be due to the instability of the averages that were created with 7 images, as previous work using 20-image averages did find that familiar face exemplars were rated as more similar to their average (White et al., 2022).

Evidence from this work supports the notion that within-person variability is idiosyncratic. A visual inspection of the dimensions of the face space also revealed that the same dimensions that represent between-people variability represent within-person variability differently. For example, the first dimension seemed to organize faces based on colour information (blonde/brunette), but the same dimension seemed to organize some identities based on factors like head rotation. It is possible that variations like head rotation provide

more information about some identities than others, and as such, contribute more to the variance in within-person similarity ratings. Indeed, I found that distinctiveness ratings (derived from inter-item similarity ratings; Light et al., 1979) varied within person as much as they varied between people, suggesting that different images capture a person's distinctiveness differently. This is consistent with Burton and Vokey's (1998) argument that most faces are distinctive in at least one dimension. This is speculative, however, as I have not systematically tested the dimensions of the face space.

The number of dimensions of the face space has been a topic of interest in previous work (e.g., Lewis, 2004). I found that a 3-dimensional face space may be sufficient to represent distinct identity clusters, and yields similar results regarding within-person and between-people distances as a function of familiarity as a 7-dimensional space. However, this was for a space that represented 8 identities, so a real psychological face space may contain more dimensions. Nevertheless, even for unfamiliar faces, a 3-dimensional space was sufficient to organize within-person variability in a way that created discriminable distances between identities.

There are several limitations in the current work in its assumptions and methodology that should be addressed in future work. First, the approach to the psychological construct of familiarity in this work is rather limited. I operationalized and measured familiarity as binary as opposed to a continuous variable, which does not reflect the range of experiences we have with familiar faces (e.g., Burton, 2025; Kramer et al., 2018). In Chapter 2, building familiarity was operationalized as exposure to within-person variability. Although face images were presented along with person-names while reporting conceptual judgments like attractiveness (Schwartz & Yovel, 2016; 2019), the faces were unfamiliar prior to the experiment and were

only familiarized within the limited context of the experiment. This may explain the divergence of findings from Chauhan et al. (2020)'s work that used personally familiar faces. In Chapters 3 and 4, familiar faces constituted famous celebrities, which again are not personally familiar. Personal familiarity involves not only richer perceptual information, but multimodal representations of conceptual and emotional information (e.g., Natu & O'Toole, 2011). Personal familiarity also involves being known, which creates a plethora of social affordances (Gibson, 1979) that I assume would immensely influence our perceptual processing. In fact, seeing personally familiar non-human objects, like "my dog", can yield neural processing similar to that elicited by familiar faces (Gordon & Tanaka, 2011). As such, the findings from this work may not generalize to personal familiarity to the same extent. I recommend future work to approach familiarity as a continuous variable, and to acknowledge the differences between processing personally familiar, experimentally familiar, and famous faces.

In the same vein, the current work here approaches face processing through an entirely perceptual lens and does not involve or make predictions about social and conceptual representation of faces. In a recent set of studies we conducted in our lab (Ueckert et al., unpublished), we found that people seen under consistent informative contexts are recognized better than people that were presented with richer perceptual variability in the absence of meaningful contextual information. This is not surprising since many difficult perceptual problems, like the problem of invariance in auditory perception, are solved through conceptually-driven processing (e.g., Pollack & Pickett, 1964). I can only assume that context plays an imperative role in establishing familiar face representations. To my knowledge, research investigating the role of context in face recognition is rather sparse, although I have

identified more recent publications (e.g., Myftaraj et al., 2025; Noad & Andrews, 2024) examining this issue.

A major assumption of this work is that similarity in the face space is calculated as a Euclidean distance. In the studies reported in Chapter 2, I assumed that a 50-50 morph between two identities would be equidistant to those identities, an assumption informed by a Euclidean face space. In Chapter 3, all distance calculations were done using the Euclidean distance formula. In Chapter 4, morph continua were created between target images and averages that lie on a Euclidean vector space. Euclidean spaces, however, are homogeneous, essentially “flat”, and do not allow for dimensions to interact with one another (Townsend et al., 2024). Townsend and colleagues recommend investigating the face space as Riemannian manifolds that are more technically dynamic due to their curved nature.

There are practical implications of this research. Faces are an important and ubiquitous aspect of our daily lives. Our day-to-day social interactions rely heavily on our ability to successfully recognize a face. People may feel insulted when we fail to recognize their face, or it may create awkward social situations if we falsely recognize a stranger as someone familiar. Further, recognizing a face not only helps to retrieve the person’s name, but also to access semantic information about the person (e.g., occupation, recent life events) that helps guide social interactions (Bruce & Young, 1986). Beyond social interactions, many important decisions in our society also rely on successfully recognizing faces. For example, our justice system can use eye-witness testimonies to decide whether to convict somebody of a crime. We use photo identification to decide whether someone truthfully represents themselves while entering a country, or to decide whether we should sell them alcohol. We develop computer algorithms to recognize faces for security reasons — most of us even have smartphones that

unlock based on face recognition. We think that capturing a face on a camera recording of a crime means that someone, somewhere will recognize the perpetrator if only the image is circulated widely enough, ignoring whether the image captured is a good likeness of the individual. Understanding how faces become familiar can help develop more effective systems to improve eye-witness testimonies, photo identifications, and security.

To conclude, I argue that the two opposing constituents of the perceptual problem of face recognition (i.e., homogeneity and variability) share the same underlying construct of similarity. As such, the face space theory, initially developed to understand the problem of homogeneity and structurally based in inter-item similarity, can reconcile these two problems of face recognition. In this work, I investigated how familiarity is represented in the face space. I found that familiarity changes how faces are organized in the face space, specifically by compressing within-person variability in identity-specific subspaces that are organized around an identity-specific average. I also found that familiarity may not influence the distance between identities. This work aims to provide a comprehensive model of face recognition that will hopefully inform our understanding of how the human mind organizes and represents information in memory.

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6 Appendix A

6.1 Table 1

Overview of literature discussed

Paper	Main points	Limitations	Implications for the research
Valentine (1991)	Proposing the face space theory as a unifying account for inversion, distinctiveness, caricature, and other-race effects.	Assumes similar properties for representing familiar and unfamiliar faces.	Provides a basis for understanding how a population of faces are represented in memory by conceptualizing a psychological space where faces are encoded based on their similarity to other faces in the face space.
Light et al., (1979)	Inter-item similarity ratings correspond to facial distinctiveness, which predicts recognition performance.	Inter-item similarity ratings were obtained only using a single image per person, which does not account for within-person variability. Stimuli were only unfamiliar faces.	Describes inter-item similarity as a basis for how faces are represented in the face space. This assumption is central to the study I propose in Chapter 2 by collecting inter-item similarity ratings using multiple ambient photos for each identity.
Vokey and Read (1992)	Distinctiveness has two orthogonal components: memorability and familiarity. A distinctive face is high in memorability and low in familiarity.	Distinctiveness was assumed to be an invariant property of a face. Familiarity and memorability judgments were collected using one image per face, which does not account for within-person variability.	Assuming distinctiveness to be composed of memorability and familiarity is most relevant considering unfamiliar faces. Familiar faces are both memorable and familiar, which raises the question of how distinctiveness interacts with familiarity.
Lee et al. (2000)	Distinctiveness judgments of caricatures, anti-caricatures, and veridical images	Judgments were collected using one image per person, which does not account for within-person	The multi-dimensional scaling (MDS) approach will be adopted from this study to model a theoretical face space using inter-item similarity ratings using multiple ambient

Paper	Main points	Limitations	Implications for the research
Lewis and Johnston (1999)	correspond to their theorized locations in the face space with caricatures in less dense regions. The Voronoi-based model was proposed as an extension of the exemplar-based model of the face space where faces are represented as distinct identity regions where multiple inputs can activate a face's representation.	variability. Stimuli were only familiar faces. The size of a region corresponding to an identity is directly determined by distinctiveness, which does not account for familiarity. Since the Voronoi-based space is entirely filled with identity regions, the model cannot give an "unfamiliar" response.	photos for each identity to understand how within-person variability is represented in the face space for familiar and unfamiliar faces. The Voronoi-based model of the face space is a step toward conceptualizing the face space in a way that can accommodate variability across different viewpoints.
Leopold et al. (2001)	Adapting to an anti-face (e.g., anti-Adam) makes it easier to identify the target face (e.g., Adam), and makes an average face more likely to appear as the target face (e.g., Adam), suggesting that the architecture of the face space is norm-based.	Only unfamiliar faces were used, and only one image per identity was used, not accounting for familiarity and within-person variability.	Adaptation paradigms using anti-faces can provide insights into how the face space is constructed. As described in Chapter 4, the methodology of this study will be followed to understand how identity-specific sub-spaces are represented.

Paper	Main points	Limitations	Implications for the research
Tanaka et al. (1998)	Proposed the attractor field model of the face space. Like the Voronoi-based model, faces are represented as identity regions that can accept multiple inputs as the face, accounting for changes in viewpoints.	The size of an attractor field is determined by distinctiveness, which does not necessarily explain the role of familiarity.	The attractor field model can account for within-person variability. Understanding how the size of an attractor field changes as a face becomes familiar would provide insights into how familiarity is represented in the face space. I propose a study based on this in Chapter 3.
Laurence et al. (2016)	An important factor of the other-race effect (ORE) is that tolerance to within-person variability is less for other-race faces, which is predicted by the attractor field model.	Faces that were used in this study were unfamiliar to the participants, so it is not clear how the attractor field model can account for tolerance to within-person variability for familiar faces.	This study provides evidence that the attractor field model can make predictions about how within-person variability can be represented in the face space.
Burton et al. (2016)	Found that a computer model trained to recognize faces make more errors recognizing a non-target face after being familiarized with a target face, providing evidence for idiosyncratic within-person variability.	Does not provide behavioral data.	This paper proposes the existence of identity-specific sub-spaces – since within-person variability is idiosyncratic, each face must be encoded with idiosyncratic dimensions. In the proposed research, I explore the properties of such identity-specific sub-spaces.

Paper	Main points	Limitations	Implications for the research
Burton and Vokey (1998)	Provided a mathematical account that truly typical faces are actually rare in the multidimensional face space, and that most faces are distinctive in at least one dimension.	Does not provide behavioral data.	The assumption that most faces are distinctive in at least one dimension raises the possibility that the perceived distinctiveness of a face can be subject to within-person variability, and that distinctiveness can be a photo-centred property as much as it is a person-centred property. The latter was the assumption in previous studies without explicit consideration of the former. I explore this in Chapter 2.
Faerber et al. (2016)	Familiar faces were rated as more distinctive as their unfamiliar anti-faces, despite being in regions with similar exemplar density in the face space.	Ratings were collected using a single photo per identity, which does not account for within-person variability.	I aim to gain further insight into perceived distinctiveness of familiar faces by collecting inter-item similarity ratings on a stimulus set that incorporates within-person variability, described in Chapter 2. Further, this research suggests that familiar faces migrate to more isolated regions in the face space due to their increased perceived distinctiveness. I propose to test this prediction in Chapter 3.
Chauhan et al. (2020)	Found that identity boundaries are narrower for familiar faces, as a 50-50 morph between familiar and an unfamiliar face was perceived to be more like the unfamiliar face despite being equidistant in the face space.	Observers were asked which of the two <i>identities</i> was more similar to the morph, which means that observers may have relied on different quality representations for the familiar and unfamiliar faces in the study.	The narrowed identity boundary account contradicts the predictions made by the attractor field model, which further raises the question of how a face's representation changes in the face space as it becomes familiar.

Paper	Main points	Limitations	Implications for the research
Laurence et al. (2022)	The magnitude of adaptation is the same for the same identity across different ages. This suggests that a single face representation hosts different ages.	A limitation identified in the context of the current research program is that it is unclear whether similar findings would be obtained for faces that change in dimensions other than age.	The assumption that there is a single representation for each familiar face that hosts different ages provides further support for the existence identity-specific sub-spaces, as it suggests that an identity is encoded with its own set of dimensions. If age was only a common dimension in the face space, there would be multiple representations for one face across different ages.
Koca and Oriet (2023)	An average representation is refined in memory as a face becomes familiar, suggesting that averaging is an underlying mechanism for face familiarization.	It is unclear how exemplars or within-person variability is represented for familiar faces. If exemplars are represented alongside the average, it is unclear why an average would be represented in the first place.	The average may serve as “the norm” for the identity-specific sub-spaces, allowing for within-person variability to be encoded as exemplars in reference to this norm. This norm-based model of the identity-specific sub-spaces are explored in Chapter 4.

Note. The papers are listed roughly in the order they are presented throughout the Introduction. I selected papers that have direct implications for the studies that are conducted.